



Big Bore Family

Version 4.8 SP

PHILIPS

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1 Introduction

The Philips CT systems are advanced continuous-rotation computed tomography systems suitable for a wide range of computed tomographic (CT) applications.

Intended User

Philips systems may be used by a trained healthcare professional.

There are three types of personnel involved in the operation of the CT and/or PET/CT scanners. The Operator interacts directly with the system, and performs the scans. The physician provides clinical interpretation of the scans and/or performs interventional imaging. The physicist is responsible for onsite quality assurance of the system, including calibrations.

About this Reference Guide

This manual is intended to assist users in the safe and secure operation of the equipment described.

This guide is intended to assist with quality assurance testing of the CT scanner. It includes physics information critical to understanding dose, sensitivity, and other scan information subjects. This document also includes classifications, ratings, and the hazard labels and symbols included on your system. You must pay special attention to all the information given, and procedures described in the SAFETY and SECURITY sections.

Before attempting to operate the equipment, you must read, note, and strictly observe all **DANGER** notices and safety markings on the CT System.

Before attempting to operate the equipment, you must read the "Instructions for Use" thoroughly, paying particular attention to all **Warnings**, **Cautions** and **Notes** incorporated in it.

NOTICE

Review this information carefully; be aware of system-specific information where applicable.

Product Family

Big Bore 4.8	6NC
Philips CT Big Bore	728242
Brilliance CT Big Bore Upgrades	728272

IEC-60601 Classification



Type of protection against electric shock	Class I equipment
Degree of protection against electric shock	Type B equipment
Degree of protection against harmful ingress of water	Ordinary equipment (IPX0) Round foot switch (at least IPX1) Continuous CT foot switch (at least IPX1)
Possible interference with other equipment	IEC 60601-1-2 Group 1 Class A Device for Radiated Emission
Mode of operation	Continuous operation (per IEC 60601-1 edition 3.1) Long time operation with momentary loading (per UL/ANSI/AAMI 60601-1 and NFPA 70)

IEC/EN Statement of Compliance

This equipment is compliant to the following standards:

IEC 60601-1:1988+A1:91+A2:95

IEC 60601-1-3:2008

IEC 60601-1:2012

EN 60601-1:2006 + C:2010 + A1:2013

IEC 60601-1-1:2000

IEC 60601-1-2:2000+A1:04

IEC 60601-1-2:2007

IEC 60601-1-2:2014

EN 60601-1-2:2015

IEC 60601-1-3:1994

EN 60601-2-28:2010

IEC 60601-1-3:2008+A1:2013

IEC 60601-1-4:1996+A1:99

IEC 60601-1-6:2006

IEC 60601-1-6:2010+A1:2013

IEC 60601-2-32:1994

IEC 60601-2-44:2001+A1:2002

EN 60601-2-44:2009

IEC 60601-2-44:2009
 IEC 60601-1-6:2010
 IEC 60601-2-44:2009+ A1:2012
 IEC 61223-3-5:2004
 EN 62366:2008
 EN 62304:2006+ Corr.:2008 + A1:2015
 EN 60601-1-6:2010 + A1:2015
 EN 62366-1:2015
 EN 50581:2012
 IEC 60601-2-44:2009+A1:2012+A2:2016
 IEC 60825-1:2014
 EN 60825-1:2014
 EN 60601-2-44 :2009/A11:2011 + A1:2012 + A2:2016
 EN 60601-1-3:2008 + A11:2016

NOTICE

Compliance to 60601-1-2:2014 (4th Ed.) applies after manufacturing date January 1st 2019.
 Refer to system label.

Electromagnetic Emissions

The scanner is intended for use in the electromagnetic environment specified below.

The customer or the user of the system should assure that it is used in a professional healthcare facility. The scanner gantry and patient table must only be used within an X-ray shielded location as specified in the accompanying product literature.

Emissions test	Compliance	Electromagnetic environment - guidance
RF emissions, CISPR 11	Group1 Class A for the scanner in combination with the shielded gantry and patient table location.	CT uses RF energy only for internal function. Therefore, its RF emissions are very low and not likely to cause any interference in nearby electronic equipment.
Harmonic Emissions EC 61000-3-2	Not Applicable	The CT system, when the gantry and patient table are installed in such a shielded location, is suitable for use in all establishments other than domestic and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Voltage fluctuations/flicker emissions IEC 61000-3-3	Not Applicable	

**WARNING**

It is essential that the actual RF shielding effectiveness of the shielded location is verified to establish that it meets the specified minimum values. This is ensured if a lead-shielded patient area is used per the accompanying literature recommendations.

The CT scanner should not be used adjacent to or stacked with other equipment and that if adjacent or stacked use is necessary, the CT scanner should be observed to verify normal operation in the configuration in which it will be used.

The use of accessories, transducers and cables other than those specified, with the exception of transducers and cables sold by Philips as replacement parts for internal components, may result in increased emissions or decreased immunity of the CT scanner.

Electromagnetic Immunity

The CT scanner is equipped for use in the electromagnetic environment specified below. Your facility should ensure that the system is placed in an environment that meets these conditions.

Immunity Test	IEC 60601 test level	Compliance level	Electromagnetic environment-guidance
Conducted RF IEC 61000-4-6	3 Vrms 150 kHz to 80 MHz	3 Vrms on a.c. and d.c. power and signal ports 150 kHz to 80 MHz 6 Vrms on a.c. and d.c. power and signal ports in ISM bands 150 kHz to 80 MHz See Note 4 for list of ISM bands	Only those cables supplied with the CT system should be used.
Radiated RF IEC 61000-4-3	3 V/m 80 MHz to 2.7 GHz	3 V/m 80 MHz to 2.7 GHz	See Note 2 below Interference may occur in the vicinity of equipment marked with the following symbol:
			
Proximity fields from RF wireless communications equipment RF IEC 61000-4-3	380 MHz - 5800 MHz	380 MHz - 5800 MHz 9-28 V/m	IEC 60601-1-2, Edition 4.0, Table - 9 Test specifications for ENCLOSURE PORT IMMUNITY to RF wireless communications equipment

Immunity Test	IEC 60601 test level	Compliance level	Electromagnetic environment-guidance
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Note 1: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.

Note 2: CT has not been tested for radiated RF immunity over the entire frequency range of 80 MHz to 5.8 GHz. It was step tested at the selected frequencies 88 to 108, 144 to 148, 151, 185, 380 to 390, 425, 430 to 470, 468, 704 to 787, 800 to 960, 902 to 928, 939, 1700 to 1990, 2400 to 2570, and 5100 to 5800 MHz respectively, at minimum of 3 V/m field strength. Test signal was either 80% modulated AM, at 2Hz, 1000 Hz modulation frequencies or Pulse modulation at 18 Hz or 217 Hz.

Note 3: Only the equipment specified in the CT Installation Manual may be used inside the gantry and patient table room.

Note 4: The ISM (industrial, scientific and medical) bands between 0.15 MHz and 80 MHz are 6765 MHz to 6795 MHz; 13553 MHz to 13,567 MHz; 26,957 MHz to 27,283 MHz; and 40.66 MHz to 40.70 MHz. The amateur radio bands between 0.15 MHz and 80 MHz are 1.8 MHz to 2.0 MHz, 3.5 MHz to 4.0 MHz, 5.3 MHz to 5.4 MHz, 7 MHz to 7.3 MHz, 10.1 MHz to 10.15 MHz, 14 MHz to 14.2 MHz, 18.07 MHz to 18.17 MHz, 21.0 MHz to 21.4 MHz, 24.89 MHz to 24.99 MHz, 28.0 MHz to 29.7 MHz and 50.0 MHz to 54.0 MHz.

Electrostatic discharge (ESD) IEC 61000-4-2	±8 kV contact ±15 kV air	±8 kV contact ±15 kV air	
Electrical fast transient/burst IEC 61000-4-4	±2 kV for power supply lines ±1 kV for input/output lines	±2 kV for power mains input ±1 kV for signal ports with 100kHz repetition frequency	Main's power quality must comply with the CT planning reference data (PRD).
Surge IEC 61000-4-5	±1 kV lines to lines ±2 kV to earth	±1 kV lines to lines ±2 kV to earth	Main's power quality must comply with the CT planning reference data (PRD).
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	0 % U_T for 5 s	0 % U_T for 5 s	Main's power quality must comply with the CT planning reference data (PRD). If the user of the CT requires continued operation during power mains interruptions, it is recommended that the CT be powered from an uninterruptible power supply (UPS).
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	30 A/m, 50/60 Hz	30 A/m, 50/60 Hz	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.

**WARNING**

Your Clinical suite may include one (or several) third-party UPS devices.

When UPS batteries are not properly maintained, or if they are held in service beyond their usable service life, failure can result in the leaking of electrolyte (sulfuric acid), overheating, and/or the emission of fumes.

To ensure continued safe and reliable performance from these devices, periodic maintenance is required, including possible battery replacement. Based on industry standards, the typical useable service life of a UPS battery is less than five years.

You may consult your local Philips Service representative for help in identifying the specific model of your UPS device(s) and available service provider options in your geography.

Electrical Ratings

Voltage (VAC)	Phase	Frequency (Hz)	Power consumption (kVA)	
			Continuous (IEC) long time (UL/NFPA 70)	Short time (IEC) momentary (UL/NFPA 70)
200	3	50/60	15	110
208	3	50/60	15	110
240	3	50/60	15	110
380	3	50/60	15	110
400	3	50/60	15	110
415	3	50/60	15	110
440	3	50/60	15	110
480	3	50/60	15	110
500	3	50/60	15	110

CT Scanner - IEC 60601-2-44: The apparent resistance of the supply mains shall not exceed the values provided in the System planning reference data (PRD).

**WARNING**

To avoid risk of electric shock, this equipment must be connected to a supply mains with protective earth. See the power requirements document for details.

Electric Output Data, per IEC 60601-2-44

Stated values refer to the complete device for the CT scanner in which part of the high voltage generator is integrated with the X-ray tube assembly.

1) NOMINAL X-RAY TUBE VOLTAGE and highest X-RAY TUBE CURRENT obtainable from the HIGH-VOLTAGE GENERATOR when operated at that X-RAY TUBE VOLTAGE	120kV; 665mA
Highest X-RAY TUBE CURRENT and the highest X-RAY TUBE VOLTAGE obtainable from the HIGH-VOLTAGE GENERATOR when operated at that X-RAY TUBE CURRENT	665mA; 120kV
Combination of X-RAY TUBE VOLTAGE and X-RAY TUBE CURRENT which results in the highest electric output power	120kV at 665mA = 80kW or 140kV at 571mA = 80kW
NOMINAL ELECTRIC POWER given as the highest constant electric output power (kW) which the HIGH-VOLTAGE GENERATOR can deliver for a LOADING TIME of 4 seconds at an X-RAY TUBE VOLTAGE of 120 kV, or nearest to 120 kV and the value of LOADING TIME nearest to but not less than 4 seconds	80kW
NOMINAL ELECTRIC POWER given together with the combination of X-RAY TUBE VOLTAGE and X-RAY TUBE CURRENT and the LOADING TIME, which are used in the CT scanner	80 kW, 140kV, 571mA, 4.1 sec

2 Labels and Symbols

System Labels

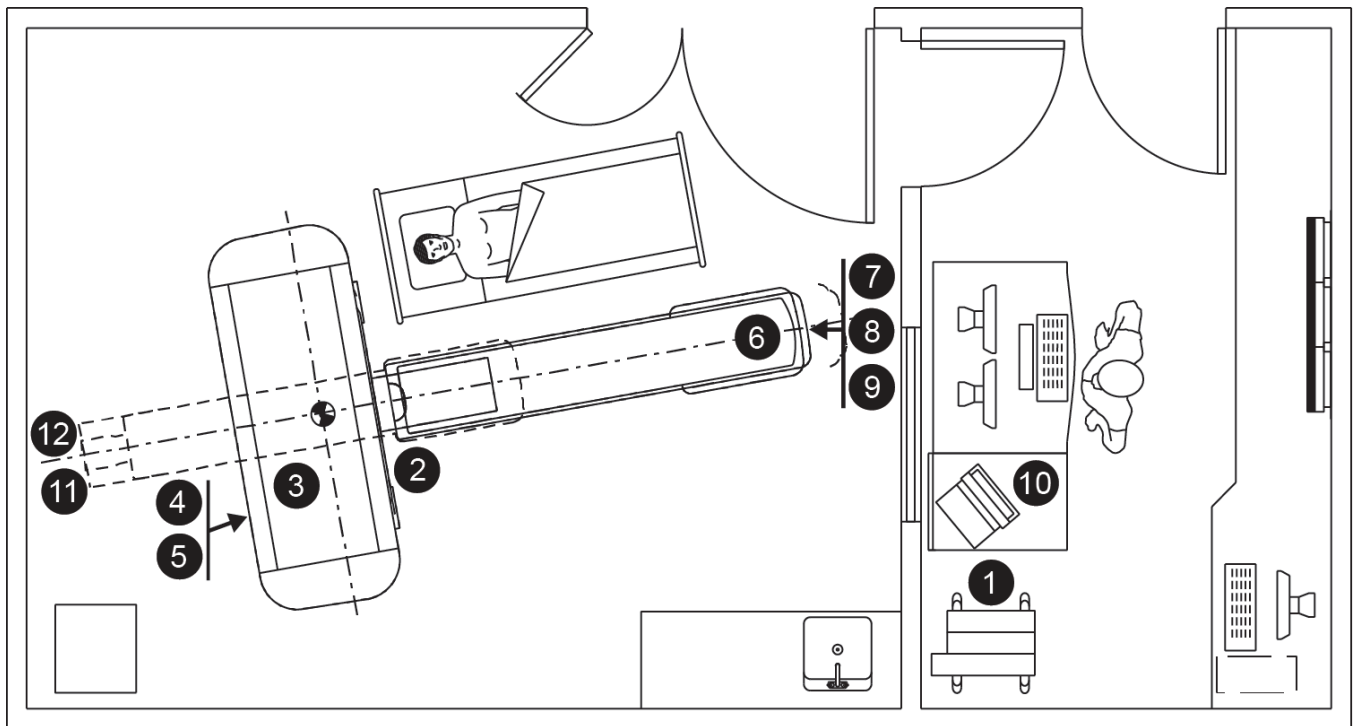
Philips Healthcare products are all designed to meet stringent safety standards. However, all medical electrical equipment requires proper operation and maintenance, particularly with regard to human safety.

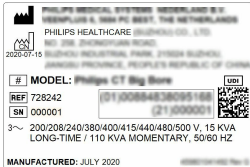
It is vital that you read, note, and where applicable, strictly observe all DANGER notices and safety markings on the CT system.

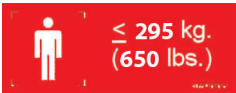
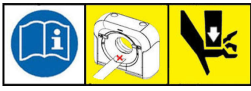
NOTICE

Depending on the system configuration and layout at your site, not all of the labels are applicable.

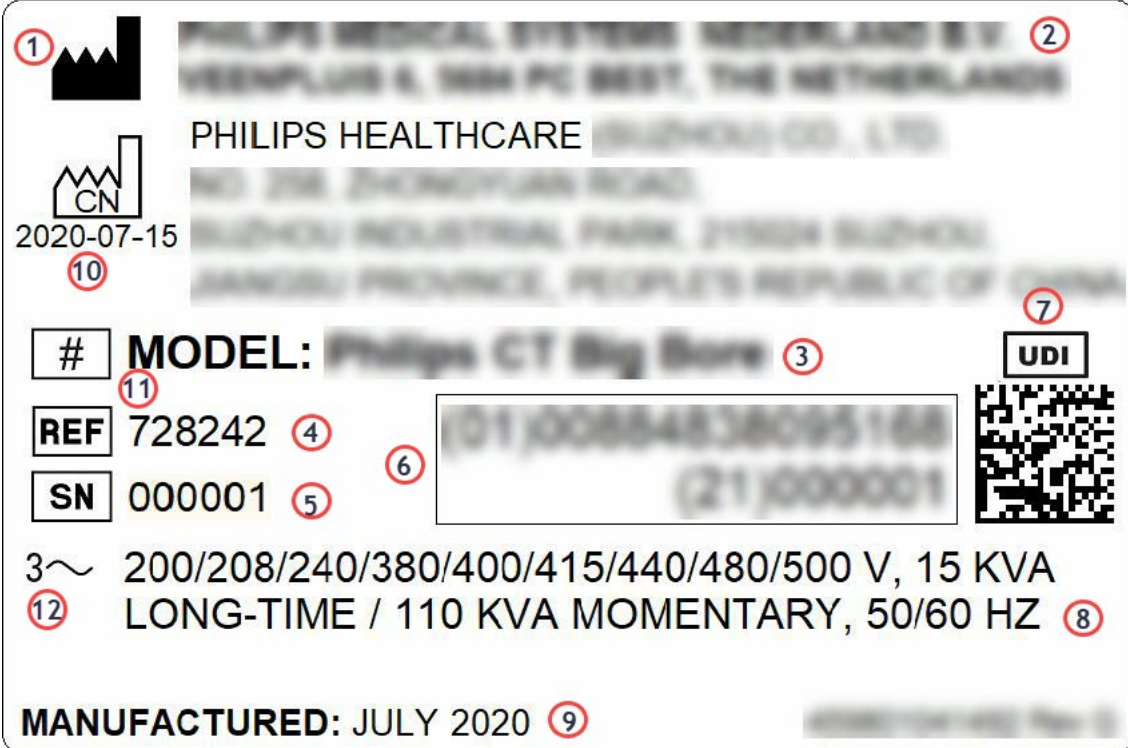
The following diagram displays the labels and signs visible in the Scanner Suite.



Number	Label/Sign	System	Location	Description
1		BR16, BR64, BR Big Bore, Ingenuity, BR iCT	Console UPS	WARNING: UPS voltage may be present when power is off. Contact may cause electric shock or burn. Turn off and lock out UPS output power before servicing.
2		Ingenuity, BR Big Bore	Front of Gantry	CAUTION - Laser Radiation. Do not stare into beam. Class 2 laser product, 1mW maximum output. Wavelength 650nm. IEC/EN 60825-1:2014
3		BR16, BR64, Ingenuity	Front and Rear of Gantry	CAUTION: Pinch point. Keep hands clear during operation.
4		BR Big Bore	Front and Rear of Gantry	CAUTION: Pinch point. Keep hands clear during operation.
5		BR16, BR64, BR Big Bore, Ingenuity	Rear of Gantry	WARNING: HAZARDOUS VOLTAGE. Electric shock hazard exists behind cover. Restrict service to qualified personnel only. Can result in electrocution or death.
6		BR Big Bore	Rear of Gantry	Unique Device Identification (UDI) label. See below for full description.
7		Ingenuity, BR Big Bore	Rear of Gantry	CAUTION - Class 3B invisible laser radiation when open and interlocks defeated. Avoid exposure to the beam and use laser safety goggles. 150 mW maximum output - wavelength 850 nm. EN/IEC 60825-1:2014
8		BR16, BR64, BR Big Bore, Ingenuity, BR iCT	Sides of Patient Table	Pressing a tape switch unlocks the table from its driving mechanism and allows for manual movement.
9		BR16, BR64, Ingenuity, BR iCT	End of Patient Table	WARNING: Maximum patient table load is 204 kg, 450 lb

Number	Label/Sign	System	Location	Description
10		BR64, Ingenuity, BR iCT	End of Patient Table	WARNING: Maximum patient table load is 200 kg, 440.8 lb
11		BR Big Bore, BR64, Ingenuity, BR iCT	End of Bariatric Patient Table	WARNING: Maximum bariatric patient table load is 295 kg, 650 lb
12		BR64, Ingenuity, BR iCT	Console Rack	WARNING: For connection of Philips Healthcare specified equipment only. Read the installation manual before making any connections.
13		BR64, Ingenuity, BR iCT	Head holder	Maximum weight of 46 kg
14		BR64, Ingenuity, BR iCT	Foot extension pad	Label location CAUTION: Pinch point. Keep hands clear during operation.

Unique Device Identification Label



①   ②
 PHILIPS MEDICAL SYSTEMS NETHERLAND B.V.
 VERENPLAAS 4, 3824 PC BEST, THE NETHERLANDS
 PHILIPS HEALTHCARE (SUZHOU) CO., LTD.
 NO. 208 ZHONGYUAN ROAD,
 SUZHOU INDUSTRIAL PARK, 215024 SUZHOU,
 JIANGSU PROVINCE, PEOPLE'S REPUBLIC OF CHINA
 2020-07-15 ⑩
 # MODEL: Philips CT Big Bore ③
 REF 728242 ④
 SN 000001 ⑤
 ⑥ (01)00884838095168
 (21)000001
 ⑦ UDI
 ⑧ 3~ 200/208/240/380/400/415/440/480/500 V, 15 KVA
 LONG-TIME / 110 KVA MOMENTARY, 50/60 HZ
 ⑨ MANUFACTURED: JULY 2020

Number	Description
1	Manufacturing site of the finished device.
2	Site of legal manufacturer, the entity responsible for placing the device on the market.
3	Model name of the system.
4	Reference or catalog number.
5	Serial number.
6	Global Trade Identification Number (GTIN) as Device Identifier (DI) and Production Identifier (PI).
7	Unique Device Identification GS1 2D DataMatrix barcode (contains information from 7).
8	Electrical rating specific to system.
9	Month and year of device manufacture as per 21 CFR.
10	Factory symbol
11	Model number
12	Three-phase alternating current

**COMPLIES WITH THE FOLLOWING STANDARDS
IN EFFECT AT THE DATE OF MANUFACTURE:**

**EN/IEC/UL/AAMI 60601-1
CAN/CSA C22.2 NO. 60601-1**



**RADIATION PERFORMANCE, PER
FDA 21 CFR SUBCHAPTER J. MINIMUM X-RAY
FILTRATION > 2.5MM AL, PER DIN 6815**

**LASER PERFORMANCE PER FDA 21 CFR 1040.10
AND 1040.11 EXCEPT FOR DEVIATIONS PURSUANT
TO LASER NOTICE NO. 56, DATED 8 MAY 2019.
CERTIFIED TO EN/IEC 60825-1.**

Rx ONLY



www.philips.com/IFU

Number	Description
1	Medical Device
2	Electronic IFU

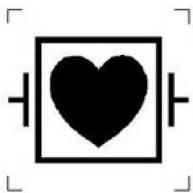
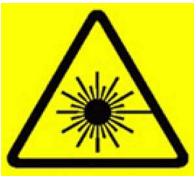
Symbols

Use the following information to interpret the symbols used on the system and accessories.

IEC 60601-1, 3rd Edition Symbols

For each symbol, the table below describes the standard in which the symbol is defined — ISO 15223-1:2012 and ISO 15223-1:2016 "Medical devices - Symbols to be used with medical device labels, labeling, and information to be supplied - Part 1: General requirements".

Symbol	Symbol Reference Number	Symbol Title	Description
	ISO 7000-6042	Caution, risk of electric shock	Identifies equipment, for example the power source, that has risk of electric shock.
	ISO 7000-5638	Emergency stop	Identifies an emergency stop control device. Found on the red buttons located on the gantry and the System Scan Control Box.
	ISO 7000-0434A	Caution	Indicates that caution is necessary when operating the device or control close to where the symbol is placed or that the current situation needs operator awareness or operator action in order to avoid undesirable consequences.

Symbol	Symbol Reference Number	Symbol Title	Description
	ISO 7000-1641	Operator's manual; operating procedures	Identifies the location where the operator's manual is stored or to identify information that relates to the operating instructions. Indicates that the operating instructions should be considered when operating the device or control close to where the symbol is placed.
	ISO 7010-M002	Refer to instruction manual/booklet	Signifies that the instruction manual/booklet must be read and the warnings and instructions therein followed.
	ISO 7000-5840	Type B applied part	Identifies a type B applied part complying with IEC 60601-1.
	ISO 7000-5336	Defibrillation-proof type CF applied part	Identifies a defibrillation-proof type CF applied part complying with IEC 60601-1.
	ISO 7000-5390	Patient, normal; person, general	Indicates a reference to a normal patient, or to indicate a person and human body.
	ISO 7000-1329	Warning, laser beam	Indicates potential harm from the presence of a laser beam or radiation from the laser apparatus.

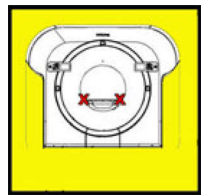
Symbol	Symbol Reference Number	Symbol Title	Description
	ISO 7000-5339	Emitting X-ray source assembly	Indicates the emission or the imminent emission of X-radiation.
	ISO 7000-2497	Date of manufacture	The date when the medical device was manufactured. The associated date is presented in YYYY-MM-DD format.
	ISO 7000-3082	Manufacturer	Indicates the medical device manufacturer (i.e. the entity placing the medical device on the market).
	ISO 7000-5665	Body weight	Identifies the control or the indicator to enter or call up the body weight of a person.
	ISO 7000-2498	Serial number	Indicates the manufacturer's serial number so a specific medical device can be identified.
	ISO 7000-2493	Catalogue number	Indicates the manufacturer's catalog number so that the medical device can be identified.

ANSI and Other Symbols

Symbol	Standard	Symbol Description
	ANSIZ535.4 (ISO 3864-2)J6737	CAUTION: Hand pinch point or crush hazard
	N/A	CAUTION: Hand pinch point or crush hazard
		
	European Union directives	Product complies with the requirements of the applicable European Union directives.
	CSA certification	This product complies with standards of CSA certification in United States and Canada.
Rx ONLY	N/A	This product is available by Prescription only.
	N/A*	CAUTION: Be aware of possible pinch points between the foot extension and the gantry.

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Symbol	Standard	Symbol Description
	N/A*	CAUTION: Be aware of possible pinch points between the therapy top and the gantry.
	ANSI Z535.2-2011	Radiation

NOTICE

This statement applies to the symbols with the following in the Standard column: N/A*.
This symbol is not included in a standard recognized by the FDA under section 514(c) of the FD&C Act. The symbol is likely to be read and understood by an ordinary individual under customary conditions of purchase and use in compliance with section 502(c) of the Act.

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Philips

3 Safety

Important Safety Directions

Philips Healthcare products are all designed to meet stringent safety standards. However, all medical electrical equipment requires proper operation and maintenance, particularly with regard to human safety.

It is vital that you read, note, and where applicable, strictly observe all **DANGER** notices and safety markings on the CT system.

It is vital that you follow strictly all safety directions under the heading **Safety** and all **Warnings** and **Cautions** throughout this document to help ensure the safety of both patients and operators.

In particular, you must read, understand and know the **Emergency Procedures** described in this **Safety** section before attempting to use the equipment for any patient examination. You should also note the following information given in the **Instructions for Use**:

- Intended purpose of the Philips CT system (Refer to the Instructions For Use)
- Training for operators of the Philips CT system (Refer to the Instructions For Use)



WARNING

The X-ray unit may be dangerous to patient and operator unless safe exposure factors, operating instructions, and maintenance schedules are observed.



WARNING

Do not use the CT system for any application until you are sure that the Image Performance Quality Assurance has been satisfactorily completed, and that the Preventative Maintenance Program is up to date. If any part of the equipment or system is known (or suspected) to be operating improperly or wrongly-adjusted, DO NOT USE the system until a repair has been made.



WARNING

Operation of the equipment or system with improperly-operating or wrongly-adjusted components could expose the operator or the patient safety hazards. This could lead to fatal or other serious personal injury.

You can find information about the image-performance quality assurance and the preventative maintenance program in the **Quality Assurance** and **User Information** sections.

**WARNING**

Do not use the CT system for any application until you have received adequate and proper training in its safe and effective operation. If you are unsure of your ability to operate this equipment safely and effectively **DO NOT USE IT**. Operation of this equipment without proper and adequate training could lead to fatal or other serious personal injury. It could also lead to clinical misdiagnosis.

For information about training, please refer to **Training** in the **Introduction** section in the **Instructions for Use**.

**WARNING**

Never attempt to remove, modify, over-ride or forcibly move any safety device on the equipment. Interfering with safety devices could lead to fatal or other serious personal injury.

**WARNING**

Do not use the CT system for any purpose other than those for which it is intended. Operation of the CT system for unintended purposes, or with incompatible equipment, could lead to fatal or other serious injury. It could also lead to clinical misdiagnosis.

Intended use of the CT system is described under the heading **Indications for Use Statement** in the **Introduction** section of the **Instructions for Use**. Compatibility is discussed under the heading **Compatibility** in the **Introduction** section of this document.

Scanner Suite

Familiarize yourself with the scanner suite at your site:

- The wall-mounted emergency stop removes the power supply for the entire CT system. Gantry movement and X-ray generation stops immediately.
- Route all system cables and patient tubing so that it does not become damaged or impede the free movement of personnel.
- If installed, the door-switch interlock helps avoid unnecessary radiation.

**WARNING**

Do not use the CT system for any application until you read, understand, and know all the safety information, safety procedures and emergency procedures contained in this **SAFETY** section. Operation of the CT system without a proper awareness of how to use it safely could lead to fatal or other serious personal injury.

Emergency Procedures

Emergency Stop

To bring scanner and Patient table movements and X-ray production to an immediate halt, press one of the red **Stop** buttons. One button is located on the Scan control box, and one on each gantry control panel.

Reset from Emergency Stop

Use this procedure to reset from emergency stop:

1. Locate the button that was pressed to initiate the stop.
2. Turn the button until it disengages from the stop position and returns to its original position.
3. Turn the key clockwise on the scan control box.

**WARNING**

After the Stop button is pressed, the table is locked in place for two seconds. Then it will be free floating with no up/down capabilities. Make sure that you maintain control of the table so that it does not move.

**WARNING**

During all movements of the gantry (automatic and manual) and the patient table, keep the patient under continuous observation to avoid pressing the patient against the gantry or between table parts, as well as to avoid disconnecting any infusion or resuscitation apparatus.

**WARNING**

Make sure that the motion of the table is in the direction that will ensure that the patient can be easily released and will not get pressed against the gantry covers.

Emergency Patient Release

If the patient's head is lying on one side of the gantry opening and the trunk and legs are lying on the other side of the opening, the patient should be released in the direction of the patient table support.

If the head is likely to touch the roof of the gantry opening, lower the head by removing the head support or the pillow, and turn the head to the side before moving the patient table.

To release the patient **in the event of a power failure or in an emergency stop situation**, either **pull the patient out** or **push the patient in** according to the procedure outlined in the applicable sections.

NOTICE

In the event of a power failure or an emergency stop, you cannot move the patient table down. Be prepared to help the patient from the table.

Pulling the Patient Out

1. Grasp the handle at the end of the patient table.
2. If the patient can safely be pulled out, pull the patient table out.
3. Help the patient dismount.

Pushing the Patient In

1. Grasp the handle at the end of the patient table.
2. If the patient can be safely pushed in, push the patient table towards the back of the gantry.
3. Help the patient dismount.

Rapid or Emergency release of the patient table can also be achieved by grabbing the tape switches along either side of the table, or by pressing one of the foot switches. This unlocks the table from its driving mechanism and allows it to be manually extended or retracted.

NOTICE

For Safety purposes, the Tape and Foot Switches act as an Emergency Clutch of the main patient table unit, preventing dangerous outward movement.

Electrical Safety & Grounding

Covers or cables should only be removed by qualified and authorized service personnel.

**WARNING**

Do not remove covers or cables from this equipment. High electrical voltages are present within this equipment. Removing covers or cables could lead to serious or fatal personal injury.

Only use this equipment in rooms or areas that comply with all applicable laws (or regulations which have the force of law) concerning electrical safety for this type of equipment.

This equipment is permanently installed, grounded equipment (IEC Class I). It employs redundant protective earth connections to maintain safe operation. The mains power supply and grounding connections must conform to the power requirements, site planning and installation documents for this product.

Do not connect this equipment or any of its components to a wall receptacle. Approved accessories (such as injectors) should be connected to grounded wall receptacles per the accessory equipment's instructions for use.

Mechanical Safety

Covers should only be removed by qualified and authorized service personnel.

**WARNING**

Do not remove covers from this equipment. Removing covers could lead to serious or fatal personal injury.

Explosion Safety

This equipment must not be used in the presence of explosive gases or vapors, such as certain anaesthetic gases. Use of electrical equipment in an environment for which it was not designed can lead to fire or explosion.

**WARNING**

Flammable or potentially explosive disinfecting sprays must not be used, since the resultant vapor could ignite, causing fatal or other serious personal injury and/or damage to equipment.

Implosion Hazard



WARNING

Do not subject the system to serious mechanical shock, as the cathode ray tube (CRT) can fracture if struck or jarred. This may result in flying pieces of glass and phosphor coating that can cause serious injury.

Fire Safety

Use of electrical equipment in an environment for which it was not designed can lead to fire or explosion.

Conductive fluids that seep into the active circuit components of the operator's console may cause short circuits that can result in electrical fires. Therefore, do not place any liquid or food on any part of the consoles or other modules of the system.

Fire regulations for the type of medical area being used should be fully applied, observed and enforced. Fire extinguishers should be provided for both electrical and non-electrical fires.

All operators of this medical electrical equipment should be fully aware of and trained in the use of fire extinguishers and other fire-fighting equipment, and in local fire procedures.



WARNING

Only use extinguishers on electrical or chemical fires which are specifically labelled for those purposes. Using water or other liquids on an electrical fire can lead to fatal or other serious personal injury.

If it is safe to do so, attempt to isolate the equipment from electrical and other supplies before attempting to fight a fire. This will reduce the risk of electric shocks.

Electromagnetic Compatibility

The Philips CT system complies with the requirements of applicable EMC standards.

Mobile Telephones and Similar Products

Portable and mobile radio frequency (RF) communications equipment can affect the CT system.

**WARNING**

You should not allow portable radio transmitting devices (such as mobile telephones) into the examination room - whether switched on or off. Such devices could exceed EMC radiation standards could interfere with the proper functioning of the CT system. This could, in extreme cases, lead to fatal or other serious personal injury or to clinical mis-diagnosis. Portable RF communications equipment (including peripherals such as antenna cables and external antennas) should be used no closer than 30 cm (12 inches) to any part of the CT system, including cables specified by the manufacturer. Otherwise, degradation of the performance of this equipment could result.

Electronic and Implanted Stimulators

Emissions from the CT system may affect other electronic equipment that does not meet the EMC immunity limits.

The FDA Preliminary Public Health Notification: Possible Malfunction of Electronic Medical Devices Caused by Computed Tomography (CT) Scanning, July 14, 2008, advises that with any CT scanner, there is a possibility that the X-rays used during CT examinations may cause some implanted and external electronic medical devices (pacemakers, defibrillators, neurostimulators, and drug infusion pumps) to malfunction.

Philips recommends that users check the device manufacturer's recommendations/precautions regarding use in a CT Scanner. Further, the following FDA recommendations should be considered.

Recommendations prior to scan:

1. Ask the patient if he/she has any implanted or external electronic medical devices.
2. Use CT Surview scans to determine if implanted or externally worn electronic medical devices are present and if so, their location relative to the programmed scan range.
3. For CT procedures in which the medical device is in, or immediately adjacent to, the planned scan range, make these adjustments:
 - Determine the device type.
 - If practical, try to move external devices out of the scan range.
 - Minimize X-ray exposure to the implanted or externally worn electronic medical device by using the lowest possible X-ray tube current consistent with obtaining the required image quality, and making sure that the X-ray beam scans over the device for less than a few seconds.

**WARNING**

For CT procedures that require scanning over the medical device for more than a few continuous seconds (as with CT perfusion or interventional exams) users should prepare to treat possible adverse reactions.

Recommendations after scan:

1. Have the patient turn the device back on if it had been turned off prior to scanning.
2. Have the patient check the device for proper functioning.
3. Advise the patient to contact his/her health care provider as soon as possible if he/she suspects the device is not functioning properly after a CT scan.

Radiation Safety

X-ray and gamma rays are dangerous to both operator and others in the vicinity unless established safe exposure procedures are strictly observed.

**WARNING**

To avoid overexposure to radiation, ensure the scan room is clear of personnel during servicing and related service scanning. Follow the procedures established for your site.

The useful and scattered beams can produce serious or fatal bodily injuries to patients and persons in the surrounding area if used by an unskilled operator. Adequate precautions must always be taken to avoid exposure to the useful beam, as well as to indirect radiation including scattered radiation from within the scanner as well as anything in the path of the beam.

Those authorized to operate, participate in or supervise the operation of the equipment must be thoroughly familiar and comply completely with the current established safe exposure factors and procedures described in publications, such as the "Diagnostic X-ray systems and their major components," section of subchapter J of Title 21 of the Code of Federal Regulations, and the National Council on Radiation Protection (NCRP) No. 102, "Medical X-ray and gamma ray protection for energies up to 10 MEV equipment design and use," as revised or replaced in the future.

Operators are strongly urged to comply with the current recommendations of the International Commission on Radiological Protection, or in Japan, the Medical Law and its enhanced regulations, or in the United States, with those of the US National Council for Radiological Protection.

- ICRP, Pergamon Press, Oxford, New York, Beijing, Frankfurt, Sao Paulo, Sydney, Tokyo, Toronto
- NCRP, Suite 800, 7910 Woodmont Avenue, Bethesda, Maryland 20814, USA

Individuals responsible for the planning of X-ray and gamma ray equipment installations must be thoroughly familiar and comply completely with NCRP No. 49, "Structural shielding design and evaluation for Medical of X-rays and gamma rays of energies up to 10 MEV," as revised and replaced in the future.

In Japan, users should refer to Medical Law and its enhanced regulations, Laws Concerning the Prevention from Radiation Hazards due to Radioisotopes and Others and its enhancement regulations, Industrial safety and Health Law, Laws Concerning the Prevention from Electrical Dissociation Radiation Hazards, Ordinance by Local Government on Fire Prevention and Dangerous Article.

Failure to observe these warnings may cause serious or fatal bodily injuries to the operator or those in the area.

Radiation Warning Lamps

The radiation warning lamps on the gantry panels, on the scan control panel, as well as site radiation warning lamps, must light up if scanning has been triggered.

If a radiation warning lamp does not light up:

- Shut down the system immediately and contact Customer Service.
- Press the **Emergency Stop** button if there is danger to you or the patient.

Installation and Environment

Except for installations requiring certification by the manufacturer per United States Federal Performance Standard, see that a radiation protection survey is made by a qualified expert in accordance with NCRP 012, Section 7, as revised or replaced in the future.

Perform a survey after every change in equipment, workload or operating conditions which might significantly increase the probability of persons receiving more than the maximum permissible dose equivalent. In Japan, report the installation to the Competent Authority.



WARNING

To avoid potential injury, do not attempt to unpack any part of the CT system. Unpacking and installation of the system must be completed by a qualified expert. Contact Philips Service for further information.

Coolant Leaks

Parts of your CT system are liquid-cooled. This is a closed-circuit, sealed system.



CAUTION

If coolant leaks are detected, shut down the scanner and immediately contact the nearest Philips field service office.

Laser Safety



WARNING

Follow these Laser Safety warning instructions:

- Do not stare into the laser beam and instruct the patient not to stare into the beam.
- The use of optical instruments, such as eyeglasses with large diopter or mirrors, with this product will increase eye hazard.
- Ensure that the Patient wears protective glasses specific to the laser radiation generated by the system in any situation where direct exposure of the patient's eyes from the laser beam is possible.



CAUTION

Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

Protection Measures

Take the following protection measures to protect both yourself and the patient.

Only the patient should be in the scan room for scanning. Anyone who has to be near the patient during scanning must wear protective clothing (lead apron), wear a PEN dosimeter and/or film badge, and stay in the zone shielded by the system (to the side of the gantry or behind a mobile protective wall).

The physician is responsible for protecting the patient from unnecessary radiation.

- Use protective shielding whenever appropriate to minimize dose to sensitive organs.
- Use the applicable exams for children.

Phantom Handling

The liquid-filled phantoms require no special precautions during normal daily use. However, they do contain a specialized aqueous solution (phantom liquid), which is mostly water, developed specifically for Philips scanners which requires personal protection.

**WARNING**

Follow these Safety warning instructions when refilling or when coming into contact with the phantom liquid:

- Handle an opened or damaged phantom with care.
- The phantom solution is a strong eye irritant and may be harmful to skin.
- Exercise care in handling: wear protective gloves and safety glasses.
- The solution contains surfactant and biocide.
- If the solution splashes on eye or skin, immediately wash with clear water for at least 10 minutes.

Residual Risks to be Considered

Residual Thermal Hazards

Though the system is designed to zeroize thermal hazards, the following residual risks must be considered:

- When covers are removed, service personnel may be exposed to thermal hazards. The system complies with the IEC60601-1 clauses for internal markings and service manuals have warning instructions for these hazards which should be followed by service personnel to ensure their safety.
- During servicing, avoid touching components that can be hot, e.g., X-ray tube, High Voltage Generator (Power Block Booster [PBB] and Power Block Unit [PBU]), DMS Assembly, Heat Exchanger, and Linear Induction Motors (LIMs). The compressor motor surface can get hot during operation. Make sure the compressor has cooled before removing air inflect filters. Be sure to allow the components listed above to cool to a safe temperature before performing any maintenance procedure on them.
- Wear safety glasses with side shields while working on the gantry with covers open and power on, in the unlikely event that an anode crack failure in the tube causes a failure of the seals, and coolant (Tyfocor®) maybe released from the tube.
- If burning smell, smoke or any sign of fire or flame or electrical sparks are detected, remove personnel from the immediate vicinity, turn off power to the system and contact Philips service.
- This equipment must not be used in the presence of explosive gases or vapors, such as certain anesthetic gases. Use of electrical equipment in an environment for which it was not designed can lead to fire or explosion.

Residual Risks Related to Moving Parts

Though the system is designed to zeroize risks because of moving parts, the following residual risks must be considered:

- For patients connected to life support systems, extra care should be taken by the operator to ensure that the all connections are positioned in a manner to avoid pulling or disconnecting during the scan. During all movements of the gantry (automatic and manual) and the patient table, keep the patient under continuous observation to avoid pressing the patient against the gantry or between table parts.
- Make sure that the motion of the table is in the direction that will ensure that the patient can be easily released and will not get pressed against the gantry covers.
- For all patient table types, take care when using attachments (such as the head and foot holders) to avoid collision with the gantry. Non-original patient supports may cause danger for the patient through collisions with the gantry. Positioning aids must be used exclusively for their intended purpose: head holder only for positioning the head, table top extension only for positioning the feet.
- Be aware of possible pinch points between the foot extension and the gantry
- Ensure that the infant cradle does not collide with the gantry during table top movement
- Make sure that the patient is strapped securely to avoid dangling of the hands.
- When using the radiation Therapy Table Top, users should be aware of possible pinch points
- During studies, the patient table and gantry movements (if applicable) are automatic. Ensure enough clearance between the patient and the gantry. Before initiating the scan, perform manual movements to check the clearance. Auto scan means that automatic motions are expected without using the enable button.
- While moving the table or Gantry, avoid placing your feet under the table side covers or between the Gantry and Patient table. Avoid inserting your fingers between the table top and the table carriage. Avoid placement of ancillary equipment (such as wheelchairs, IV pumps or beds) under the table. The table could collide with these items during movement.
- Use caution when opening or closing the front cover. When lifting the front cover make sure to stand aside and let the cover lift to the fully open position. The locking pins **MUST** be engaged whenever the front cover is fully opened for servicing. Failure to comply may result in serious injury to service personnel.
- There are multiple pinch points underneath the gantry covers. Pay attention to all safety labels and follow service instructions to minimize risk of injury.
- Verify that the rotor is not spinning, by viewing through the gantry cone before opening the covers. Use caution when opening or servicing the scanners. Never service the rotating frame when or if rotational movement is enabled. In case of a power failure or fault condition the rotor can spin (approximately 25 minutes or longer) after the power is removed from the system. Wait until the rotor stops before opening covers Failure to comply may result in equipment damage, serious injury or death to service personnel.
- Check for rotational interference between rotor and cones (by hand rotation). Be aware and careful of pinch points during service operations.
- The Rotor contains heavy parts, such as the X-ray Tube, DMS, etc., and is perfectly balanced. When this balance is disturbed by removing a part from it, it will start an uncontrolled mechanical rotational motion to reach a new equilibrium point. This motion cannot be stopped and may injure the service engineer.

- Wear proper clothing to reduce the risk of injury due to hair, clothing or jewelry getting caught in parts of the system during servicing.
- Install the vertical safety support brace whenever personnel are working under the table. It is especially CRITICAL when performing repair or replacement on the vertical drive system. If the motor/brake assembly is removed without it, the patient support will free fall to the ground. The patient support scissors support a weight of ~600 pounds (272 kg); a fall may result in personnel injury or death

Residual Laser Radiation Risks

Though the system is designed to zeroize laser radiation risks, the following residual risks must be considered:

- The system has class 2 lasers that may be exposed to users and patients of the system during clinical use and care should be taken to avoid staring into the laser beam. Approved patient eye protection should be used for all head exams to minimize risk; the use of optical instruments (such as eyeglasses with large diopter or mirrors) with this product will increase the risk of eye injuries
- Service personnel can access class 3 lasers. Care needs to be taken to avoid looking directly into the laser beam or at its reflection on smooth, mirror-like surfaces like waveguides or plated metal. Service personnel are advised to remove power from the transmitter electronics box when working in areas of the gantry where there is a risk of eye exposure to laser energy

Residual Mechanical Gravity Related Hazards

Though the system is designed to zeroize mechanical gravity related hazards, the following residual hazards must be considered:

- If coolant leaks are detected, shut down the scanner and immediately contact the nearest Philips field service office. The floor may be slippery and present a risk of slipping or tripping.
- During maintenance procedures, check the entire X-segment cooling circuit for any obvious leakage
- All cables should be routed between the injector, the patient, the table and the CT scanner so they do not impede the free movement of personnel. Route all cables between the PIM, the patient, the patient table, and the CT scanner so that they do not impede the free movement of personnel. Route cables in existing troughs, ducts, or adjacent to system components to prevent obstructions that can cause personnel to trip and fall
- If the monitor is located on a cart, make sure that the cables connected to the device are not in the way of the patient or the personnel in the scan room. The additional monitor cart inside the scanner room should not be used to hold anything but the original monitor. The 21-inch monitor-base should always be on top of the stand and secured properly. When not in use, the cart and its cables should be moved to a corner of the room so they do not interfere with routine activities in the scanner room. Care must be taken not to collide with the monitor stand or trip on the monitor cables.

- Make sure that the patient is strapped securely to avoid dangling of the hands. Ensure that the patient is placed securely on the patient table and is not in danger of falling. During all movements of the gantry and the patient table (automatic and manual), keep the patient under continuous observation to ensure safety of the patient.
- If a head holder or support is not engaged securely, it can come loose causing injury to the patient. Positioning aids must be used exclusively for their intended purpose: head holder only for positioning the head, table top extension only for positioning the feet.
- Follow all service instructions when installing or servicing the system. Pay attention to clearance requirements to prevent excessive strain on wheel castors to minimize the risk of rollover and toppling. Minimize the clearance between the Gantry/Patient Table/CIRS Rack base and the floor when transporting these components.. Use assistance wherever specified (e.g. CIRS rack removal from pallet). Wheel castors should be secured to minimize tipping hazard. Ensure components are attached to floor as specified during service and replacement procedures. Failure to comply may result in serious injury or death to service personnel.
- Handling equipment without proper tools, training, and adherence to all warning labels, etc., can cause damage to equipment or harm to personal.
- Make sure the gantry and other system components are on level ground. The system may move/roll on uneven ground. Block/brace the wheels on gantry as instructed to avoid undesired movement

Residual Lifting and Ergonomics Related Risks

Though the system is designed to zeroize lifting and ergonomics related risks, the following residual risks must be considered:

- System components such as gantry caster, jun-air compressor, CIRS racks, patient support and battery are very heavy. Caution should be used when moving these items and two people should be involved whenever instructed. Failure to comply with instructions can result in injury.
- While parts of the system are designed to take handling into account, care should be taken by users when handling heavy phantoms or accessories to avoid loss of control or unstable footing that could result in dropping the part which may cause injury

Residual Loss of Communication and Noise Related Risks

Though the system is designed to zeroize loss of communication and noise related risks, the following residual risks must be considered:

- The operator should watch the patient at all times during system operation to monitor patient status and avoid patient distress in case of failure of communication.

Residual Risks Related to Mechanical Expelled Parts

Though the system is designed to zeroize risks related to mechanical expelled parts, the following residual risks must be considered:

- Compliance with standards for design combined with periodic maintenance and inspection of the system according to detailed service instructions are designed to detect any problems in the system such as structural breakdown or human error that might lead to the possibility of a part coming loose and, if system enclosures are breached, becoming a projectile that could injure a person.
- Service personnel should wear protective glasses whenever instructed. The S-clips on the shipping crates are under tension and can fly off during removal. Failure to comply can result in injury to personnel.

Residual Risk of Accidental Radiation

Though the system is designed to zeroize risks accidental radiation, the following residual risks must be considered:

- All personnel that work with ionizing radiation must understand the key principles of occupational and public radiation protection.
- The radiation warning lamps on the gantry panels, on the scan control panel, as well as site radiation warning lamps, must light up if scanning has been triggered. If the radiation warning lamps do not light up shut down the system immediately and contact Customer Service. Press the Emergency Stop button if there is danger to you or the patient.
- If there is any indication that X-rays are not turned off after releasing the foot pedal switch, press one of the STOP buttons on the gantry control panels or the Pause button on the scan control box. This stops the generation of X-rays, scanner rotation, and patient table motion.
- Only the patient should be in the scan room for scanning. When occupancy of the scanner room is unavoidable, attention should be paid to the zones of occupancy as documented in the Technical Reference Guide. Anyone who has to be near the patient during scanning must wear protective clothing (lead apron), wear a PEN dosimeter and/or film badge, and stay in the zone shielded by the system (to the side of the gantry or behind a mobile protective wall). The physician is responsible for protecting the patient from unnecessary radiation. Use protective shielding whenever appropriate to minimize dose to sensitive organs.
- Use the applicable exams for children. Use pediatric exams based on the age, weight, and indications to avoid over exposure. Philips recommends the use of the infant mode for newborns up to 18 months of age
- When performing Advanced Interventional procedures, prepare the appropriate radiation shielding equipment and materials to avoid accidental radiation exposure.
- For CT procedures that require scanning over the medical device for more than a few continuous seconds (as with CT perfusion or interventional exams) users should prepare to treat possible adverse reactions.
- Properly center all patients in the gantry. Patients not properly centered may be under or over exposed to radiation if the table height is set too high or too low
- To avoid overexposure to radiation, ensure the scan room is clear of personnel during servicing and related service scanning. Follow the procedures established for your site.

- The useful and scattered beams can produce injuries to patients and persons in the surrounding area if used by an unskilled operator. Adequate precautions must always be taken to avoid exposure to the useful beam, as well as to indirect radiation including scattered radiation from within the scanner as well as anything in the path of the beam.
- Follow Safety warning instructions when refilling or transporting phantoms

Residual Risk of Potential Electrical Hazards

Though the system is designed to zeroize risks potential electrical hazards, the following residual risks must be considered:

- Compliance with standards for design combined with planned maintenance and inspection of the system according to service instructions are designed to detect any problems with the system that might lead to potential electrical hazards, such as damaged couch or gantry covers exposing electrical components or improper system grounding. However, hazardous voltages are still present within the system that can cause serious injury. The operator should not remove covers or override safety locks present on the system. If covers are damaged or removed, the operator should not operate the system until repaired by a qualified service personnel.
- Before installing and prior to any service or maintenance activity, make sure to switch off the system at the main power supply and the UPS (ensure no power is applied to the Ghost) as a precaution against electrical hazards.
- Never remove or connect the Gantry cone cables with power ON to the rotor and/or stator as a precaution against electrical hazards.
- Front end electronics is energized through the slip rings even when you Power OFF the system. Therefore, before you remove the rear cover, to avoid any electrical danger, switch off all circuit breakers on the gantry left column, and on the main power supply to the scanner, and wait at least ten minutes for energy to be discharged as a precaution against electrical hazards.
- Batteries can present a risk of electrical shock or burn from high short circuit current. Observe proper precautions. Servicing should be performed by qualified service personnel knowledgeable of batteries and required precautions. Keep unauthorized personnel away from batteries. When servicing the battery, wear rubber gloves, electrically insulated footwear, and insulated tools
- The Philips CT system complies with the requirements of applicable EMC standards. Emissions from the CT system may affect other electronic equipment that does not meet the EMC immunity limits. There is a possibility that the X-rays or other electromagnetic radiation from the CT may cause some implanted and external electronic medical devices (pacemakers, defibrillators, neurostimulators, and drug infusion pumps) to malfunction. Philips recommends that users check the device manufacturer's recommendations/ precautions regarding use in a CT scanner prior to exposing a patient to a scan.
- Touch current can reach the patient in the patient area by any chance contact through various paths (i.e. operator touching the patient and accidentally coming into contact with exposed electrical components of the system). The patient area is defined as any area less

than 15ft away from the table. The operator should not make contact with potential voltage sources and the patient at the same time in the patient area to avoid potential electromagnetic harms to a patient from leakage current.

Residual Risk of Misrepresentation

Though the system is designed to zeroize misrepresentation, the following residual risks must be considered:

- The system is intended for use by trained users that understand the technological limitations and the types of artifacts that can be caused inherently by the CT scanner technology and by impacts from techniques used to generate images.
- Users are expected to perform image analysis to verify that measurement results when making critical measurements are correct.
- The system requires routine calibration and maintenance. Users are expected to perform daily image quality checks and periodic calibrations and maintenance as specified in the user manuals. Failure to do so can result in image artifacts or inaccurate measurements.
- Errors in RTP may occur when there are problems with geometrical accuracy of the system for RTP (e.g., table sag over distance while table is loaded, image perpendicularity). The overall treatment planning accuracy is dependant on a quality assurance plan by the clinician that takes into account the accuracy of the CT Scan data.
- Spatial positioning errors in RTP and final treatment may occur if the user does not use the oncology radiation therapy flat table top ("Therapy Table Top") and its compatible accessories. To reduce these errors in final treatment, offsets between the lasers and CT center should be measured by scanning a phantom designed for this specific purpose, and then the measured values should be entered into the RTP system. The overall treatment planning accuracy is dependant on the quantification of any geometric inputs used by the system.
- Errors in RTP may occur if positional accuracy is compromised. In order to maintain positional accuracy, users should evaluate Therapy Table Top alignment immediately following any possible events which could cause misalignment (e.g. forceful table contact, system service). Any malfunction or damage should be evaluated by qualified Philips service personnel, and quality assurance procedures should be repeated after any subsequent repairs or adjustments.
- Patient positioning lasers should not be used for absolute marking in RTP as they are not designed for this purpose and may compromise accuracy. Instead, high accuracy external lasers (supplied by a third party) designed for marking patients for therapy should be used for this purpose.
- RTP accuracy may be compromised if the system is used for RTP prior to completion of both Image Performance QA procedures and any preventive maintenance to be conducted in accordance with the maintenance schedule provided for the system. The system is intended for use by trained personnel. Errors in geometric accuracy can impact the treatment location targeted by the RTP system, which could then potentially harm

surrounding healthy tissue. Additionally, errors in determining the CT tissue density can impact the calculations used for therapy beam strength, which can thereby reduce the effectiveness of the treatment.

- Systems used for radiation therapy planning are expected to be maintained and calibrated consistent with AAPM Report No. 083 - Quality assurance for computed-tomography simulators and the computed-tomography-simulation process: Report of the AAPM Radiation Therapy Committee Task Group No. 66.
- List of Commonly Recognized CT Artifacts and their Causes:
 - Patient-based artifacts: motion artifact, transient interruption of contrast, clothing artifact, and jewelry artifact.
 - Physics-based artifacts: beam hardening, cupping artifact, streak and dark bands, metal artifact/high-density foreign material artifact, partial volume averaging, quantum mottle (noise), photon starvation, and aliasing in CT.
 - Hardware-based artifacts: ring artifact, tube arcing, out of field artifact, and air bubble artifact.
 - Helical and multichannel artifacts: windmill artifact, cone beam effect, multiplanar reconstruction (MPR) artifact, zebra artifact, and stair step artifact.

Residual Risks Related to Biocompatibility

Though the system is designed to zeroize risks related to biocompatibility, the following residual risks must be considered:

- The clinical suite may include one (or several) third-party UPS devices. When UPS batteries are not properly maintained, or if they are held in service beyond their usable service life, failure can result in the leaking of electrolyte (sulfuric acid), overheating, and/or the emission of fumes. To ensure continued safe and reliable performance from these devices, periodic maintenance is required, including possible battery replacement. Based on industry standards, the typical useable service life of a UPS battery is less than five years. You may consult your local Philips Service representative for help in identifying the specific model of your UPS device(s) and available service provider options in your geography.
- Blood and bodily fluids from patients may leak onto surfaces of the system which present potential health risks. Take appropriate health and safety precautions when cleaning and disinfecting the system to minimize the risk of cross contamination. To prevent transmission of biological infection hazard and to prevent damage to the system, cleaning should be performed using materials and methods as described in the system instructions for use. Cleaning should be done using a commercial biocide approved by your governing authority to clean the surface of the system and should ensure that no remnants of cleaning materials remain on the system surface including the console, gantry, table, and accessories.
- Applied parts, which patients and users, may come into contact with during normal operation are designed for biocompatibility according to ISO 10993, however an individual may still have a reaction to such contact. Service personnel may be exposed to additional internal parts and surfaces during their normal duties and should follow all instructions and precautions to minimize their exposure to possible irritants.

- Coolant or other substances that leak from the system may cause harm. Field service should be contacted immediately upon detection
- Service personnel may be exposed to chemical substances during service operations and should take proper precautions to avoid risk.
- The system contains hazardous materials. Incorrect disposal of any of these materials may lead to serious environmental pollution. This system may contain devices that contain mercury, which must be recycled or disposed of in accordance to local, state, or federal laws. Within this system, the backlights in the monitor display contain mercury.

Residual Risk of Sharp Edges

Though the system is designed to zeroize risks related to sharp edges, the following residual risks must be considered:

- Sharp and rough edges are present inside the gantry and on other non-accessible surfaces and service tools. All service instructions should be followed to minimize risk of injury.
- The system and its accessories are designed to withstand day to day usage. However, if carts or other equipment collide with the system or its accessories, they may be damaged and sharp edges can become exposed
- The system and its accessories are designed to withstand cleaning and disinfection as required for clinical usage. Care should be taken to follow the instructions regarding recommended cleaning materials, frequency and process to avoid damaging the equipment and to ensure effectiveness.

Undesirable Side Effects

Undesirable side effect is a form of harm that may affect any person near or in the device, due to a natural reaction arising from exposure to a device, during normal conditions of use, caused by a factor other than a failure of control.

Undesirable side effects identified are as follows:

- Radiation Effects.
*Radiation exposure used for image creation. This type of ionizing radiation is essential for acquisition of an x-ray image suitable for clinically relevant diagnostic interpretation. The effects of this type of harm may manifest over an extended period of time (long-term effect of radiation exposure).
- Emotional Trauma/Anxiety.

Compatibility with Other Devices

CT/AMI Philips is performing compatibility testing for combinations of CT/AMI systems and accessories. The compatibility test results are verified to support device intended use in combination with other devices. Ask your Philips Representative or go to www.philips.com/IFU for the available compatibility statements."

Safe De-Installation of the CT System

Detailed instructions for dismantling the CT system is included in the CT System De-Installation document intended for Philips/trained technicians who dismantle the system.

For any information regarding the system dismantling procedure, please refer to the customer service contact information.

Customer Service Contact Information

local Philips Healthcare representative. Alternatively, contact:

Philips Healthcare

PO Box 10 000

5680 DA BEST

The Netherlands

Facsimile: +31 40 276 2205

4 System and Data Security

Philips Healthcare is dedicated to helping you maintain the confidentiality, integrity, and availability of electronic protected health information and the hardware and software products that create and manage these data.

Maintaining security of Philips Healthcare's products should be an important part of your facility's security-in-depth strategy. You should implement a comprehensive, multi-layered strategy (including policies, processes, and technologies) to protect information and systems from external and internal threats. Your security strategy should follow industry-standard practices, addressing physical security, personnel security, procedural security, risk management, security policies, and contingency planning.

The practical implementation of technical security elements varies by site and may employ a number of technologies, including firewalls, virus scanning software, authentication technologies, etc. As with any computerbased system, protection must be provided such that firewalls and/or other security devices are in place between the medical system and any externally accessible systems. These perimeter and network defenses are essential to good security practice.

This information provides guidelines to help the operator and owner understand some of the possible ways security can be compromised, and then insure that safeguards are in place to prevent this from happening. For specific information about security within their institutions, operators and owners can consult with the following officers at your location:

- Information Systems Security Officer
- Chief Information Officer
- HIPAA Officer (in the U.S.A.)
- Safety Officer

Regulatory Controls

Protect Patient's Health Information

One of the most important assets to protect with security measures is the patient's health related information. Many governments require maintaining the confidentiality of this information. Therefore, strict security measures must be taken to guard this protected information.

NOTICE

De-identify patient studies when exporting via network or removable media, in compliance with your local privacy policies. See "Anonymize All Patients" in the Instructions for Use for more information.

(Users in the U.S.A. may find guidelines at <http://www.hhs.gov/ocr/privacy/>.)

Prevent Unauthorized Device Modification

Philips Healthcare sells highly complex medical devices and systems. We are required to follow government-regulated quality assurance procedures to verify and validate modifications to the operation of our medical devices.

Operators and owners of this medical equipment must permit only Philips-authorized changes to be made to these systems, either by Philips' personnel or under Philips' explicit published direction.



CAUTION

Although the Philips CT operates on a personal computer (PC) platform, the installation of PC software not specified in the Philips system documentation may adversely affect the operation and security of the system, as well as the networks to which the system is connected. These adverse effects may not be immediately apparent to the user. Users should therefore not install unauthorized software onto their system.

Security Issues and Guidelines

In addition to the patient information and device integrity needs discussed in the preceding section on regulatory requirements, the following topics, issues, and guidelines should be understood and addressed by operators and owners.

Network Security

The CT system must be placed on a secure local computer network that has protections against viruses and other harmful computer system intruders. Make sure the equipment is connected to a local network that uses appropriate protection, such as a firewall, network access controls, and network virus scanner. Clinical data transferred across the network is not encrypted.

- The Host system can connect to IPv6 enabled remote nodes such as ISP systems and PACS systems.
- The Host system time can be configured to synchronize with an NTP server.



CAUTION

Connection of the System to a network that includes other equipment could result in unidentified risks to privacy of patient, user, operator, and other data. You should identify, evaluate, and control these data-privacy risks on-site. Changes to a network (including configuration, additional connections, disconnections, updates, and upgrades) may introduce new data-privacy risks and require further analysis.

Hard Drive Encryption

Encryption can be enabled on the Host system hard drive using an additional license. To enable encryption, contact Philips Service.

NOTICE

If encryption is enabled on the hard drive, the system performance might be impacted.

It is recommended to start using the system once disk encryption is completed.

NOTICE

When you transfer data to an USB drive, you can either select the compression mode as compressed or non-compressed.

Remote Service

Philips Healthcare has a global, remote service network for connecting many of your Philips systems to our advanced service resources. This secure tunnel approach provides your equipment with a single point of network access to on-site Philips equipment using Virtual Private Network (VPN) and iSSLink point-to-point connection technologies. The remote service function is a secure connection through explicit authorization and authentication control which includes the encryption of data.

Data Disaster and Recovery Planning

If not already in place, it is recommended that your facility develop a data disaster and recovery plan for the system. The plan should specify:

- system and patient-data backup plan
- safeguards in place to store protected health information and backup data
- procedures for restoring system and patient data in the event of a local disaster
- If the removable media is to be stored for safekeeping, protect the data from “fading” loss by storing it in a suitable environment and performing media renewal as recommended by the media manufacturer.
- If removable media is used to store patient data, protect the information from media obsolescence by planning and performing data migrations to newer storage technologies.

Access Control

Room Access Control

Local procedures should be put in place to limit physical access to medical equipment, to prevent accidental, casual, or deliberate contact by unauthorized individuals.

Access to the room containing the CT should be controlled by policy and procedures that identify who is authorized to occupy specific areas. Check with your Safety and Security Office for more information on what measures are in place or how to implement room access controls.

Local Administrator

The Local Administrator user account has access to system management options.

The Local Administrator can change his/her own password if the previous password is known.

If the password for the Local Administrator user account requires a reset, please contact your Philips Service Representative.

Individual Access Control/User Accounts

Individual clinical user accounts are created and managed by the Local Administrator account. All authorized clinical users have the same read/write permissions to perform scans, reviews, analysis, and other standard tasks.

NOTICE

All login credentials (such as user names and passwords) should be kept confidential. It is recommended to regularly change user passwords.

Once an account is created by the Local Administrator, enter the credentials at login. Only one user can login to the system at any time: concurrent clinical users are not supported.

User Account Roles

The following user account roles are supported by the system:

- Clinical
- Philips Service
- Local Administrator
- Third-party Service

The ability to enable or disable access to external input and output devices such as USB drives or DVD drives is available only to Philips Service and Third-party Service user accounts.

Manage Clinical User Accounts

Use the Local Administrator account to create and manage clinical users. The Local Administrator does not have permissions to perform scans, reviews, analysis, and other clinical tasks.

To access the Local Administrator account, enter the user name and password (provided by your Philips representative) on the login screen. The default Local Administrator login name and password will require change upon first login, and the new login information should be stored in a secure location.

Create User

When creating user accounts, it is recommended that:

- the default clinical user account be disabled (see chapter “Manage Users” on page 51); and
- a unique user account is created for each person using the system.

To create a user:

1. Enter the Local Administrator user name and password on the login screen.
2. Select **Create User**.
3. Enter the clinical user name, select **Clinical** from the drop-down list, then enter and re-enter password.
4. If required for the user, select **Access to Exam Card Manager**.
5. If required for the user, select **Access to Dose Management**.
6. Click **OK** and then **Logoff**.

NOTICE

It is recommended to take regular backup of user accounts. Record all clinical user logon information and store it in a safe place.

Manage Users

Once logged on as the Local Administrator, you can create users, remove users, reset a user password, enable or disable user accounts. You can also provide access to the following options:

- Exam Card Manager
- Dose Management options in the Preferences window
- View Audit Trail Logs (applicable only for Local Administrator role)

Remove a user

Only non-factory users can be removed. Select a user from the Users list. Click **Remove User** and confirm the permanent deletion of the account.

Reset a user password

Select a user from the Users list. Click **Reset Password**. Give and confirm the new password. The password will require change upon first login after reset.

It is recommended to regularly change user passwords. See chapter “Password Complexity Rules” on page 52.

Enable or Disable a user account

Select the user and check or uncheck the box to disable or enable the selected user account. One active user should be available in the selected role to disable the user account.

Grant or remove access to Audit Trail Logs

Select a user account with the Local Administrator role and check or uncheck the box to grant or remove access to the Audit Trail Logs.

Grant or remove access to the Exam Card Manager

Select a Clinical or Philips Service user account and check or uncheck the box to grant or remove access to the Exam Card Manager.

Grant or remove access to the Dose Management

Select a Clinical or Philips Service user account and check or uncheck the box to grant or remove access to Dose Management options in the Preferences window.

NOTICE

It is recommended to take regular backup of user accounts. Record all clinical user logon information and store it in a safe place.

Password Complexity Rules

- Password cannot contain the user's user ID.
- New password should not be same as old password.
- By default, the password must contain at least 8 characters and must satisfy at least 3 rules from below:
 - 1 uppercase letter
 - 1 lowercase letter
 - 1 number;
 - 1 special character: !@#\$%&*().
- Password is case sensitive
- By default, passwords must be reset every 180 days.
- By default, after 5 consecutive failed login attempts, the account is locked.

Password character length, reset period, and number of failed login attempts allowed can be configured by the Local Administrator.

Change the Password Policy

Once logged on as the Local Administrator, you can change the default password policy for all users.

NOTICE

Only Philips Service user can reset the Local Administrator user password.

To change the default password policy:

1. Login as local administrator user.
2. Click the **Password Policy** button on the left.
3. Enter the login attempts allowed. The value can be between 3 and 10. The default value is 5 attempts.
4. Enter the number of days after which the password should expire. The value can be between 30 and 180 days. The default value is 180 days.
5. Enter the number of days before which the password expiry message should be displayed. The value can be between 1 and 30 days. The default value is 5 days.
6. Enter the length of the password. The value can be between 4 and 14 characters. The default length is 8 characters.
7. Select the check box to enable password complexity. When this check box is selected, all the password complexity rules are applied. Refer to Password complexity rules.
8. Click **Save**.

NOTICE

Click **Restore Factory Settings** to revert to the default factory remote set password policy.

User Account Backup and Restore

User account information (excludes the password) can be backed up and restored to any of the Philips CT systems running software version 4.8 by the Local Administrator or Philips Service Representative.

Local Area Network Access

Local administrator or Philips service users can access the Local Area Network (LAN) administrative tool for DICOM configuration. For more information, contact your Philips representative.

Positioning of Display Monitors

Unauthorized visual access to protected information can be minimized by positioning the system's display monitor to prevent viewing from doorways, hallways and other traffic areas. To help in limiting unauthorized visual access, an unattended CT display automatically goes blank after a set period of time.

Emergency Login

An emergency access option available at the login screen. By default, the Emergency user login is not enabled. The Local Administrator needs to enable the user and can also set a password if required.

To log in to the system, when there is no password set for your emergency user name:

1. Enter the user name as **Emergency**.
2. Enter your user name in the **Emergency User** field. The user name may consist of letters, numbers, and underscores and special characters.

The name can be between 2-53 characters long consisting of alphabets, numbers and special characters "_ " ' and ".".

3. Click **Emergency Login**.
4. Follow the on screen prompts.

To log in to the system, when there is a password set for your user name:

1. Enter the user name as **Emergency**.
2. Enter the password that the Local Administrator provided you. The Emergency login password can be changed only by using the Reset Password option.
3. Enter your user name in the **Emergency User** field. The user name may consist of letters, numbers, and underscores only.

The name can be between 2-53 characters long consisting of alphabets, numbers and special characters "_ " and ".".

4. Click **Emergency Login**.
5. Follow the on screen prompts.

After using the emergency login, the console will display **Emergency User** and **Emergency_<Emergency user name>** in the upper right corner of the console screen while the system is in use. After 5 exams under the emergency login, the user is required to login again using the same user name that was used earlier.

Note: An appropriate number of authorized clinical users should be maintained in order to avoid the use of the emergency clinical user login.

System Logoff

The system does not support automatic log off in order to avoid unexpected system log off during extended clinical exams. The system must be manually logged off and shutdown by the user if desired. However, to protect sensitive data, the screen will go blank if no user activity is detected after a set period of time.

Automatic Screen Blanking

The system is capable of automatically blanking the console displays after a set period of time in which no mouse or keyboard input occurs. By default, this period is 60 minutes. The Local Administrator can enable or disable this feature, or change the time period.

System Hard Drive

Clinical images that reside in the system hard drive are encrypted if the Philips-Service enables encryption on the hard drive based on Hospital request. Even then, it is recommended to restrict access and establish local access controls to only authorized users.

System Backup Media

The clinical data that is backed up on removable media is not encrypted and should be stored in a secure location to avoid unauthorized access.

Removable and Portable Media

When using removable media (CD-ROMs, DVDs, and USB drives) be aware of these security issues:

- Inserting removable media (such as a USB drive or CD-ROM) can introduce a virus to the medical device. Be certain to scan the portable media for malware before inserting the media into the scanner.
- Patient data that is transferred to removable media is not encrypted, handle and store media according to your privacy protection policies.
- If the media is to be discarded it must be destroyed or disabled so that the data can no longer be accessed.
- If removable media is used to store patient data, protect the information from media and technical obsolescence by planning and performing data migrations to newer storage technologies.
- If the removable media is to be stored for safekeeping, protect the data from “fading” loss by storing it in a suitable environment and performing media renewal as recommended by the media manufacturer.

**CAUTION**

Removable media that contains images and/or other medical information should be stored in a secure area that is not accessible by unauthorized individuals.

User Logging and Audit Trails

When any of the following events occur, the system logs the event in an ePHI audit log:

- Application activity
- Audit log used
- Transferring DICOM instances begins
- DICOM instances are accessed or a study is deleted
- DICOM instances transferred
- Patient-record event
- PHI-export
- PHI-import
- Query information
- Security alert
- User authentication

ePHI audit logs are accessible only to the Local administrator through the use of a special viewer.

The Local administrator can, however, provide this privilege to another Local Administrator that he or she has created on a case by case basis. ePHI audit logs are available up to 8 months from the initial date of the log. Logs older than 8 months are deleted from the archive. ePHI audit logs can be backed up and restored.

The system also supports a number of system-level event logs to assist with system troubleshooting and repair, including successful and unsuccessful user log in.

NOTICE

It is recommended to backup the audit trail logs in a secure location.

Data Integrity Checks

- The system supports basic system level data and database integrity checks.
- If you suspect improperly altered or destroyed clinical data, notify your local IT security office or notify Philips service.

System Application Control

The system utilizes McAfee Application Control, a whitelist file verification method. This is an alternate approach to preventing virus and malware infection. The whitelist allows only trusted software provided by Philips to run, and blocks untrusted software.



CAUTION

Whenever media is inserted into the CT system, be sure that the media has not been previously exposed to potential viruses, worms and trojans that infect desktop PCs.

In the event that an unauthorized application is opened, or an infected file is found, the whitelist will prevent it from execution. The system will display an error detailing the disallowed file activity and log this information. If you see this error or notice unfamiliar system behavior or performance changes, especially after the system has been restarted, contact a Philips Field Service Engineer to have the system checked.

Performing Data Sanitization on Hard Drive

You can clear the hard drive contents by using the Secure Erase BIOS feature or a third-party application that, ideally, is U.S. Department of Defense (DOD) 5220.22-M approved.

To run Secure Erase, enter the F10 Bios Setup menu by powering on the system and pressing F10 as soon as the HP logo appears.

NOTICE

It is recommended to backup all data before performing data sanitization.

1. Select the **Security** menu and scroll down to the **Hard Drive Utilities** menu.
2. Select **Secure Erase**.
3. Select the desired drive.
4. Select **Continue**. The estimated time to complete Secure Erase will be displayed along with a final warning not to remove power.
5. Select **Continue**. The elapsed time will be displayed until "Secure Erase Complete" is displayed.
6. Press **Esc** twice to return to the drive selection menu. Repeat step 3 to Secure Erase additional drives or exit BIOS Setup, if done.

NOTICE

The Secure Erase process will take a long time. The amount of time varies based on the hard drive capacity and write speed.

Data Sanitization can be performed by qualified and authorized IT personnel or Philips service representative. No clinical persons are allowed to perform Data Sanitization.

Third Party Software used with the System

The following third party software are supplied with the Philips CT system:

- Adobe Acrobat Reader DC 19.012.20034
- McAfee Solidifier 8.2.1.143
- Microsoft Windows 10 Enterprise 2016 LTSB (64-bit) Version: 10.0.14393
- Techsmith Snagit 19.1.2
- Tevfik Karagulle CopSSH 6.4.0
- Wind River Systems VxWorks 5.4.2

Open Source Software used with the System

The following opens source software are used with the CT system:

- 7-Zip 4.57
- 7-Zip 64-bit 9.2
- ALogic CeasarFTP 0.99E
- Boost C++ library and templates 1.65.1
- CLIPS 6.3
- Free Software Foundation GNU zip 1.2.4
- ICSsharpCode.SharpZipLib.dll (#ZipLibrary) 0.85.5
- Joe Richards CPAU.exe-CPAU 01.10.00cpp
- Json.net 8.0.3
- Log4net 1.2.10.0
- NPlot Charting Library for .NET 9.9.2
- OpenCV 2.0 2
- Philippe Jounin TFTP32 3.03
- Prism 6.2.0
- RedHat Cygwin (Support Telnet Commands over SSH) 2.87
- SharpZipLib (ICSsharpCode.SharpZipLib) 0.85.5
- Simon Tatham Putty 0.67.0.0

- TightVNC 2.8.11
- WeifenLuo.WinFormsUI.Docking 2.7.0.0

ePHI De-Identified Items

The following items are **blanked** as part of ePHI de-identification:

- INSTANCE CREATOR UID
- ACCESSION NUMBER
- INSTITUTION NAME
- INSTITUTION ADDRESS
- REFERRING PHYSICIANS NAME
- REFERRING PHYSICIANS ADDRESS
- REFERRING PHYSICIANS TELEPHONE NUMBERS
- STATION NAME
- STUDY DESCRIPTION
- INSTITUTIONAL DEPARTMENT NAME
- PHYSICIANS OF RECORD
- PERFORMING PHYSICIANS NAME
- NAME OF PHYSICIANS READING STUDY
- OPERATORS NAME
- ADMITTING DIAGNOSES DESCRIPTION
- DERIVATION DESCRIPTION
- OTHER PATIENT IDS
- OTHER PATIENT NAMES
- MEDICAL RECORD LOCATOR
- MEDICAL ALERTS
- ETHNIC GROUP
- OCCUPATION
- ADDITIONAL PATIENTS HISTORY
- PREGNANCY STATUS
- PATIENT COMMENTS
- DEVICE SERIAL NUMBER
- PROTOCOL NAME
- IMAGE COMMENTS
- REQUESTING PHYSICIAN

- REQUESTING SERVICE
- REQUESTED PROCEDURE DESCRIPTION
- ADMISSION ID
- SPECIAL NEEDS
- CURRENT PATIENT LOCATION
- PATIENT STATE
- SCHEDULED PROCEDURE STEP DESCRIPTION
- REQUESTED PROCEDURE ID
- NAMES OF INTENDED RECIPIENTS OF RESULTS
- REQUESTED PROCEDURE COMMENTS
- IMAGING SERVICE REQUEST COMMENTS
- STORAGE MEDIA FILE SET UID
- ELSCINT1_PATIENT_LANGUAGE
- REFERENCED FRAME OF REFERENCE UID
- RELATED FRAME OF REFERENCE UID
- Main Header (Imagio) structure
- Sub-header (Imagio) structure
- Worklist Blob File

The following items are **removed** as part of ePHI de-identification:

- REFERENCED STUDY SEQUENCE
- REFERENCED PATIENT SEQUENCE
- REQUESTED PROCEDURE CODE SEQUENCE
- SCHEDULED PROCEDURE STEP SEQUENCE
- REQUEST ATTRIBUTES SEQUENCE
- CONCEPT_SEQUENCE

The following items are **modified to 1** as part of ePHI de-identification:

- STUDY DATE
- SERIES DATE
- ACQUISITION DATE
- CONTENT DATE
- PATIENTS BIRTH DATE

The following item is **rounded** as part of ePHI de-identification:

- PATIENTS AGE

The following item is **rounded the nearest 10** minutes as part of ePHI de-identification:

- PATIENTS BIRTH TIME

The following items are **changed to one** entered by operator as part of ePHI de-identification:

- PATIENTS NAME
- PATIENT ID

The following item is **rounded as per logic described below** as part of ePHI de-identification:

- PATIENTS SIZE

The rounding is according to the following guidance:

Actual Height	Round to the Nearest
H < 50 cm	1 cm
50 cm < H < 100 cm	2 cm
H > 100 cm	5 cm

Tab. 1: Metric:

Actual Height	Round to the Nearest
H < 2 ft	0.5 inch
2 ft < H < 4 ft	1 inch
H > 4 ft	2 inch

Tab. 2: English:

The following items are **retained with no change** as part of ePHI de-identification:

- SERIES DESCRIPTION
- REFERENCED SOP INSTANCE UID
- PATIENTS WEIGHT
- STUDY ID
- FRAME OF REFERENCE UID
- SYNCHRONIZATION FRAME OF REFERENCE UID
- SPECIFIC CHARACTER
- STUDY TIME
- SERIES TIME
- ACQUISITION TIME
- CONTENT TIMESET

The following item is **retained only if it is "M" or "F" for "Other" it will be set to blank** as part of ePHI de-identification:

- PATIENTS SEX

The following items are **newly generated** as part of ePHI de-identification:

- STUDY INSTANCE UID
- SERIES INSTANCE UID
- ELSCINT1_PATIENT_DATA_MODIFIED

The following item is **set to "Yes"** as part of ePHI de-identification:

- PATIENT_IDENTITY_REMOVED

The following item is **set to "Basic Application Confidentiality Profile"** as part of ePHI de-identification:

- DE_IDENTIFICATION_METHOD

Software Distribution

Software Distribution option allows you to download and install software patches released by Philips. The software patches are automatically downloaded on the scanner, when the system is connected to Philips Remote Service (PRS). The download happens only when the scanner is not in use. The software patches can be downloaded manually too from PRS by Philips Service, Third Party Service users. For the hospitals not connected to PRS, the software patches can be downloaded from Philips site and copied on the scanner using removable media by Philips Service, Third Party Service users.

Notes:

- The automatic download of software patches may affect the system performance. Philips Service Personnel, Third Party Service or Local Administrator can disable the automatic download of the software patches using the Software Distribution option.
- The software patches require more than 11GB free space for the download.
- Once the download is complete, the clinical users will receive a patch availability notification during login. Philips Service Personnel, Third Party Service or Local Administrator can install the downloaded software patches.
- The software patch which requires post-installation intervention would only be available for installation by Philips Service user.
- Any failed software patch installation prevents clinical user login with a notification on the screen.

5 Quality Assurance

Read this section carefully and follow all instructions regarding scheduling and performance of **Quality Assurance** (QA) checks. These instructions represent the manufacturer's required QA and constancy performance checks. If additional testing is required by your national or local authorities, please contact your Philips Service Engineer.

- **Short Tube Conditioning** brings the tube to normal operating temperature. This process is required daily before any scans are performed on patients after 8 hours of scanner inactivity.
- **Air Calibration** is part of normal system maintenance. The test should be performed once a week.
- **Head IQ Check** must be run at least once a week (follow local regulations).
 - For radiation oncology users, this test is recommended to be run daily. The water (phantom liquid) CT number and noise should be checked.
- **Body IQ Check** must be run at least once a month (follow local regulations).
- **Constancy Test** must be run at least once in half year and is used for both Head and Body Quality Assurance.

Note: Body Quality Assurance Is tested on head section for Constancy test.

- **CT Number Linearity and In Plane Spatial Integrity Check** must be run monthly to check the CT number accuracy of 5 different materials and the in-plane spatial integrity. This test is only suggested for radiation oncology users.
- The infant phantom is used to check the performance in the infant scanning mode. The test only applies to systems with the optional infant phantom.
- **Monitor Calibration** must be run daily if the DIN 6868-157 standard is followed.

NOTICE

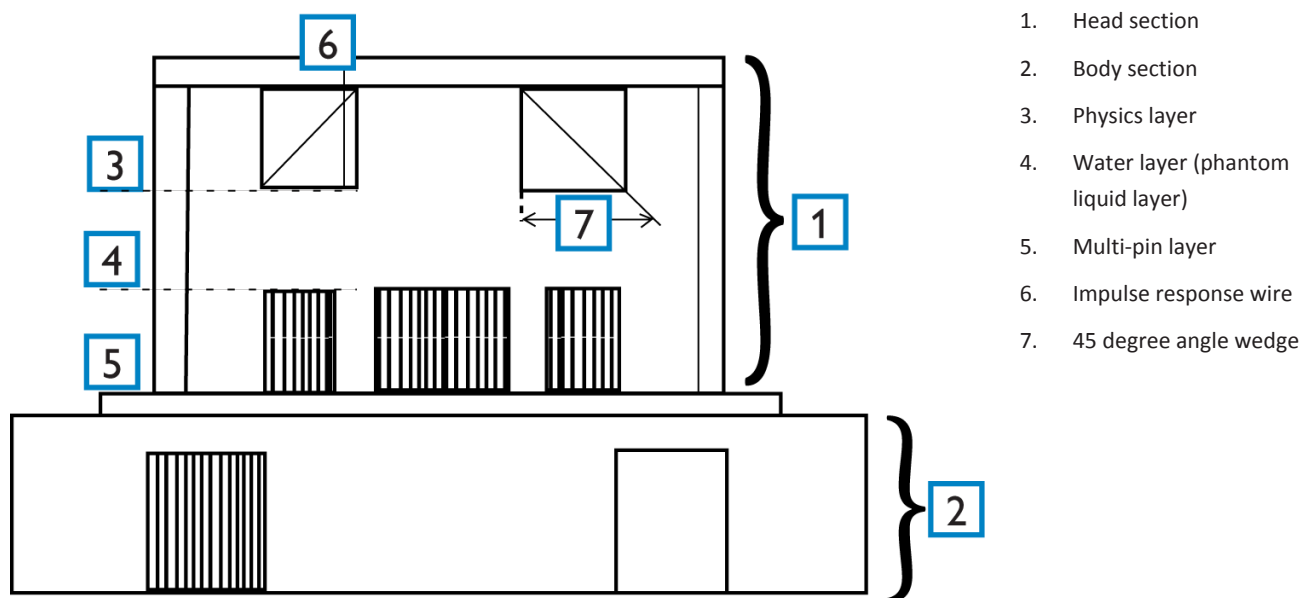
Prior to testing image quality, the system should be fully calibrated. This should be done by ensuring that all Calibrations have been run at the recommended frequency.

NOTICE

If table looseness is detected when performing Quality Assurance (QA) checks or when positioning a patient on the table, report the findings to your local Service Representative.

System Performance Phantom

The phantom consists of two portions which cover the aspects of head and body scans. This section covers the specifications of both the head and body portions of the phantom. Familiarize yourself with this information before you scan either portion. The illustration below shows the entire phantom.



Head Section

The head phantom is a Polyvinyl chloride (PVC) shell filled with water (phantom liquid). It is 200 mm outer diameter and consists of three layers:

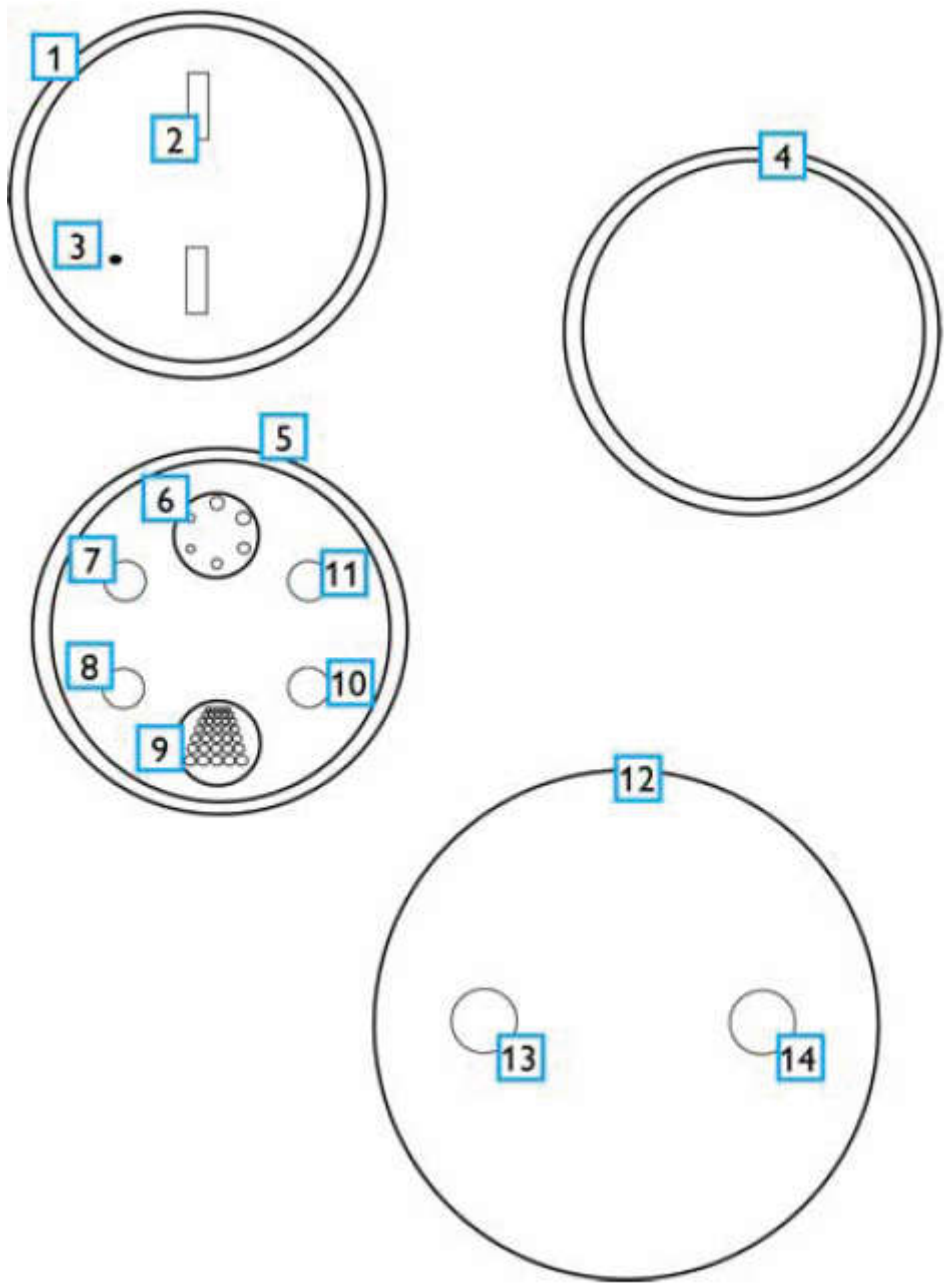
- Physics layer for resolution test and tomographic section thickness (slice width) measurements
- Water layer (phantom liquid layer) for measuring noise and uniformity
- Multi-pin layer for checking contrast scale

Body Section

The body phantom is a nylon cylinder 300 mm outer diameter. It includes these features:

- Teflon pin
- Water (phantom liquid) pin

Phantom Composition



Item	Description
1	Physics Layer
2	Aluminum strips embedded at 45 degrees
3	0.18 mm copper wire for impulse response measurement
4	Water Layer (phantom liquid layer)
5	Multi-pin layer

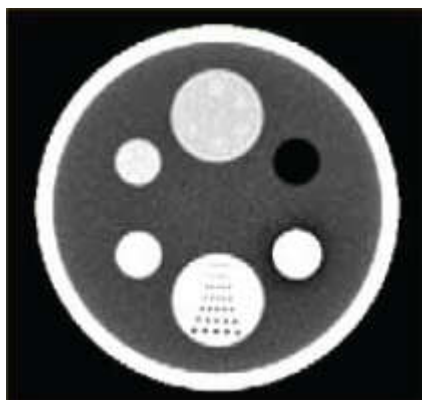
Item	Description
6*	Nylon (Aculon) body with six smaller Lexan pins of 3 mm, 4 mm, 5 mm, 6mm, 7 mm and 8 mm respectively
7	Lexan pin
8	Acrylic pin
9	Acrylic with seven rows of holes of different diameters. Each row has five equidistant holes of the same diameter: • Row 1 - 1.00 mm holes, 2.00 mm apart • Row 2 - 1.25 mm holes, 2.50 mm apart • Row 3 - 1.50 mm holes, 3.00 mm apart • Row 4 - 1.75 mm holes, 3.50 mm apart • Row 5 - 2.00 mm holes, 4.00 mm apart • Row 6 - 2.50 mm holes, 5.00 mm apart • Row 7 - 3.00 mm holes, 6.00 mm apart
10	Teflon pin
11	Polyethylene pin
12	Body layer
13	Water (phantom liquid) hole
14	Teflon pin
* Not all phantoms include this item.	

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Representative Quality Assurance Images

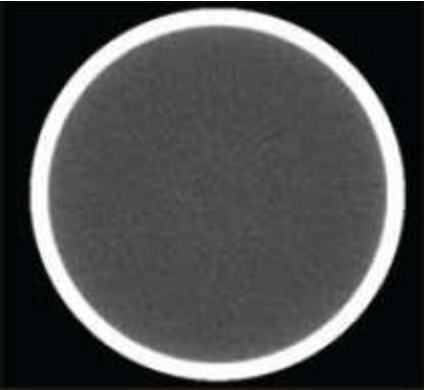
Quality assurance images acquired during the weekly and monthly checks can be stored on CD or DVD if desired. The digital data of the following images are stored on the Reference QA Images CD.

Multi-pin layer, QA Axial Head 2D

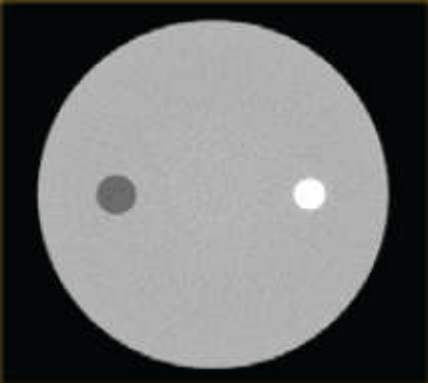


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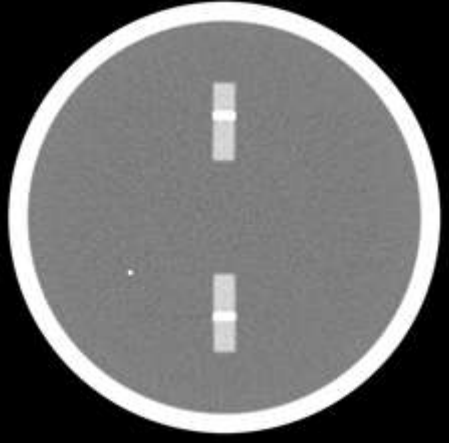
Water (phantom liquid) layer, QA Axial Head 2D



Body layer, QA Axial Body 2D

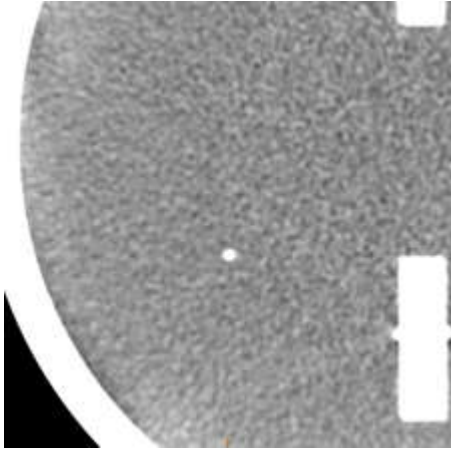


Physics layer (slice width image, QA Axial Head 2D)



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Impulse response (modified-100 FOV, QA Axial Head 2D)

System Performance Harmonized Phantom

Either the Harmonized or System Performance Phantom is supplied as part of every CT system. Familiarize yourself with this information before you scan either portion.

The system phantom is used for CT numbers calibrations and Quality Assurance tests.

The phantom kit consists of the following parts:

- Harmonized phantom - Head section
- Harmonized phantom - Body section
- Harmonized phantom - Physics section
- Harmonized phantom - Infant head section is optional. Head adult section can be used for Infant Body IQ testing.

The system phantom is used for the image quality tests and calibrations which can be accessed from the Quality Assurance menu including the automatic phantom centering.

1. IQ check (available for all users)
2. Constancy test (do not have to install the Body section)
3. Acceptance test (available for service users)
4. HCOR calibration (available for service users)
5. Performance test (available for service users and used in special cases)

The graphic displays the phantom sections that are installed on the phantom holder.

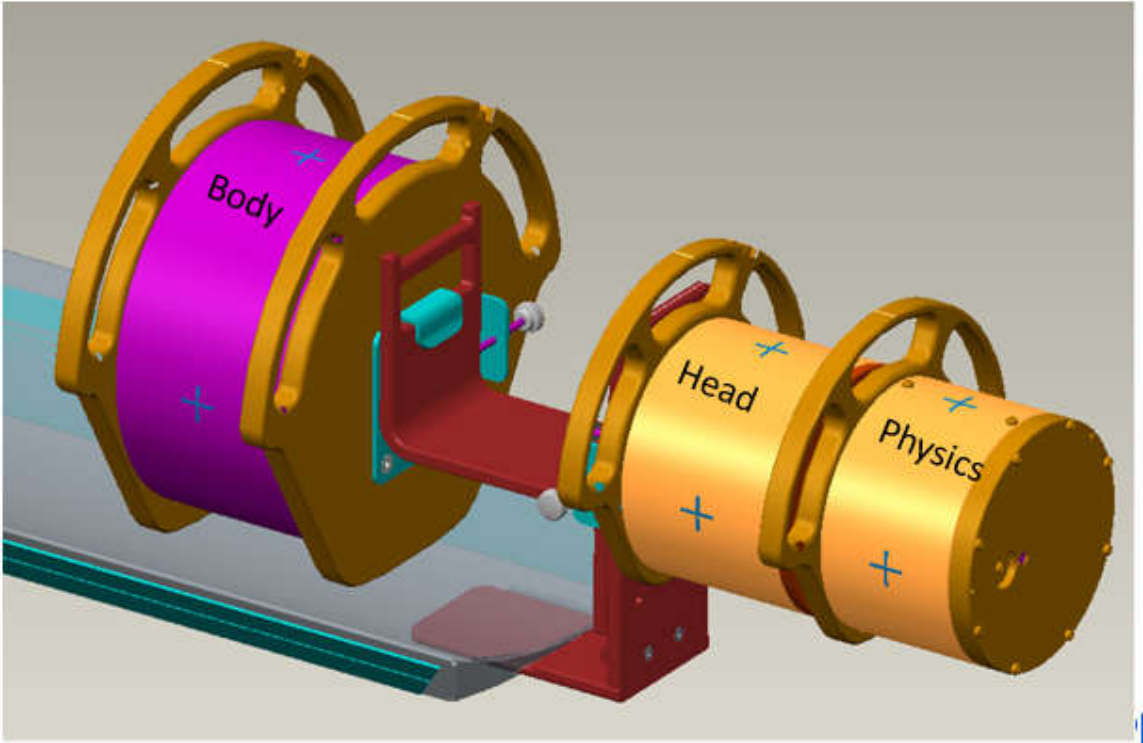


Fig. 1: Harmonized Phantom sections

The graphic displays the possible setups of system phantom.

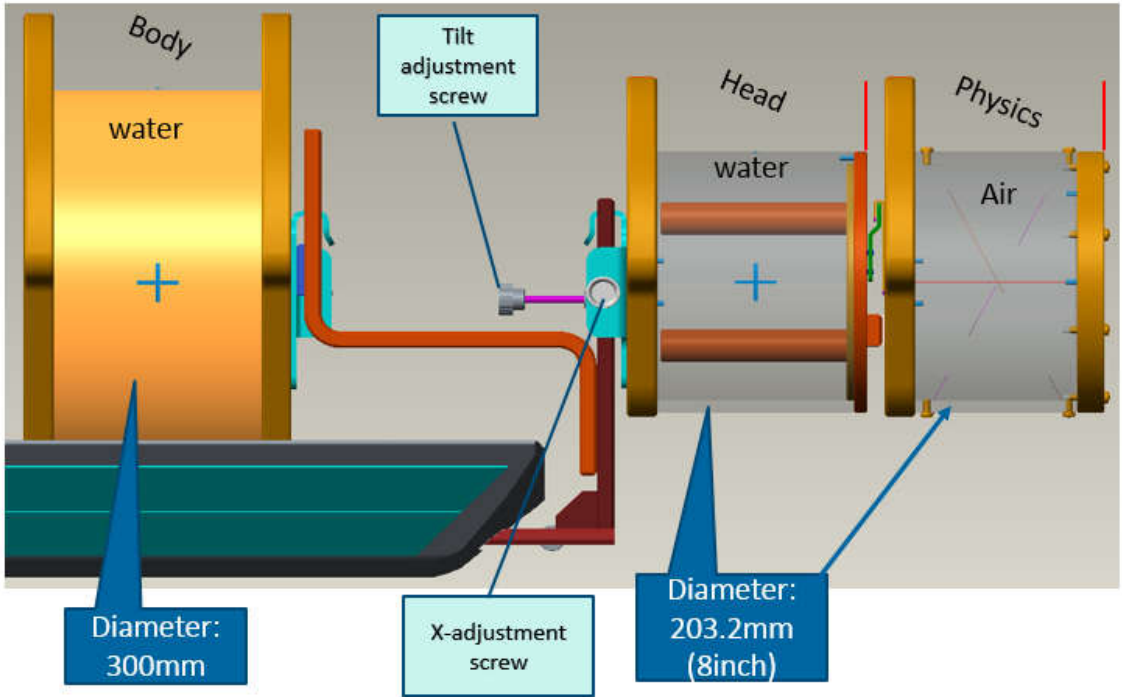


Fig. 2: Harmonized Phantom setup

Using Phantom with the Radiology Flat Top

For systems using the Radiology Flat Top for therapy planning, the following setup is used for Body section. You do not need to remove the Flat Table Top for IQ testing and calibrations.

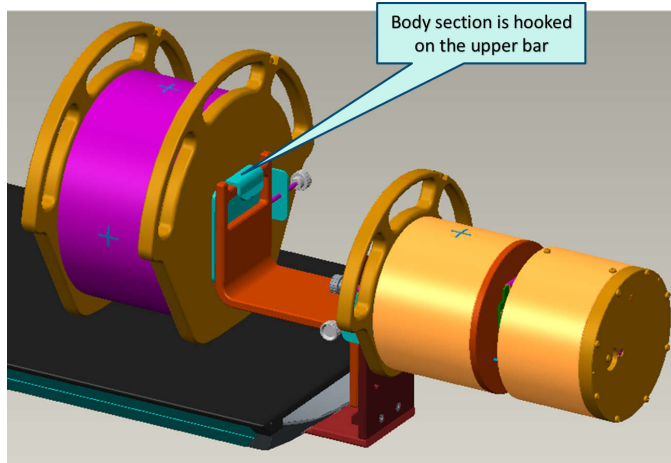
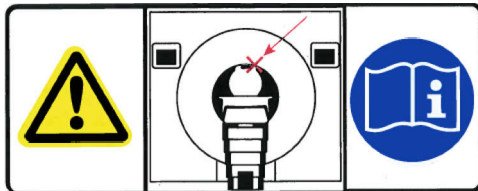


Fig. 3: Harmonized Phantom with Radiology Flat Table Top



CAUTION

Do not raise the system table to its maximum vertical height when the head or body phantom is mounted to it. The phantoms may collide with the gantry covers.



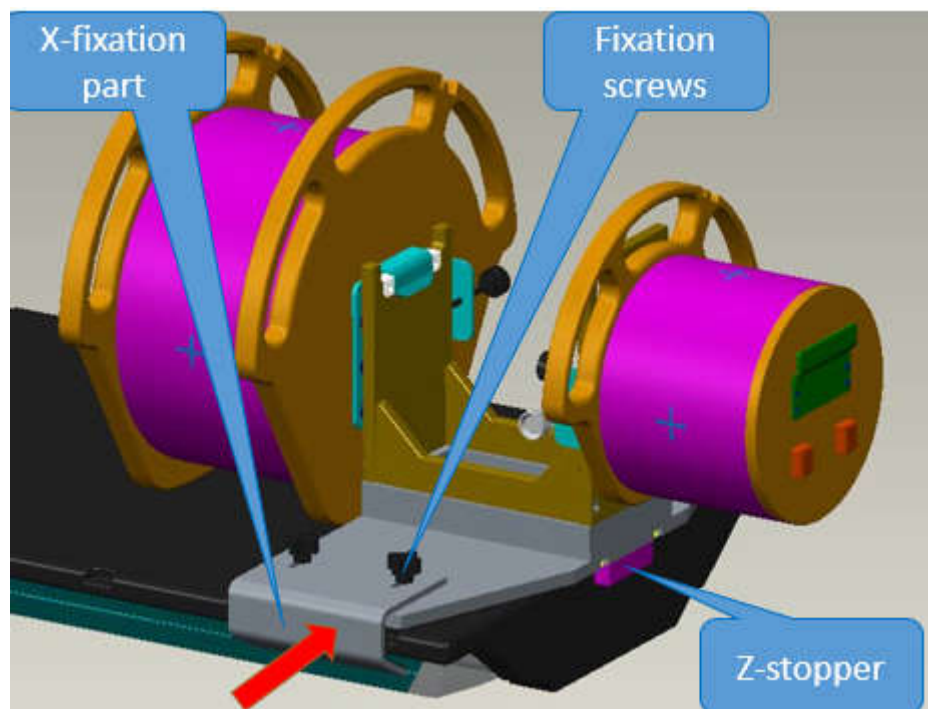
Instructions for Phantom Installation

1. Insert the system phantom holder by sliding it into the couch table top bayonet. Lock the bolt at the left hand side of the holder to prevent it from sliding out.
2. Insert the Head section and ensure that the X-adjustment bolt is not interfering by shifting the phantom to your right until it is hooked securely. Gently shift it to the left until the adjusting x-screw touches the holder.
3. Install the Body section in the same way on the rear branch of the holder hooking it on the lower holder bar:
 - in case the Flat Top is installed, the Body section is hooked on the upper bar of the rear holder branch.
 - in both cases above the end opposite to hooking mechanism of the body section lays freely either on the table top or on the flat top.

4. To install the Physics section used for Resolution and Slice thickness measurement, slide it down onto the interconnecting mechanism of the Head section. Verify it is securely connected to head section and cannot fall off. To verify this check if both sections are concentric to each other. Refer to chapter “Using Phantom with the Radiology Flat Top” on page 70.
5. To install the Infant Head section, remove the physics section (if installed) and hook it on the Head section in the same way as described above.

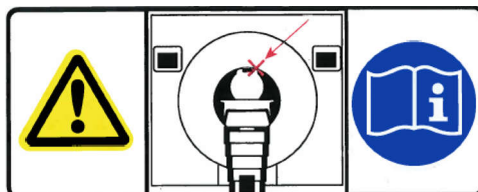
Using Phantom with the Therapy Top

For systems using the Therapy Top for therapy planning, the following setup is used for IQ testing. You do not need to remove the Therapy Top for IQ testing and HCOR calibrations.



CAUTION

Do not raise the system table to its maximum vertical height when the head or body phantom is mounted to it. The phantoms may collide with the gantry covers.



Instructions for Phantom Installation on Therapy Top option

1. Insert the system phantom holder by sliding it on top of the Therapy Top and push in it until it is stopped by Z-stopper and then push it to the right until it stops. Slide the X-fixator part to the left (see red arrow in picture above) and lock two Fixation screws to prevent it from sliding out.
2. Insert the Head section and ensure that the X-adjustment bolt is not interfering by shifting the phantom to your right until it is hooked securely. Gently shift it to the left until the adjusting x-screw touches the holder.
3. Install the Body section in the same way on the rear branch of the holder:
 - the end opposite to hooking mechanism of the body section lays freely either on the therapy top.
4. To install the Physics section used for Resolution and Slice thickness measurement, slide it down onto the interconnecting mechanism of the Head section. Verify it is securely connected to head section and cannot fall off. To verify this, check if both sections are concentric and parallel to each other.
5. To install the Infant Head section, which is optional and can be used for IQ testing, remove the physics section (if installed) and hook it on the Head section in the same way as described above.

Instructions to run HCOR calibrations using Therapy Top option

The HCOR calibration is normally performed by service. It is possible to run HCOR calibration without removing the Therapy top.

Harmonized Phantom - Head Section

Harmonized Head phantom has only one layer.

Head section is enclosed in clear shell of 203 mm outer diameter, filled with phantom liquid.

It is used for Head HCOR calibrations and Image Quality (IQ) tests, measuring uniformity, CT accuracy and noise, and measuring and calculating low contrast resolution during the IQ Check.

It contains two plastic pins made of following materials: Polyethylene and Acrylic used to measure linearity and provides contrast scale together with water and air measured out-side the phantom.

NOTICE

Poly (methyl methacrylate) (PMMA) is also known as acrylic or acrylic glass.

Harmonized Phantom - Body Section

The body phantom is a single 300 mm outer diameter PMMA shell, filled with phantom liquid. The body section is used for Body HCOR calibration and IQ tests measuring noise, uniformity and CT accuracy.

Harmonized Phantom - Physics Section

The physics phantom or section is a single 203 mm outer diameter PMMA shell, filled with air. It contains:

- A steel wire parallel to z-axis for resolution measurement using Impulse Response algorithm followed by MTF calculation.
- Two steel slant wires with opposite slope of 1:2 for Slice thickness measurement.

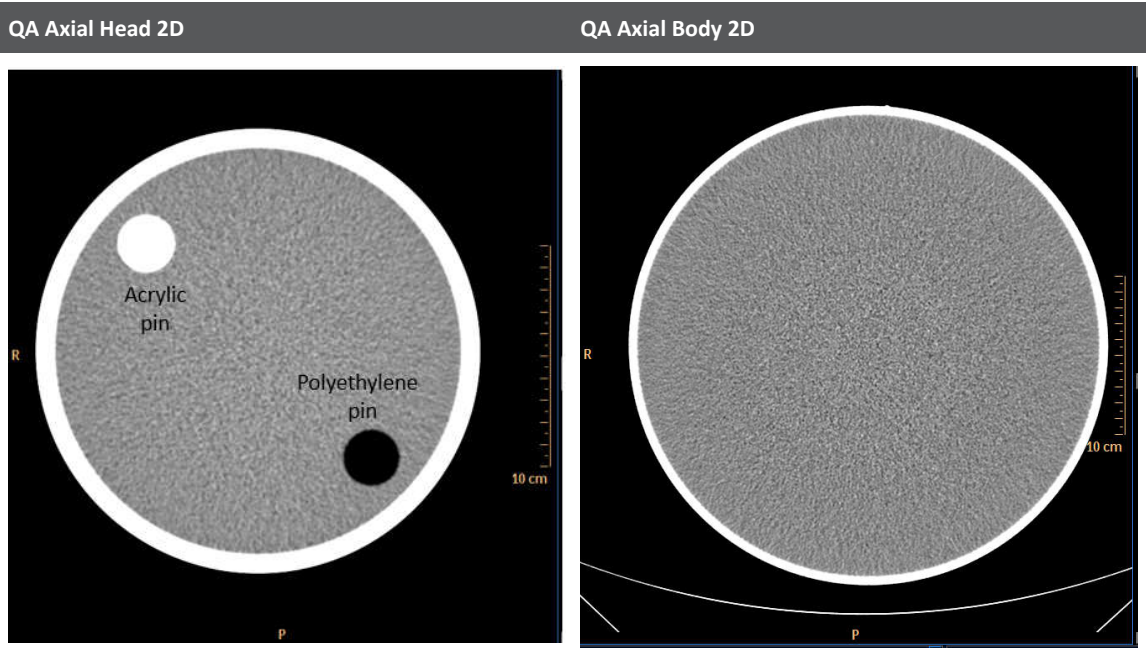
Harmonized Phantom - Infant Section (Optional)

The Infant phantom is a single 110 mm outer diameter PMMA shell, filled with phantom liquid. The Infant section is used for Infant Head HCOR calibration and IQ tests measuring uniformity and CT accuracy.

The Infant Body HCOR calibration is performed on the (adult) Head section phantom.

Representative Quality Assurance Images

Quality assurance images acquired during the weekly and monthly checks can be stored on CD and DVD if desired. The digital data of the following images are stored on the Reference QA Images CD.

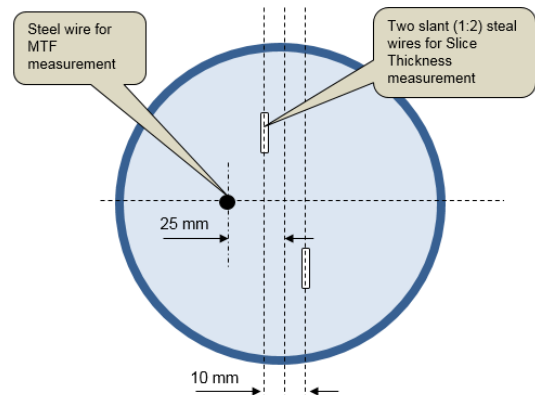
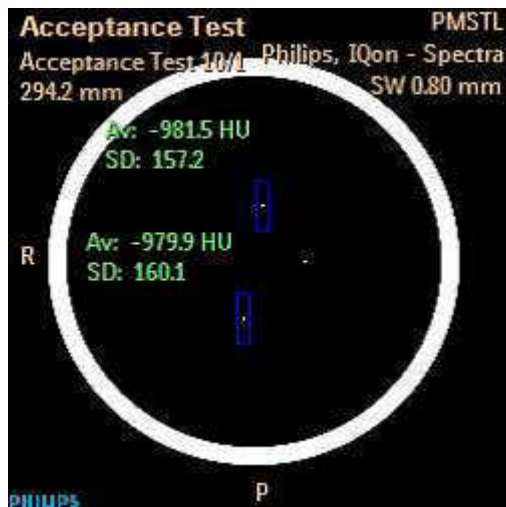


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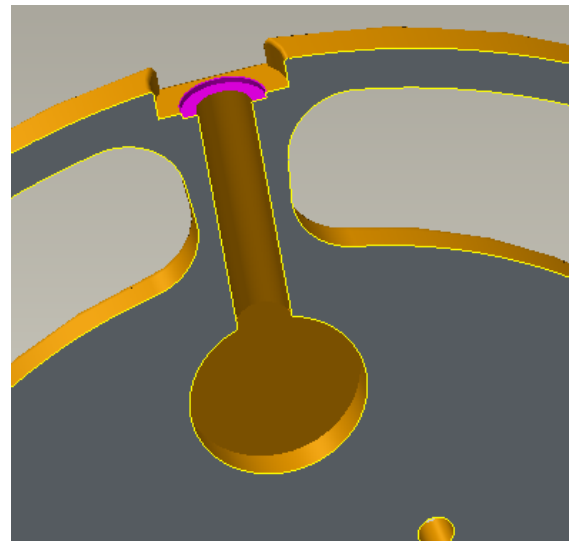
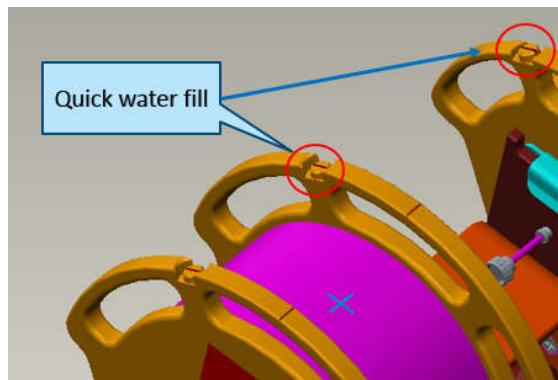
Physics layer for Resolution and Slice Thickness measurement

(QA Axial Head 2D)



Phantom Maintenance

The system phantom is certified during scanner manufacturing process and does not require routine calibrations. Adding phantom liquid (water) solution will possibly be needed because of air outgassing. Before use, verify the cylindrical volumes of the phantom are filled with phantom liquid. Take care to allow small air bubbles to escape to the bubble trap chamber by tilting slightly the phantom. The scanned areas should be free of bubbles not to interfere with measurements. Adding water is performed using specially designed opening on the phantom handle.



Daily Short Tube Conditioning

Short Tube Conditioning brings the tube to normal operating temperature. This process is required daily before the first scan and after 8 hours of scanner inactivity.



WARNING

Do not perform Short Tube Conditioning when there is a person in the scanning room.

1. Check the scan room to ensure no people are present, and that the table is positioned up and into the gantry.



2. Click the **Tool** icon.
3. Click **Quality Assurance**.
4. From the **Quality Assurance** dialog box, click **Short Tube Conditioning**. The **Procedure** column lists the additional tests that may be performed at the same time (for example, Constancy Test, Air Calibration, and so on).
5. Click **Start** and follow the screen prompts.
6. After procedure is complete, click **Exit** to close the program.

Note: If artifacts are observed even after performing short tube conditioning, contact Philips Service.

NOTICE

Perform long tube conditioning under the following scenario:

- during first time CT System operation before starting any scan operation.
- any time the tube has not been used for seven days or more.
- after any replacement of the X-Ray tube assembly before starting any scan operation.
- In case of X-ray Tube arcing as a service measure.
- If short tube conditioning is not successful after three attempts.

For More details, contact Philips Service.

Weekly Tests

Air Calibration and the **Head IQ Check** must be performed at least once a week. If time permits, both tests can be performed at the same time; however, it is not required to run the tests together.

Weekly Air Calibration

Air Calibration is part of normal system maintenance. The procedure helps to reduce ring artifacts.

- To ensure proper operation of the scanner, conduct this procedure at least once per week.
- Because this procedure must be done at stable, operating temperature, perform at midday after a number of patients have been scanned for best results. Allow about 20 minutes for completion.
- Ensure that no objects are in the scan field (in the gantry).

NOTICE

If you discover any objects in the scan field (for example, the head holder) after you begin **Air Calibration**, remove the objects from the scan field and restart the process.



WARNING

Do not perform Air Calibration if there is a person in the scanning room.

1. Check the scan room to ensure no people are present.
2. Ensure the table (and/or phantom) does not extend into the gantry.



3. Click the **Tools** icon.
4. Click **Quality Assurance**.
5. From the **Quality Assurance** dialog box, click **Air Calibration**. The **Recommended In** column lists the additional tests that may be performed at the same time (for example, Short Tube Conditioning, Air Calibration, and so on).
6. Click **Next** and follow the screen prompts.
7. After calibration is complete, click **Exit** to close the program.

Weekly Head IQ Check

Head IQ Check should be performed weekly to ensure best possible image quality from your scanner.

NOTICE

For radiation therapy planning, **Head IQ Check** is suggested to be run daily. The water (phantom liquid) CT number and noise should be checked.

**WARNING**

Do not perform the IQ Check if there is a person in the scanning room.

1. Attach phantom holder to the end of the patient table.
2. Check the scan room to ensure no people are present.
3. Place the system Head and Body phantom(s) or Harmonized Head Phantom on the holder.



4. Click the **Tools** icon to open the system utilities.
5. From the **Tools** menu, click **Quality Assurance**. The Quality Assurance dialog displays.
6. Select **IQ Check** and any other tests to be run. The **Procedure** column lists the additional tests that may be performed at the same time (for example, Short Tube Conditioning, Air Calibration, and so on).
7. Click **Next**. Select the appropriate IQ Check procedure.
8. Click **Next**. Follow the on-screen instructions for all the tests, including phantom placement. After completing the steps, click **Start** to automatically move the table and begin the check.

NOTICE

Do not skip the Automatic Centering step. This vertically aligns the phantom and correctly positions the table for the scan.

9. When the check is complete, the **Quality Assurance** dialog box displays.

NOTICE

IQ Check and **Constancy Test** report data may be recorded manually but may not be electronically exported to an external device (such as USB). If you need to export an electronic copy of a report, contact your Philips service representative.

**WARNING**

If ring artifacts are observed in the acquired images, perform Air Calibration. If ring artifacts persist, contact your Philips Service Representative.

If any Test Failed

1. Click on the **Report** icon to view the failure or failures.
2. Ensure the phantom is properly aligned and level.
3. Check the images for any foreign objects (such as pins from another section of the phantom), ring artifact, or band artifact.
4. Perform Air Calibration procedure.
5. Repeat the test.

If any test fails again or if ring or band artifacts persist, report the findings to your local Service representative before scanning patients in order to ensure safe operation.

Monthly Constancy Test**NOTICE**

Constancy Test, Body IQ Check, CT Number Linearity and In Plane Spatial Integrity Check must be performed at least once a month. It is not required to run the tests together.

**WARNING**

Do not perform the Constancy Test if there is a person in the scanning room.

The following test should be performed once a month:

1. Attach phantom holder to the end of the patient table.
2. Place the system Head and Body phantoms or Harmonized Head and Body Phantoms with Physics layer on the holder.
3. Check the scan room to ensure no people are present.
4. Click the **Tools** icon and then select **Quality Assurance**.



5. Select **Constancy Test**. The **Procedure** column lists additional tests that may be performed at the same time (for example, Short Tube Conditioning, Air Calibration, and so on).
6. Click **Next**.
7. Click **Next**. Follow the on-screen instructions for all the included tests, including phantom placement. After completing the steps, click **Next** again.

NOTICE

Ensure that you are running all the tests that make up the **Constancy Test**. Do not deselect any of the sub-tests.

8. The **Instructions** page opens, with instructions to correctly set up the phantom. After completing these instructions, click **Start**.

NOTICE

Do not skip the Automatic Centering step. This vertically aligns the phantom and correctly positions the table for the scan.

9. Type your name in the **User Name** field and the reason for the test (for example, monthly constancy test). Click **OK**.

Click **Start**. If a temperature stabilization message displays, click **Close** to proceed.

10. When the test is complete, the **Quality Assurance** dialog box displays.

NOTICE

IQ Check and **Constancy Test** report data may be recorded manually but may not be electronically exported to an external device (such as USB). If you need to export an electronic copy of a report, contact your Philips service representative.

If any Test Failed

If a test fails, click on the **Report** icon to view the failure or failures.

Ensure the phantom is properly aligned and level. Repeat the test.

NOTICE

Check the images for any foreign objects such as pins from another section of the phantom or possibly ring artifacts. If ring artifacts are seen, allow the scanner to warm up for 10 to 15 minutes by repeating the **Short Tube Conditioning** procedure multiple times.

When prompted: **Do you want to continue the last attempt and perform only the failed modes?** Click **Yes** to repeat only the failed tests. Click **No** to repeat all the tests.

If Any Test Failed a Second Time



If the test fails a second time, click on the **Report** icon to view the failures and report the findings to your local Service Representative before scanning patients in order to ensure safe operation.

Monthly Body IQ Check

Body IQ Check should be performed monthly to ensure best possible image quality from your scanner.



WARNING

Do not perform the IQ Check if there is a person in the scanning room.

1. Attach phantom holder to the end of the patient table.
2. Place the system Head and Body phantom or the Harmonized Body Section on the holder.
3. Check the scan room to ensure no people are present.



4. Click the **Tools** icon to open the system utilities.
5. From the **Tools** menu, click **Quality Assurance**. The Quality Assurance dialog displays.
6. Select **IQ Check** and any other tests to be run. The **Procedure** column lists the additional tests that may be performed at the same time (for example, Short Tube Conditioning, Air Calibration, and so on).
7. Click **Next**. Select the appropriate IQ Check procedure.
8. Click **Next**. Follow the on-screen instructions for all the tests, including phantom placement. After completing the steps, click **Start** to automatically move the table and begin the check.

NOTICE

Do not skip the Automatic Centering step. This vertically aligns the phantom and correctly positions the table for the scan.

9. When the check is complete, the **Quality Assurance** dialog box displays.

NOTICE

IQ Check and **Constancy Test** report data may be recorded manually but may not be electronically exported to an external device (such as USB). If you need to export an electronic copy of a report, contact your Philips service representative.

Monthly CT Number Linearity and In Plane Spatial Integrity Check (System Phantom)

This check should be performed monthly to ensure the CT number accuracy of 5 different materials and the spatial integrity in the image plane. This check is suggested for users using the system for radiation therapy planning.



WARNING

Do not perform the CT Number Linearity and In Plane Spatial Integrity Check if there is a person in the scanning room.

1. Attach phantom holder to the end of the patient table.
2. Check the scan room to ensure no people are present.
3. Place the System Performance Phantom in the holder and center it in the scan circle.
4. Click **Patient**. Enter the demographic information. Select **Adult** from the **Age** group.
5. Click **Reference Exam Cards**.
 - Click **Abdomen** exam card group. Select the **Survview** exam card.
 - Click **IQ Test** exam card group. Select the **Head Axial 2D** exam card.
 - Click **OK**.
6. Set the survview to scan the entire System Performance Phantom.
7. Using the survview, plan the **Head Axial 2D** scan at the multi-pin layer of the head section. Perform the scan with the following parameter values:
 - **Resolution**: Standard
 - **Collimation**: 16 x 1.5
 - **Scan Angle**: 360°
 - **Rotation Time**: 0.75s
 - **Reconstruction Filter**: UB
 - **FOV (mm)**: 250
 - **Storage**: Local
 - **Thickness (mm)**: 3.0
 - **Increment (mm)**: 0

- **kVp:** 120
 - **mAs:** 200
 - **Number of Scans:** 1
8. Select one middle image from the series and measure the CT number of different pins and regions (different materials) by positioning a small ROI (radius of about 10mm) well within each of the checked pins and regions. The CT number of the different pins must be as follows (values in HU):
 - **Water (phantom liquid):** 0 ± 4
 - **Polyethylene:** -62 ± 25
 - **Acrylic:** 130 ± 25
 - **Lexan:** 105 ± 25
 - **Teflon:** 890 ± 75
 9. Select one middle image from the series, use the **Line** tool to measure the diameter of the large Acrylic pin. The diameter measured should be 50 ± 1 mm.

Monthly CT Number Linearity and In Plane Spatial Integrity Check (Harmonized Phantom)

This check should be performed monthly to ensure the CT number accuracy of 5 different materials and the spatial integrity in the image plane. This check is suggested for users using the system for radiation therapy planning.



WARNING

Do not perform the CT Number Linearity and In Plane Spatial Integrity Check if there is a person in the scanning room.

1. Attach phantom holder to the end of the patient table.
2. Check the scan room to ensure no people are present.
3. Place the System Performance Phantom in the holder and center it in the scan circle.
4. Click **Patient**. Enter the demographic information. Select **Adult** from the **Age** group.
5. Click **Reference Exam Cards**.
 - Click **Abdomen** exam card group. Select the **Survview** exam card.
 - Click **IQ Test** exam card group. Select the **Head Axial 2D** exam card.
 - Click **OK**.
6. Set the survview to scan the entire System Performance Phantom.
7. Using the survview, plan the **Head Axial 2D** scan at the head section. Perform the scan with the following parameter values:

- **Resolution:** Standard
 - **Collimation:** 8 x 3
 - **Scan Angle:** 360°
 - **Rotation Time:** 0.75s
 - **Reconstruction Filter:** UB
 - **FOV (mm):** 250
 - **Storage:** Local
 - **Thickness (mm):** 3.0
 - **Increment (mm):** 0
 - **kVp:** 120
 - **mAs:** 200
 - **Number of Scans:** 1
8. Select one middle image from the series and measure the CT number of different pins and regions (different materials) by positioning a small ROI (radius of about 10mm) well within each of the checked pins and regions. The CT number of the different pins must be as follows (values in HU):
 - **Water (phantom liquid):** 0 ± 4
 - **Polyethylene:** -100 to -40
 - **Acrylic:** 104 to 164
 - **Air:** -1010 to -990
 9. Select one middle image from the series, use the **Line** tool to measure the diameter of the large Acrylic pin. The diameter measured should be 25 ± 1 mm.

Test Failure

If any test fails, follow the procedure below.



WARNING

If ring artifacts or band artifacts are observed in the acquired images, perform the Air Calibration procedure. If the ring artifacts persist, contact your Philips Service Representative and report this issue before scanning patients in order to ensure safe operation.



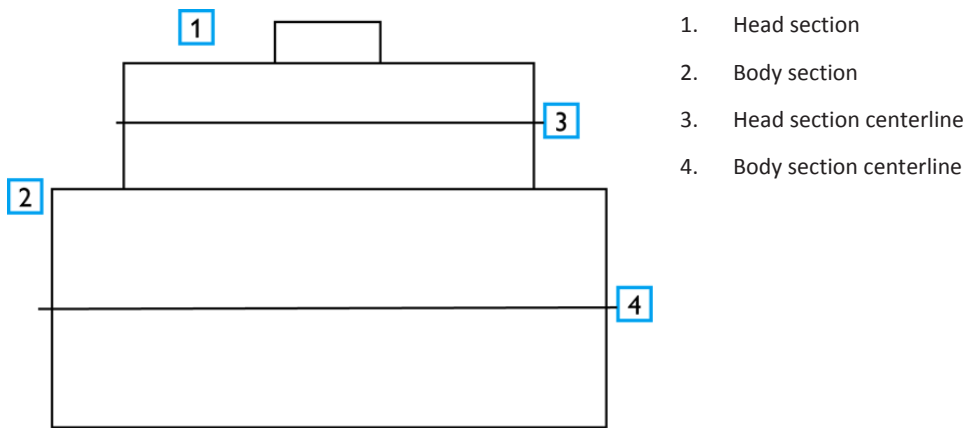
1. Ensure the phantom is properly aligned and level.
2. Check the images for any foreign objects (such as pins from another section of the phantom), ring artifact, or band artifact.
3. Perform Air Calibration procedure.
4. Repeat the test.

If any test fails again, report the findings to your local Service Representative before scanning patients in order to ensure safe operation.

Infant Phantom Testing

The infant scanning mode is recommended for scanning infants under eighteen months of age. This section provides instructions for using your infant phantom to check the performance in the infant scanning mode.

The Infant Phantom is intended to radiographically simulate the infant imaging condition.



This phantom consists of a head section and a body section.

- The Body section consists of an approximately 15 cm diameter cylindrical acrylic shell.
- The Head section is made of an approximately 11.4 cm outer diameter cylindrical PVC shell with an approximately 1.5 mm wall thickness.

Before use, fill the cylindrical volumes of the phantom with the phantom liquid. Take care to allow all air bubbles to escape the chamber through the top fill box.

The Harmonized Phantom set uses

- Adult Head Section for Infant Body testing
- Infant Head Section mounted on the Adult Head Section, for Infant Head testing

Attach & Scan Phantom

Before you can use the infant phantom for quality assurance testing, you must attach it to the patient table. Use this procedure to install the phantom.

1. Attach phantom holder to the end of the table.

2. Move the patient table away from the gantry (OUT).
3. Check the mounting bracket to ensure it is tightly attached.
4. For the one-piece Infant Phantom, mount the phantom onto the holder. For the Harmonized Phantom set, mount the Head phantom onto the holder and then insert the infant phantom by sliding its hooks onto the Head Phantom and make sure it is securely attached to the Adult Head section.
5. Turn on the laser markers.
6. Move the patient table to position the phantom in isocenter.
7. Move the patient table towards the gantry (IN).
8. Position the table so that the laser markers line up with the centerline on the phantom holder. For the Harmonized phantom set, position the table so that the laser markers line up with the white line on the phantom holder. Zero the table.
9. Click **Patient**. Enter the demographic information. Select **Infant** from the Age Group.
10. Click **Reference Exam Cards**.
 - Click the **Head** exam card group. Select the **Brain Axial 0-18m** exam card.
 - Click the **Abdomen** exam card group. Select the **Abd/Pel 0-10 kg** exam card.
 - Click **OK**.
11. Set the survview to scan the entire Infant Phantom(s).
12. Using the survview, plan the Infant Brain Axial scan to (at minimum) cover the Head part of the Infant phantom (see the parameters in the table). Perform the scan.
13. Using the same survview, plan the Infant Body Helical scan to (at minimum) cover the Body part of the Infant phantom (see the parameters in the table). Perform the scan.
14. The phantom liquid (water) CT values should yield these results:
 - For the Head Section, using an ROI of $100 \pm 20 \text{ mm}^2$, the CT number should measure 0 ± 4 CT numbers.
 - For the Body Section, using an ROI of $195 \pm 30 \text{ mm}^2$, the CT number should measure 0 ± 6 CT numbers.

NOTICE

Your Service Engineer can adjust the CT number level if it is out of specification.

1. Perform Air calibration and recheck.
2. If still out of limits call Field Service Engineer.

Infant Scan Protocols		
	Head	Body
Resolution	Standard	Standard
Collimation	16 x 0.75	16 x 1.5
Scan Angle	420°	n/a
Rotation Time	0.75	0.5
Reconstruction Filter	UB	B
FOV (mm)	200	250
Storage	Local	Local
Thickness (mm)	3.0	3
Increment (mm)	n/a	1.5
kVp	100	100
mAs	300	165
Number of Scans	1	1

Monitor Calibration test

This procedure is for use with both standard and Deutsche Industrie-Norm (DIN) 6868-157 standard Acceptance Test for Image Display Systems with diagnostic quality only.

If you are unsure if your monitor follows DIN 6868-157 standard, contact your local Philips service representative.

1. On the left monitor, click the wrench icon on the top-right corner.
2. From the panel opened on left side of the monitor, select **Monitor Calibration**.
3. Select the desired image type or pattern type.
4. Select the image position as center, clinical or full screen.
5. Click on the required test pattern.
6. Follow the DIN 6868-157 procedure for the Acceptance Test for Image Display Systems (For DIN monitors only).
7. Select **Close** button to terminate the **Monitor Calibration** application.

NOTICE

Monitor Calibration is not available during active scan.

**WARNING**

If an error occurs during the test, the system displays a red X. Contact Philips Service for assistance.

Advanced Quality Assurance Checks - System Phantom

These procedures can be useful for resolving problems revealed by the daily checks. They are intended for use as advanced applications for Physicists and your Philips Service Specialist.

Spatial resolution or MTF measurements

1. Position the physics layer of the head phantom in the center of the bore.
2. Perform a scan using the Head STD-QA protocol, with the parameter values listed below for Standard resolution:
Axial 2D, Head, 120kV, 64x0.625, Standard Resolution, 0.75s, 200mAs, Slice thickness 1.25mm, YB, FOV 50, 1024x1024, X=25, Y=-47.
3. After the images arrive in the archive, select the images from the archive and use the Image Tests menu (or Application) in the Analysis menu. Use one of the middle reconstructed slices for the measurement.

You may need to adjust the window level to see the wires.

- Press on Resolution Test button (“Position the pin” icon) and place a pin on or near the Resolution (straight) wire.
- Press on the Calculate resolution button (next to the Resolution Test button).
- Look at the Results window that pops up.

4. The 10% MTF should be 8.9 ± 1.0 lp/cm and 50% MTF should be 4.3 ± 0.7 lp/cm
5. Repeat steps above for High resolution using the Head STD-QA protocol, with the parameter values listed below setting High resolution:
Head Impulse Response, 120kV, 16x0.75 High Resolution, 1.5s, 400mAs, slice Thickness 3.0mm, Filter E, FOV 50, 1024x1024, X=25, Y=-47.
6. The 10% MTF should be 11.9 ± 1.0 lp/cm and 50% MTF should be 5.5 ± 0.7 lp/cm.

Tomographic section thickness measurements

This function is used to check the Slice Thickness (width) of the scanner during monthly tests. Available protocols depend on your scanner type.

Setup

Before activating the Slice Thickness function, be sure perform the preliminary procedures in steps 1-4 below.

1. Install the head system phantom used for slice width measurements.
2. Perform a 90° Surview scan.

3. Plan an axial slice on the Surview scan, centered on the slice width wires. You may need to adjust the window level to see the wires.
4. Adjust In/Out position of phantom to obtain better precision.

Checking the 0.75 mm slice thickness

This section contains information for conducting advanced quality assurance procedures.

1. Adjust the patient table so the center of the Physics layer on the Head Phantom is aligned with the laser lights.
2. Set up a standard patient scan using the workflow bar. See for more details.
3. Perform the first Axial scan with the following parameters: Use Body Axial scans:
Axial 2D, 16x0.75, 0.75 sec, 120 kVp, 200 mAs, Standard Resolution, Standard Reconstruction, Slice Thickness 0.75 mm, Window Center -1000, Window Width 400, Filter E, FOV 250, Matrix 512x512.
4. Access the Archive window.
5. Select the first scan series.
 - Select the Image tests application from the Analysis menu.
 - Select one of the middle images of the first scan.
 - Press on Slice Thickness button and position the rectangle on the upper Aluminum ramp.
 - Press on the Calculate button (next to the Slice Thickness button).
 - Look at the Results window that pops up. Use the Slice Width at 50% (FWHM) measurement.
6. Repeat the measurement on the upper Aluminum ramp.
7. The average of the two measurements should satisfy $1.2 \text{ mm} \pm 0.3 \text{ mm}$.
Note: The tomographic section thickness of narrow slices may appear wider than nominal because of the width of the measuring ramp.
8. Repeat steps 3-7 using the either first or last slice visible on one of the images.

Checking the 1.5 mm slice thickness

1. Adjust the patient table so the center of the Physics layer on the Head Phantom is aligned with the laser lights.
2. Set up a standard patient scan using the workflow bar. See for more details.
3. Perform the first Axial scan with the following parameters: Use Body Axial scans:
Axial 2D, 16x1.5, 0.75 sec, 120 kVp, 200 mAs, Standard Resolution, Standard Reconstruction, Slice Thickness 1.5 mm, Window Center -1000, Window Width 400, Filter E, FOV 250, Matrix 512x512.
4. Access the Archive window.
5. Select the first scan series.
 - Select the Image tests application from the Analysis menu.
 - Select one of the middle images of the first scan.

- Press on Slice Thickness button and (position the rectangle on the upper Aluminum ramp).
 - Press on the Calculate button (next to the Slice Thickness button).
 - Look at the Results window that pops up. Use the Slice Width at 50% (FWHM) measurement.
6. Repeat the measurement on the upper Aluminum ramp.
 7. The average of the two measurements should satisfy $1.5 \text{ mm} \pm 0.5 \text{ mm}$.
Note: The tomographic section thickness of narrow slices may appear wider than nominal because of the width of the measuring ramp.
 8. Repeat steps 3-7 using the either first or last slice visible on one of the images.

Advanced Quality Assurance Checks - Harmonized Phantom

These procedures can be useful for resolving problems revealed by the daily checks. They are intended for use as advanced applications for Physicists and your Philips Service Specialist.

Spatial resolution or MTF measurements

1. Position the Physics layer in the center of the bore.
2. Perform a scan using the Head STD-QA protocol, with the parameter values listed below for Standard resolution:
 Axial 2D, 120kV, 16x1.5, Standard Resolution, 0.75s, 200mAs, Slice Thickness 3.0mm, Filter E, FOV 50, 512x512, X=25, Y=0.
3. After the images arrive in the archive, select the images from the archive and use the Image Tests menu (or Application) in the Analysis menu. Use one of the middle reconstructed slices for the measurement.

You may need to adjust the window level to see the wires.

- Press on Resolution Test button (“Position the pin” icon) and place a pin on or near the Resolution (straight) wire.
 - Press on the Calculate resolution button (next to the Resolution Test button).
 - Look at the Results window that pops up.
4. The 10% MTF should be $8.9 \pm 1.0 \text{ lp/cm}$ and 50% MTF should be $4.2 \pm 0.8 \text{ lp/cm}$
 5. Repeat steps above for High resolution using the Head STD-QA protocol, with the parameter values listed below setting High resolution:
 Axial 2D, 120kV, 16x0.75, High Resolution, 1.5s, 200mAs, Slice Thickness 1.5mm, Filter E, FOV 50, 1024x1024, X=25, Y=0.
 6. The 10% MTF should be $12.3 \pm 2.0 \text{ lp/cm}$ and 50% MTF should be $6.2 \pm 1.5 \text{ lp/cm}$.

Tomographic section thickness measurements

This function is used to check the Slice Thickness (width) of the scanner during monthly tests. Available protocols depend on your scanner type.

Setup

Before activating the Slice Thickness function, be sure perform the preliminary procedures in steps 1-4 below.

1. Install the Physics section onto the head system phantom.
2. Perform a 90° Surview scan.
3. Plan an axial slice on the Surview scan, centered on the slice width wires. You may need to adjust the window level to see the wires.
4. Adjust In/Out position of phantom to obtain better precision.

Checking the 0.75 mm slice thickness

This section contains information for conducting advanced quality assurance procedures.

1. Adjust the patient table so the center of the Physics layer on the Head Phantom is aligned with the laser lights.
2. Set up a standard patient scan using the workflow bar.
3. Perform the first Axial scan with the following parameters: Use Body Axial scans:
Axial 2D, 120kV, 16x0.75, 0.75s, 200mAs, Standard Resolution, Standard Reconstruction, Slice Thickness 0.75mm, Filter E, Window Center -1000, Window Width 400, FOV 250, 1024x1024.
4. Access the Archive window.
5. Select the first scan series.
 - Select the Image tests application from the Analysis menu.
 - Select one of the middle images of the first scan.
 - Press on Slice Thickness button and position the rectangle around the upper steel wire.
 - Press on the Calculate button (next to the Slice Thickness button).
 - Look at the Results window that pops up. Use the Slice Width at 50% (FWHM) measurement.
6. Repeat the measurement around the upper steel wire.
7. The average of the two measurements should satisfy $0.75 \text{ mm} \pm 0.5 \text{ mm}$.

Note: The tomographic section thickness of narrow slices may appear wider than nominal because of the width of the measuring ramp.

8. Repeat steps 3-7 using the either first or last slice visible on one of the images.

Checking the 1.5 mm slice thickness

1. Adjust the patient table so the center of the Physics layer on the Head Phantom is aligned with the laser lights.
2. Set up a standard patient scan using the workflow bar.
3. Perform the first Axial scan with the following parameters: Use Body Axial scans:

Axial 2D, 120kV, 16x1.5, 0.75s, 200mAs, Standard Resolution, Standard Reconstruction, Slice Thickness 1.5mm, Filter YA, Window Center -1000, Window Width 400, FOV 250, 512x512.

4. Access the Archive window.
5. Select the first scan series.
 - Select the Image tests application from the Analysis menu.
 - Select one of the middle images of the first scan.
 - Press on Slice Thickness button and (position the rectangle around the upper steel wire).
 - Press on the Calculate button (next to the Slice Thickness button).
 - Look at the Results window that pops up. Use the Slice Width at 50% (FWHM) measurement.
6. Repeat the measurement around the upper steel wire.
7. The average of the two measurements should satisfy $1.5 \text{ mm} \pm 0.5 \text{ mm}$.

Note: The tomographic section thickness of narrow slices may appear wider than nominal because of the width of the measuring ramp.
8. Repeat steps 3-7 using the either first or last slice visible on one of the images.

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6 Harmonized System Phantom



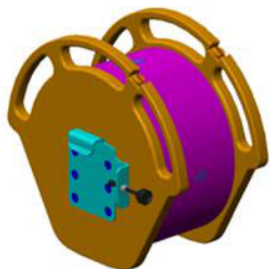
CAUTION

When setting up the harmonized phantom, do not damage any of the phantom components.

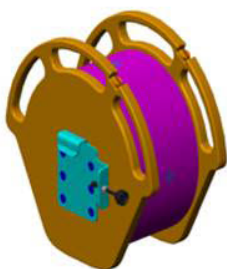
Sections in this help file:

- Harmonized Phantom Configuration Matrix
- Example of Phantom Configurations
- Harmonized System Phantom Holder Setup
- Adjusting the Phantom Left/Right Alignment
- Adjusting the Harmonized Phantom Tilt (Z-direction)
- Harmonized Body Section Installation
- Installing Harmonized Physics Section
- Installing Harmonized Infant Head Section
- Assemble Harmonized Head Phantom to Previously Adjusted Position
- Performing a Manual Scan
- Therapy Top (optional)

Harmonized Phantom Sections



Body 80mm Section



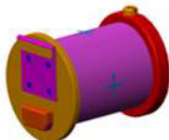
Body 40mm Section



Physics Section



Head Section



Infant Section



Eye phantom for ICT/IQ

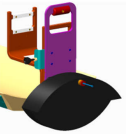
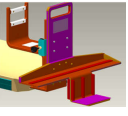
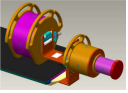
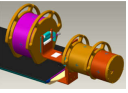
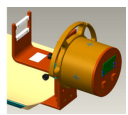
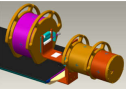
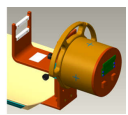
Harmonized Phantom Configuration Matrix

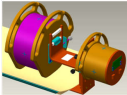
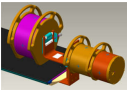


CAUTION
Perform all calibrations and image quality checks with the gantry covers closed.

The following table defines the System Harmonized Phantom usage for different applications:

Calibration	Eye	SOF	Adult Head	Body	Physics	Infant	Figures
Tube Alignment	✓ + Slant XRT pin	N/A	N/A	N/A	N/A	N/A	
Focal spot X-position	✓ + Slant XRT pin	N/A	N/A	N/A	N/A	N/A	

Phantom	✓	N/A	N/A	N/A	N/A	N/A	
UHR Comb Align	✓	N/A	N/A	N/A	N/A	N/A	
+ Straight XRT pin							
UHR Focal Spot X Position	✓	N/A	N/A	N/A	N/A	N/A	
+ Straight XRT pin							
SFS Position	✓	N/A	N/A	N/A	N/A	N/A	
+ Straight XRT pin							
DFS Position	✓	N/A	N/A	N/A	N/A	N/A	
+ Straight XRT pin							
UHR DFS Position	✓	N/A	N/A	N/A	N/A	N/A	
+ Straight XRT pin							
Off-focal	N/A	✓	N/A	N/A	N/A	N/A	
DDC	N/A	✓	N/A	N/A	N/A	N/A	
Steps	N/A	✓	N/A	N/A	N/A	N/A	
HCOR Basic	N/A	N/A	✓	✓	N/A	✓	
HCOR Slice	N/A	N/A	✓	✓	N/A	✓	
Performance	N/A	N/A	✓	✓	N/A	✓	
Acceptance	N/A	N/A	✓	✓	✓	N/A	
Constancy	N/A	N/A	✓	✓	✓	N/A	
IQ Test, Head only	N/A	N/A	✓	N/A	N/A	N/A	

IQ Test, Head and Body	N/A	N/A	✓	✓	N/A	N/A	
Phantom centering	N/A	N/A	✓	✓	N/A	N/A	

Example of Phantom Configurations

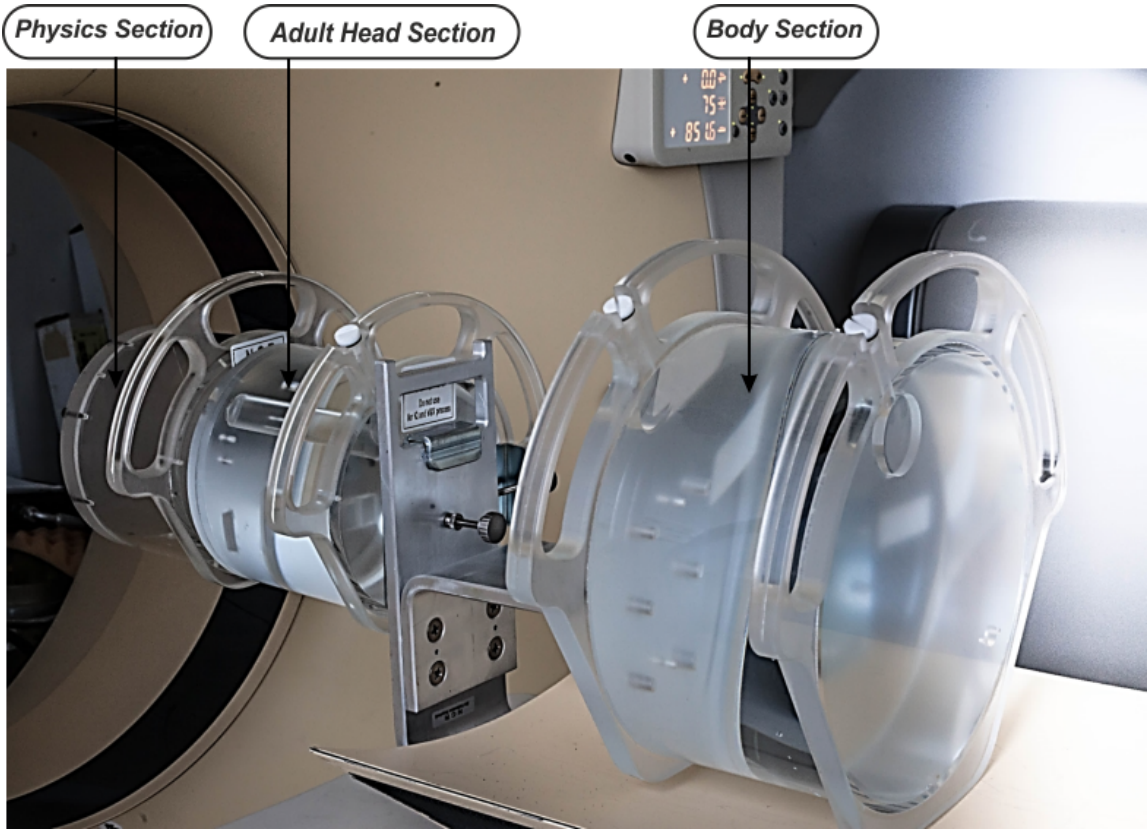


Fig. 4: Harmonized System Phantom Setup for Adult Acceptance and Constancy Test

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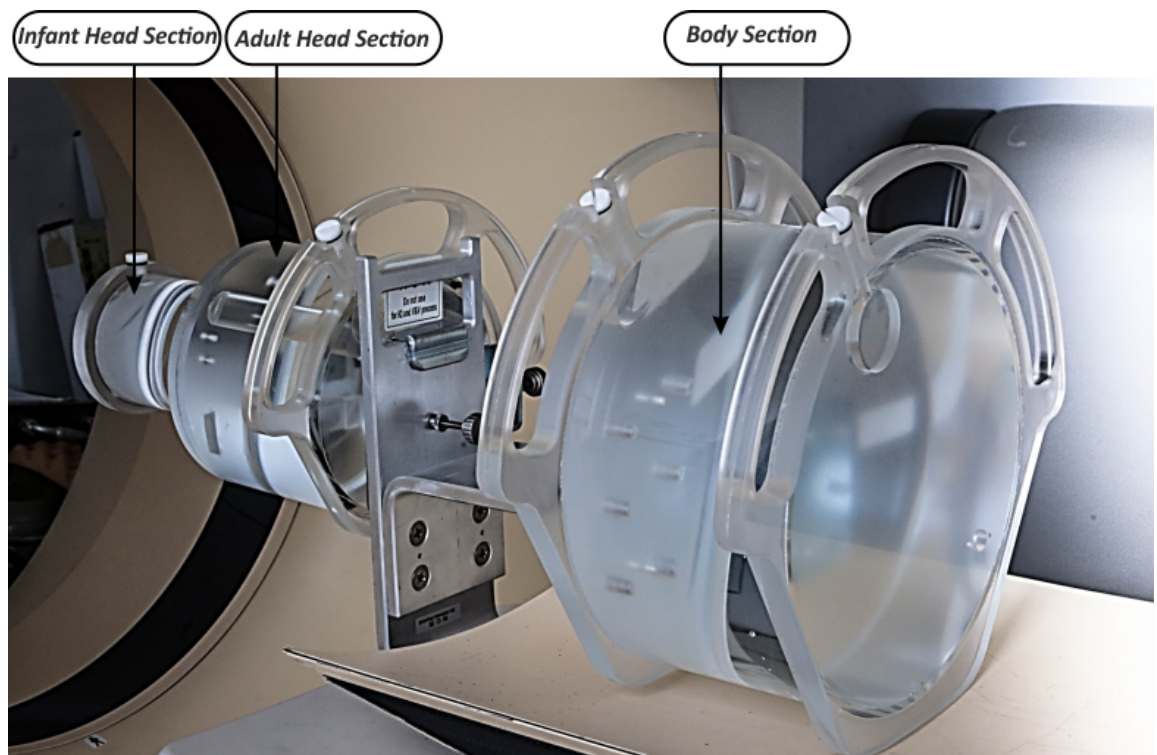


Fig. 5: Harmonized System Phantom Setup for HCOR Slice/Basic and IQ Head Performance Tests

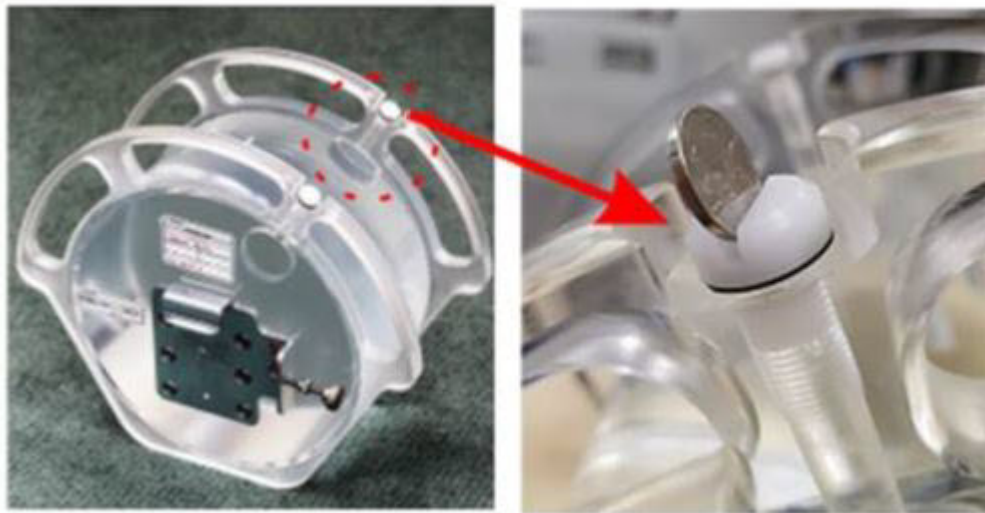
Harmonized System Phantom Holder Setup

1. Perform a visual check of the Harmonized system phantom:
 - No air bubbles larger than 2 mm (refer to the System Installation manual for your system type.)
 - Make sure there are no cracks or water leakage from water filled sections.

Step 1: Check phantom for air bubbles and cracks.

Step 2: If air bubble exist, open the air trap and fill the phantom with phantom water.





2. Install the phantom holder on the patient table.

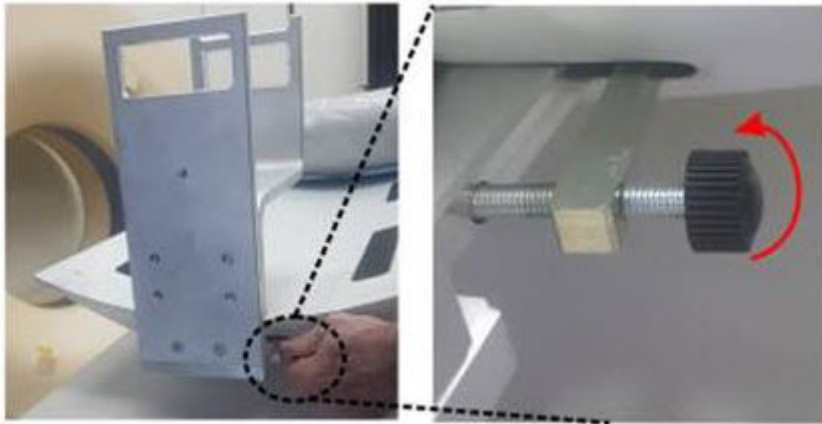
Step 1: Insert the holder into the slot as far as possible.



Step 2: Turn on the lasers and using white alignment lines on the phantom holder.



Step 3: Tighten the locking thumb screw.



Adjusting the Phantom Left/Right Alignment

After starting a test or calibration, if phantom adjustments are needed, instructions appear on the screen with the number of thumb screw turns required to adjust the phantom alignment.

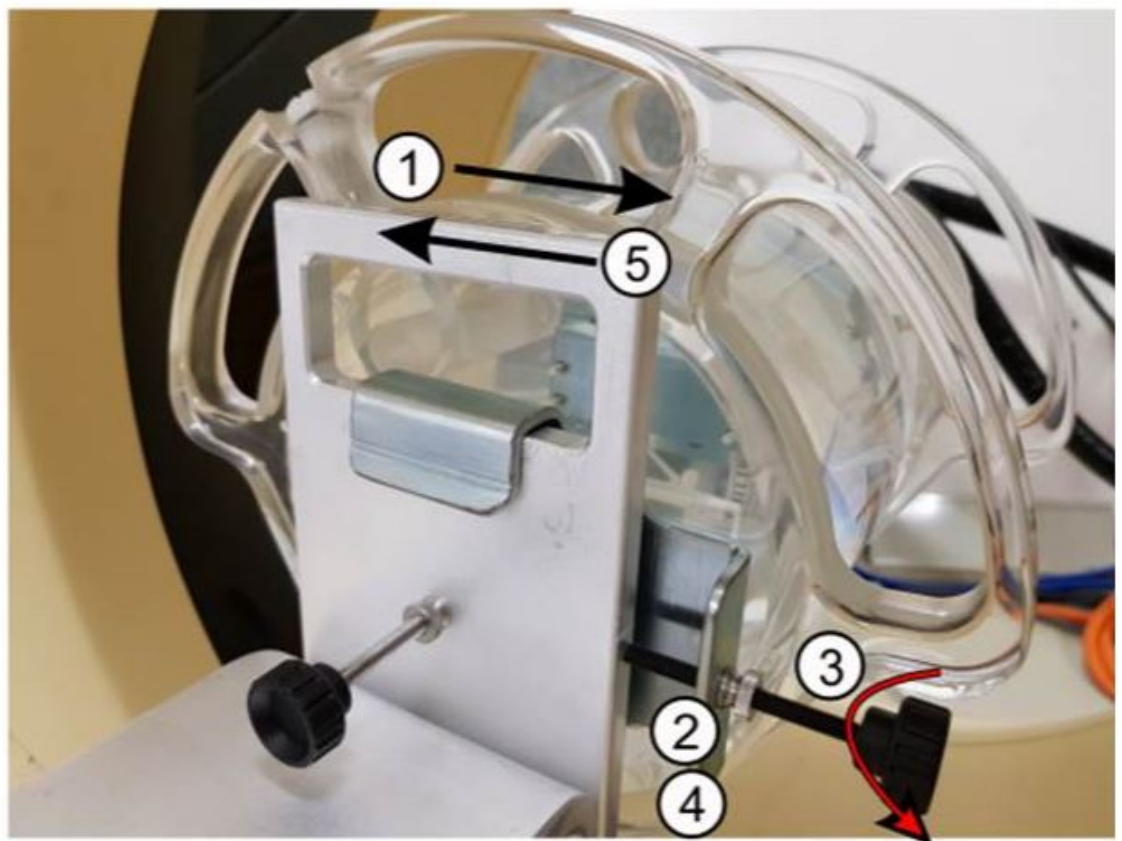


Fig. 6: X-adjustment to Shift Head Section to the Left

1. Shift phantom to the right; more than required.
2. Turn thumb nut to release thumb screw.

3. Turn thumb screw the number of full turns CCW as instructed.
4. Turn the thumb nut to lock the thumb screw.
5. Shift phantom to the left until it is stopped by the screw.

NOTICE

1 turn of thumbscrew = 1 mm

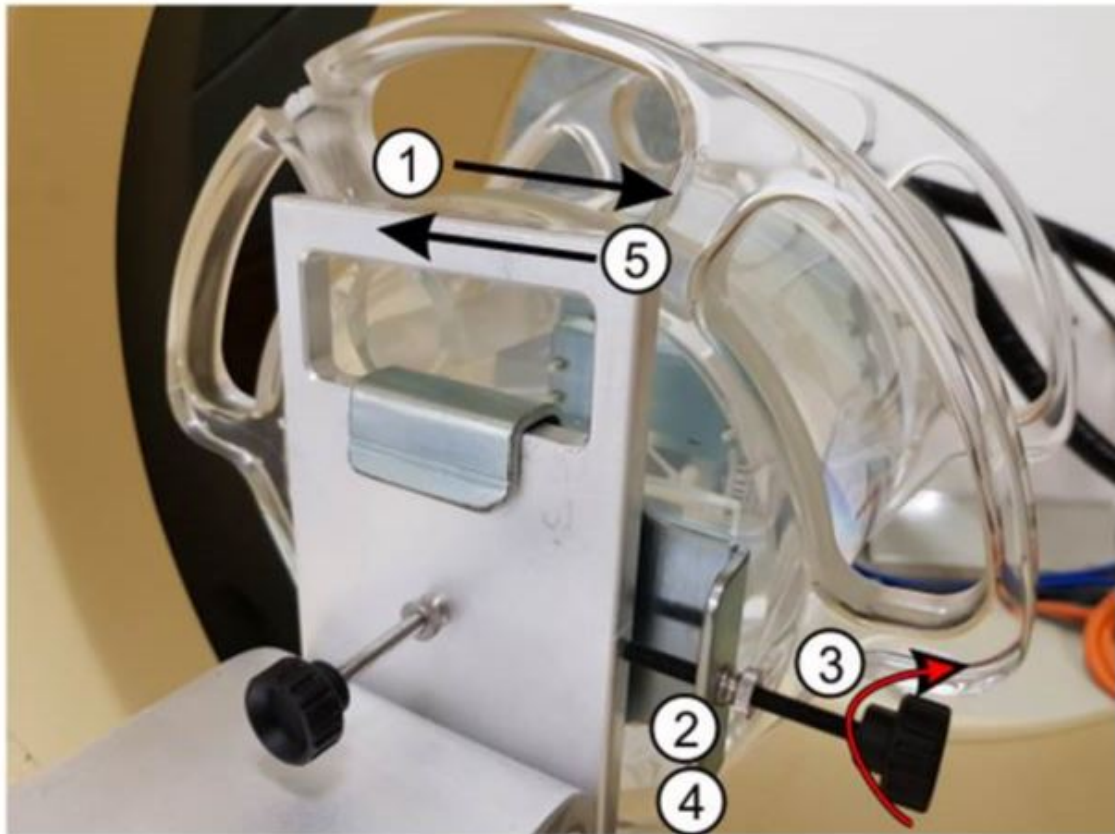


Fig. 7: X-adjustment to Shift Head Section to the Right

1. Shift phantom to the right; more than required.
2. Turn thumb nut to release thumb screw.
3. Turn thumb screw the number of full turns CW as instructed.
4. Turn the thumb nut to lock the thumb screw.
5. Shift phantom to the left until it is stopped by the screw.

NOTICE

1 turn of thumbscrew = 1 mm

Adjusting the Harmonized Phantom Tilt (Z-direction)

1. Release the thumb nut on the adjustment thumb screw.
2. Turn the adjustment screw the number of turns as instructed:
 - Clockwise - to shorten the screw and the phantom will tilt up.
 - Counter-Clockwise - to extend the screw and the phantom will tilt down.
3. Tighten the thumb nut to lock the adjustment screw.
4. Continue with the calibration/test.



When correct tilt is achieved, turn thumb nut CW to lock adjustment screw.

1 turn of thumbscrew = 1 mm

NOTICE

One turn of Tilt Adjustment thumbscrew = **0.8 deg.**

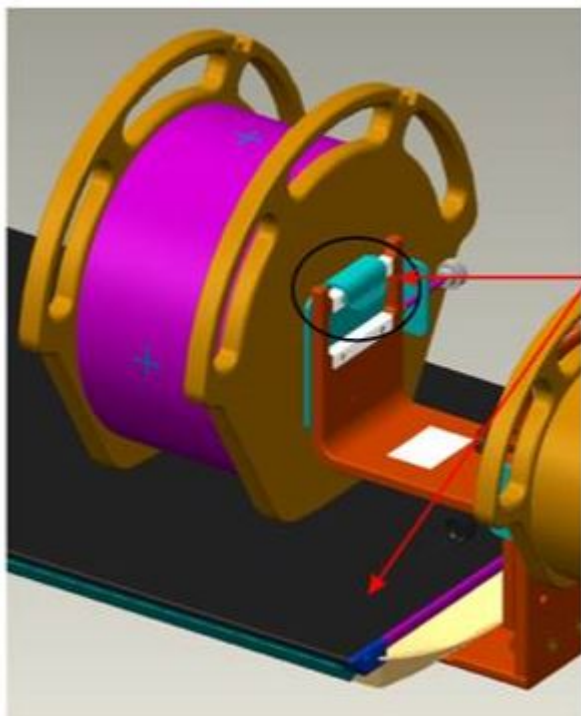
Harmonized Body Section Installation



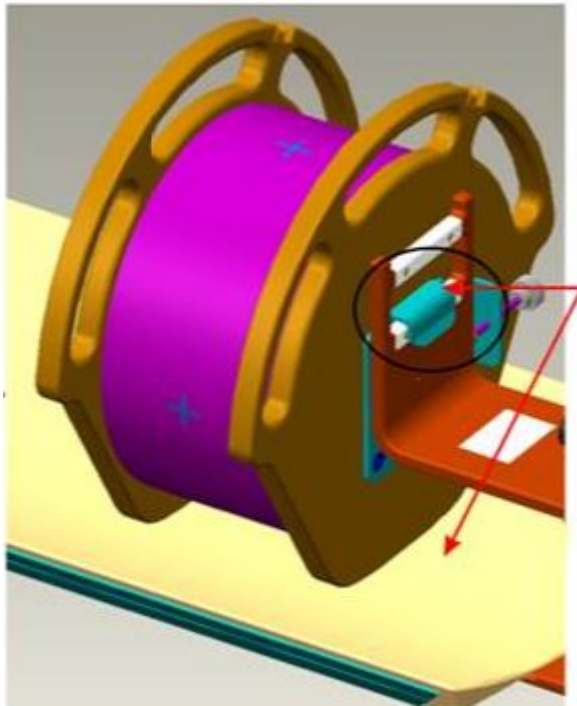
CAUTION

When setting up the harmonized phantom, do not let the adjustment screw hit the phantom holder.

1. Install the Body section on the rear part of the holder in the same manner as the Head phantom:
 - Use the top bar of the rear part of the holder, if the therapeutic top option is installed (see figure below).
 - Use the lower bar of the rear part of the holder, for all other table tops (see figure below).
2. To remove the body section, shift the phantom to the left and lift the body section.



Upper bar with flat top patient table.



Lower bar with curved top patient table.

NOTICE

The Harmonized body section rests on the standard (extended, bariatric) table top and the therapeutic (flat) top.

Installing Harmonized Physics Section

1. Install the Physics section: slide the section's hook bracket over the Head section bracket.
2. Align the Physics section with the Head section using the alignment guides.
3. Make sure both Head and Physics sections are flush and fixed securely.

NOTICE

The brackets on the sections, by design, secure the Head and Physics sections together. The brackets have guides that ensure that the sections are aligned. Additional alignment between sections is not required.

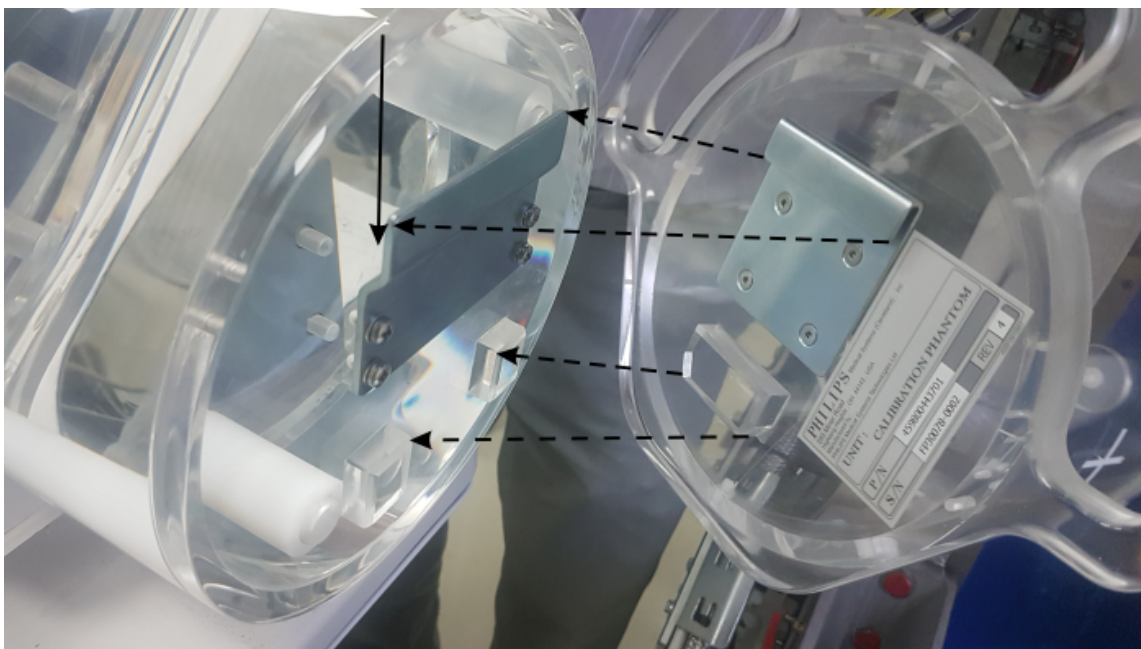


Fig. 8: Physics Section to Harmonized Body Section Assembly Alignment Slot

Installing Harmonized Infant Head Section

1. Remove the Physics section if installed.



CAUTION

When setting up the harmonized phantom, do not let the adjustment screw hit the phantom holder.

2. Install the Adult head section on the Harmonized phantom holder.
3. Slide the Infant Head section hook bracket over the Adult Head section bracket.
4. Make sure both Head and Infant sections are assembled securely together; verify the bracket guides are aligned together.

Assemble Harmonized Head Phantom to Previously Adjusted Position



CAUTION

When setting up the harmonized phantom, do not damage any of the phantom components.

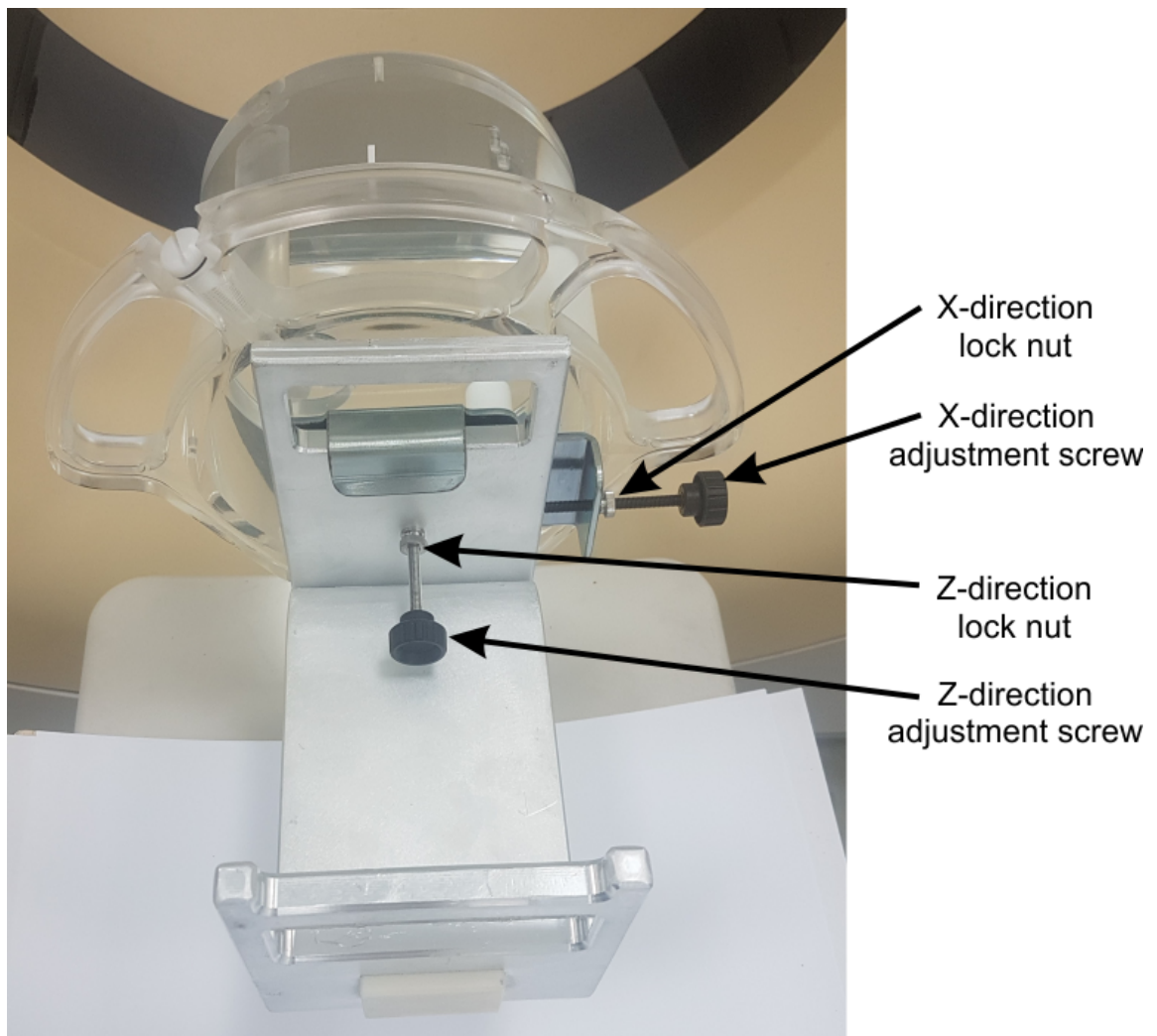
1. Keep the phantom on the right side of the phantom holder and away from the X-adjustment screw.
2. Hook the Head phantom on the phantom holder bar so that the adjustment screw is does not hit the holder.



CAUTION

Do not let the adjustment screw hit the holder. Hook the head phantom on the holder, keeping the phantom off center and to the right.

3. Slide the phantom left until the X-adjustment screw is touching the holder side. This will bring the phantom Z-alignment to the previously adjusted position.
4. To remove the head phantom, shift the phantom to the right and lift the phantom head phantom.



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Performing a Manual Scan

1. Turn On the lasers from the Gantry Control Panel.
2. Adjust the phantom holder Left/Right position on the holder so the sections are exactly aligned.
3. Press the Reset to Zero button on the gantry control panel to reset the PT position to zero.

NOTICE

When using automatic centering, there is no need to press Reset to Zero.

4. Start the test.

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Therapy Top (optional)

If the phantom is installed on the Therapy Top (optional) do the following:

1. Use Slice laser and zero the couch In/Out (Z-position) on the white line on the phantom holder.
2. Use Axial lasers to manually center the Head and Body phantom sections using the Gantry control panel and Head section Tilt using the tilt screw.
3. Click on “Skip centering” checkbox below to skip centering.
4. Click OK to perform scan and check that the phantom is centered Up-Down ± 8 mm and Right-Left within ± 8 mm if not Return to step b.

7 User Information

Technique Factors - Maximum Deviations

Peak X-ray Tube Voltage

The peak X-ray voltage displays on the Operator workstation screen. The actual x-ray voltage during scan is within $\pm 5\%$ of the displayed value.

Peak X-ray voltage

80, 100, 120, or 140 kVp

The peak X-ray voltage is measured on a resistive divider, which is calibrated during the manufacturing process.

Tube Current Exposure Time Product

The actual current exposure time product (in mAs) during a scan is within $\pm 23\%$ of the value displayed on the Operator console. The tube current exposure time product is measured by a dosimeter calibrated in mAs. The dosimeter is calibrated by measuring the tube anode current on an accurate resistor between X-ray on and off during a long exposure.

Linearity of Radiation

The maximum deviation of linearity of radiation is $\pm 10\%$.

Gantry Laser Alignment Lights

The Gantry has two sets of laser alignment lights. One set is on the outside surface of the gantry, and the other on the inside, on the scan plane.

The outside laser lights are useful for positioning the patient with respect to the axis of rotation. The inside laser alignment light is useful for defining the position of the actual X-ray beam.

The alignment light inside the gantry bisects the width of the proposed X-ray beam scan plane during patient set-up. During the scan initialization, the table moves so that the middle of the first image slice coincides with the laser line (within 2.0 mm). Isocenter is within ± 3 mm of the indication from the Sagittal and Coronal lasers.

NOTICE

For more details, refer to the "Table and Gantry Movements" section in "Chapter 3 (Preparing for an Exam)" in the Instructions For Use.

Preventive Maintenance

Routine preventive maintenance for the whole CT system is scheduled every six months and should be performed by qualified Philips personnel.

As part of routine maintenance, the Service Engineer will use a diagnostic program to check these items:

- Cathode voltage
- Emission current
- Exposure time

Cleaning and Disinfection of the System

Cleaning and disinfection are critical to minimize the risks of transmission of infectious agents. Cleaning is the removal of contaminants. It consists of the removal, usually with detergent and water, of adherent soil (e.g. blood, protein substances, and other debris) from the surfaces, crevices, serrations, joints, and lumens of a medical device. Disinfection is the process to reduce the number of viable microorganisms. This product is classified as non-critical device that is intended to contact with intact skin, therefore low to intermediate level disinfection is required. Cleaning and disinfection should follow recommendations for low to intermediate level disinfection as defined by the government agencies, e.g. CDC, using the products approved and registered with your governing authorities, e.g. EPA and VAH.



CAUTION

Wear proper Personal Protective Equipment (PPE), e.g. gloves and glasses, for cleaning and disinfection.



CAUTION

Follow the cleaner/disinfectants manufacturer's instructions for cleaning and disinfection.

The following can be used for cleaning the system including the console, gantry, table, and accessories:

- Distilled water
- Methylated spirit
- Bleach and water solution or disinfectant wipes at ratios of up to 1:10

The following disinfectants can be used for disinfecting the system including the console, gantry, table, and accessories. Only the products approved and registered with the governing authorities, e.g. EPA and VAH, should be used.

- 1:10 bleach equivalent spray cleaner or wipes

- Low- or intermediate-level disinfectant Germicidal Wipes or liquid
- 3% Hydrogen Peroxide
- Ethanol
- Quaternary ammonium compounds
- Benzyl-C12-18-alkyldimethyl

Based on how the parts are exposed to the patients, the extent of the exposures and the frequency of the exposures, the following cleaning/disinfection frequencies are recommended:

- Parts that the patients make direct contact with during normal scans shall be cleaned/disinfected for every patient;
- Parts the patients could touch or the patient body fluid (blood or other potentially infectious materials) could reach shall be cleaned/disinfected daily
- Noncritical environmental parts that the patients do not touch or the patient fluid (blood or other potentially infectious materials) is not expected to reach, shall be cleaned/disinfected weekly or as needed



CAUTION

After each occurrence of spill of contrast medium or patient body fluid (blood or other potentially infectious materials), immediately remove any residual contrast medium and/or patient body fluid, followed by cleaning and disinfection. Contact service engineers if the contrast medium or patient body fluid get inside the equipment.



CAUTION

Blood and contrast medium are health risks. Take appropriate health and safety precautions when removing blood or residual contrast medium.

Tips:

- After cleaning and disinfection, inspect for any damages to the parts, e.g. cracks on the covers, degraded labels, or torn/broken parts etc. Contact Philips Service Representative immediately if any parts are damaged.
- Clean and disinfect the system after installation and before first clinical use.
- When cleaning the front and rear covers on the scanners, cover the microphones to avoid leaking the cleaning solution inside.
- When cleaning the buttons, mouse and the inside of the Gantry opening, take care to avoid leaking the cleaning solution inside.
- When cleaning the monitor screens, use soft cloth, if necessary, moistened with water or LCD cleaner. Do not use any corrosive agents or abrasive agents. Damp cloth can be used but never use wet cloth.

- Activate the clean screen function when cleaning the touch panels (if equipped). Use soft cloth, if necessary, moistened with water or LCD cleaner. Do not use any corrosive agents or abrasive agents. Damp cloth can be used but never use wet cloth.
- The patient restraints can be machine washed or dry-cleaned. Wash the restraints closed to protect the Velcro parts. Remove contaminations with wet cloth. On site cleaning can be performed using specified cleaners/disinfectants, followed by cleaning with water as needed. Make sure the patient restraints are completely dry before using or storing them.
- Apply solution on lint-free wipes if solutions are used. Do not apply solutions directly on the device.
- Rinsing, when needed, should be done with a damp lint-free wipes. Wipes can be damped with distilled water.
- Drying, when needed, should be done with lint-free wipe.

**CAUTION**

Do not use detergents or organic solvents to clean the system. Strong detergents, alcohol, and organic cleaners may damage the finish and also cause structural weakening.

X-ray System Specifications

X-ray Tube

The X-ray tube, mounted on the gantry, has a 8.0 MHU rotating anode with a variable focal spot size of 0.5 x 1.0 and 1.0 x 1.0 (W x L). In accordance with IEC 60336 Ed 4.

Leakage Radiation

Leakage of x-ray tube assembly with beam limiting device: ≤ 0.88 mGy/hr @ 1 meter.

Loading factors 140 kV, 39 mA continuous.

Filtration

Permanent filtration of x-ray tube assembly: 1.1 mm Al / 75 kV (IEC 60522/1999).

NOTICE

The beam limiting device includes an additional 1.2 mm titanium filter.

Cooling Curves

See chapter "X-ray Tube Housing Assembly Information" on page 114.

Power Rating Charts

See chapter “Tube Power Ratings - Big Bore” on page 116.

Maximum Continuous Heat Dissipation

Nominal Continuous Input Power of the X-ray tube assembly 6.1 kW maximum.

X-ray Power Supply**Rated Line Voltage and Regulation**

480 VAC $\pm 10\%$

NOTICE

Line-matching transformer supports additional mains voltages.

X-ray System Maximum Line Current

112 A rms at loading factors 120 kV, 500 mA

160 A rms at loading factors 120 kV, 665 mA

Maximum Power

60 kW (120 kV, 500 mA or 140 kV, 428 mA) for 60 kW tube

80 kW (140 kV, 571 mA or 120 kV, 665 mA) for 80 kW tube

X-ray System Loading Factors

The following system loading factors are applicable to Big Bore CT per IEC 60601-2-44 SEC 201.7.9.2.9.

For 60 kW Tube

- Nominal X-ray tube voltage: 140 kV at 400 mA X-ray tube current.
- X-ray tube voltage corresponding to highest obtainable X-ray tube current:
 - 80 kV at 500mA
 - 100 kV at 500 mA
 - 120 kV at 500 mA
- Highest electric output power: 120 kV X-ray tube voltage and 500 mA X-ray tube current.
- Nominal Electric Power: 60 kW at 120 kV X-ray tube voltage and 500 mA X-ray tube current.
Loading time: 4.1 seconds.

kV	mA
80	600
100	500
120	500
140	400

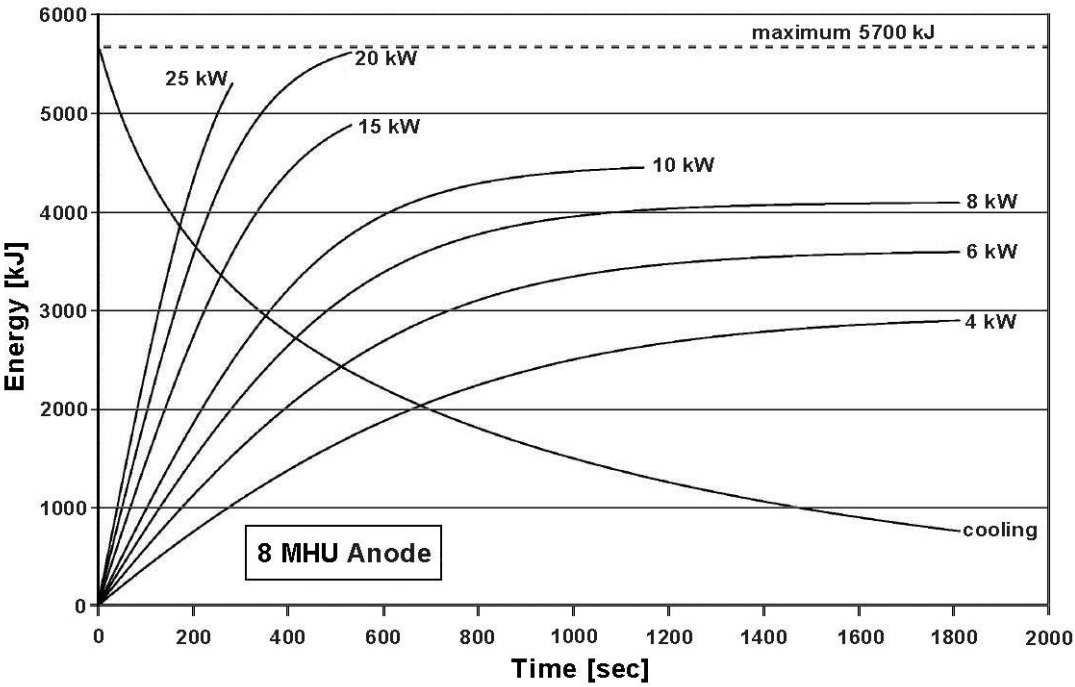
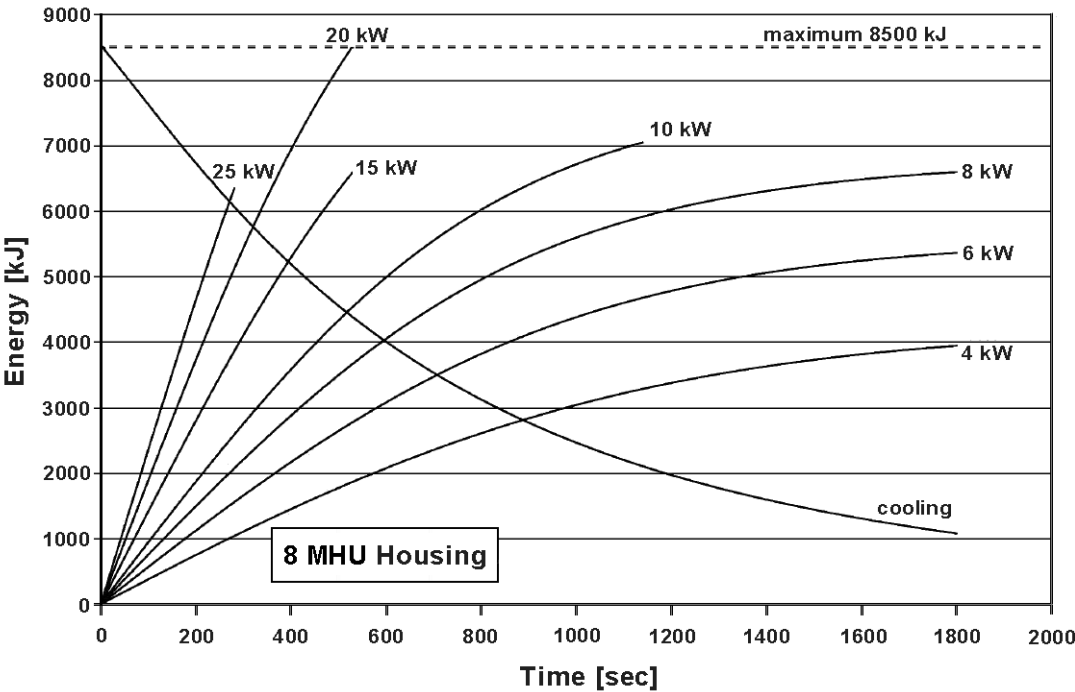
For 80 kW Tube

- Nominal X-ray tube voltage: 140 kV at 571 mA X-ray tube current.
- X-ray tube voltage corresponding to highest obtainable X-ray tube current:
 - 80 kV at 600 mA
 - 100 kV at 665 mA
 - 120 kV at 665 mA
- Highest electric output power: 140 kV X-ray tube voltage and 571 mA X-ray tube current.
- Nominal Electric Power: 79.8 kW at 140 kV X-ray tube voltage and 570 mA X-ray tube current. Loading time: 4.0 seconds.

kV	mA
80	600
100	665
120	665
140	571

X-ray Tube Housing Assembly Information

The following are graphs of the MRC 600, 800 and 880 tube housing assembly heating and cooling curves, and the anode heating and cooling curves.



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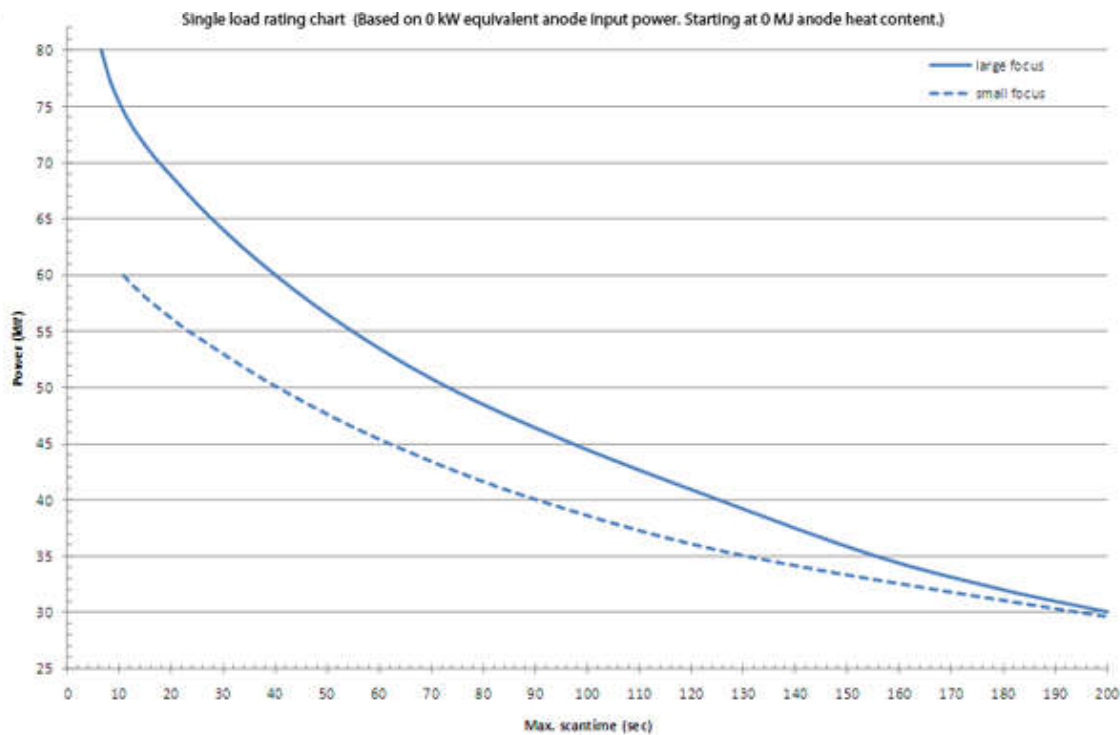
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Tube Power Ratings - Big Bore

NOTICE

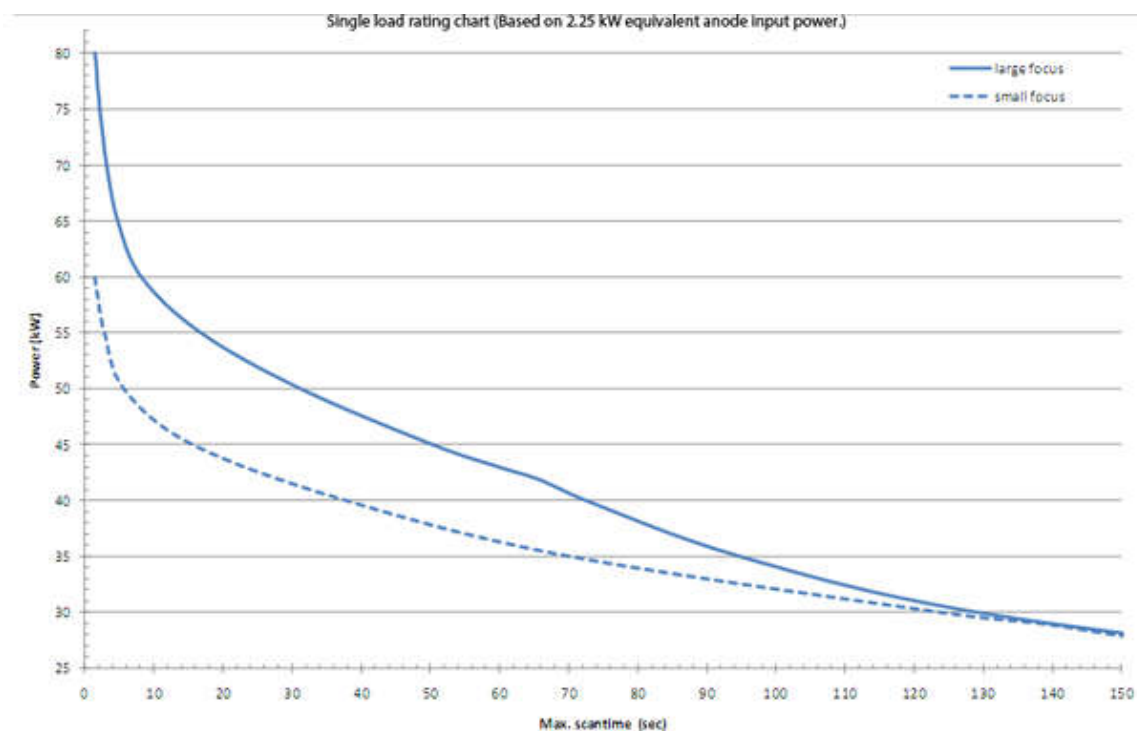
Maximum power should not exceed 60 kW, for 60 kW tube.

Maximum power should not exceed 80 kW, for 80 kW tube.



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User Dose & Imaging Information

The scanner is designed for scanning the head and the body. Therefore, dose and image quality information are provided separately for head and body scans according to the Code of Federal Regulations (21 CFR).

NOTICE

This information is provided as required by the US DHHS, pursuant to 21CFR, Chapter 1, Subchapter J, paragraph 1020.30, 1020.33 and 2013/59/EURATOM.

Phantoms & Measurement Methods

Dose Phantoms

The CT Dosimetry Phantom is the phantom used for determining the dose delivered by a CT X-ray system. The phantoms are right circular cylinders of polymethyl methacrylate, at least 14 cm long. Their density is 1.19 ± 0.01 grams/cc. The phantom for testing CT imaging of the body has a diameter of 32.0 cm, and the phantom for the head has a diameter of 16.0 cm.

The phantom provides means for the placement of dosimeter(s) along its axis of rotation and along a line parallel to the axis of rotation, 1.0 cm from the outer surface and within the phantom.

Dose Profiles & Dose Measurements

The dose profiles were measured using an X-ray sensitive, film-type media. Actual dose values were measured with a 10 cm long, pencil-shaped ionization chamber.

CTDI Definition

Computed Tomography Dose Index (CTDI) is the integral of the dose profile along a line perpendicular to the tomographic plane divided by the product of the nominal tomography section thickness and the number of tomograms produced in a single scan, as follows:

$$CTDI_{100} = \frac{1}{nT} \int_{-50mm}^{+50mm} D(z) dz$$

with these definitions:

- $D(z)$ = Dose to Air ($CTDI_{100}$) at position z
- T = Nominal tomographic section thickness
- n = Number of tomograms produced in a single scan

The CTDI definition assumes that, for a multiple tomogram system, the scan increment between adjacent slices is nT .

The $CTDI_w$ is computed from $CTDI_{100}$ as follows:

$$CTDI_w = [1/3 CTDI_{100 \text{ (center)}} + 2/3 CTDI_{100 \text{ (periphery)}}]$$

OR

$$CTDI_w = 1/3[CTDI_{100 \text{ (center)}} + 2 CTDI_{100 \text{ (periphery)}}]$$

The volume CTDI ($CTDI_{vol}$) describes the average dose over the total volume scanned for the selected CT conditions of operation. The $CTDI_{vol}$ is defined as follows:

- for axial scanning

$$CTDI_{vol} = \frac{N \cdot T}{\Delta d} CTDI_w$$

with these definitions:

- N = the number of tomographic sections produced in a single axial scan of the X-ray source
- T = the nominal tomographic thickness
- Δd = the patient support travel in z-direction between consecutive scans.
- for helical scanning

$$CTDI_{vol} = \frac{CTDI_w}{CT_{pitch \ factor}}$$

- for scanning without movement of the patient support

$$CTDI_{vol} = n \times CTDI_w$$

- n = the maximum number of pre-programmed rotations

The value for $CTDI_{vol}$ is expressed in milligrays (mGy).

Image Noise - System Performance Phantom

There are two sections of the system phantom used for noise measurement:

- The Head phantom is 184 mm in diameter, water-filled (phantom-liquid filled) with a PVC shell (8 mm wall thickness).
 - The CT number of water (phantom liquid) is 0 ± 4 .
 - The CT number of PVC shell is $+1200 \pm 200$.
- The body phantom is 300 mm in diameter and made of Nylon (Aculon).
- The CT number is 110 ± 25 .

Noise is measured using these ROIs:

- $5024 \pm 500 \text{ mm}^2$ area for head section.
- $11300 \pm 1000 \text{ mm}^2$ area for body section.

NOTICE

See the ROI information in the Graphics section of the Instructions for Use.

The SD as displayed on the screen is divided by [average CT number + 1000] and multiplied by 100 to obtain the percent noise.

The maximum deviation from the stated noise is 10%.

Image Noise - System Performance Harmonized Phantom

There are two sections of the system phantom used for noise measurement:

- The Head phantom is 187 mm in diameter, water-filled (phantom-liquid filled) with an acrylic shell (8 mm wall thickness).
 - The CT number of water (phantom liquid) is $0 \pm 4 \text{ HU}$.
- The body phantom is 284 mm in diameter, water-filled (phantom-liquid filled) with an acrylic shell (8 mm) wall thickness.
- The CT number is $0 \pm 6 \text{ HU}$.

Noise is measured using these ROIs:

- $5024 \pm 500 \text{ mm}^2$ area for head section.
- $11300 \pm 1000 \text{ mm}^2$ area for body section.

NOTICE

See the ROI information in the Graphics section of the Instructions for Use.

The SD as displayed on the screen is divided by [average CT number + 1000] and multiplied by 100 to obtain the percent noise.

The maximum deviation from the stated noise is 10%.

Modulation Transfer Function

The impulse response and the tomographic thickness (slice thickness) are not dependent upon the phantom dimensions. They are therefore, measured on the physics layer of the system phantom.

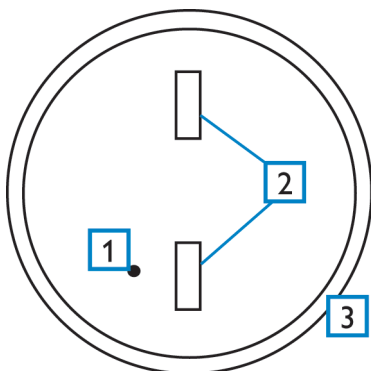
The phantom physics layer diameter is 200 mm (184 mm diameter of water [phantom liquid] in a 8 mm PVC shell).

The impulse response and associated MTF curve is measured on a 0.2 mm copper wire using the **Impulse QA** head or body exam card and the **Resolution Test** Image Tests program (in the Analysis options, accessible from the Directory window).

The maximum deviation of the MTF curve is 15%.

Tomographic Thickness Measurement

The phantom contains two aluminum strips at 45° which give projections of the sensitivity profile in the image plane.



1. 0.2 mm copper wire for impulse response measurement
2. aluminum strips embedded at 45 degrees
3. PVC shell

The profiles of the projections are equivalent to the sensitivity profiles and the FWHM (full width at half maximum) of the profile is the nominal tomographic thickness.

The profile can be measured using the **Slice Thickness** Image Tests program (in the Analysis options, accessible from the Directory window).

The maximum deviation of the derived thickness should be as follows:

Thickness ≥ 2 mm	± 1 mm
Thickness >1 to <2 mm	$\pm 50\%$
Thickness < 1 mm	± 0.5 mm

NOTICE

The 0.75 mm and 1.5 mm results appear wider due to the limited resolution of the image and the thickness of the measuring ramp.

Display CTDI Phantom Size

The 16 cm diameter CTDI phantom is used for head scans. The 32 cm diameter CTDI phantom is used for all body scans.

The phantom size used for reporting CTDI_{vol} or DLP displays in the Main Parameters.

The phantom size is also listed on the Dose information page displayed after the end of the study.

To convert a CTD_{vol} measurement which is displayed for a 32 cm (or Body) phantom to a value measured with a 16 cm (or Head) phantom, multiply the 32 cm value by the constants listed in the table below.

To convert 32 cm CTDI to 16 cm CTDI	80 kVp	100 kVp	120 kVp	140 kVp
Multiply by:	2.073	2.009	1.965	1.928

To convert from 16 cm CTDI to 32 cm CTDI, divide the 16 cm CTDI value by the constant value appropriate for the kVp.

The only Head Exam cards are Head, IAC (Inner Ear) and some Standard QA exam cards. All other Exam Cards are considered to be Body Exams on this scanner.

System Imaging Geometric Accuracy

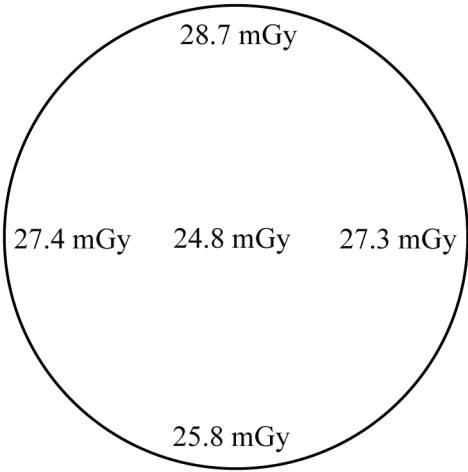
The system imaging geometric accuracy is better than ± 1 mm in the gantry plane across 50 mm and is better than ± 5 mm at 600 mm in both X and Y directions.

Head-scan Information

Head dose - typical head CT scan conditions of operation:

Main	
FOV	250
Storage	Local
Scan	
Collimation	16 x 1.5
Thickness	6
Increment	0
Rot Time	0.75
Voltage	120
mAs	250
Recon	
Filter	UB
SP Filter	No
Adaptive	No

CTDI100 Head



The maximum dose is delivered at the 12 o'clock position.

The 16 cm diameter phantom is placed in the center of the Gantry opening, on the head holder.

Maximum deviation from the values shown below is $\pm 30\%$. $CTDI_{vol} = 26.5 \text{ mGy}$.

The $CTDI_{100}$ is not dependent on the scan diameter.

The maximum deviation from the reported $CTDI_{vol}$ for 2×0.6 is $\pm 35\%$.

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Tube Current - Exposure Time Product (mAs) Dependence

The dose increases linearly with the tube current - exposure time product. The $CTDI_{100}$ in the center location of the head phantom, normalized to the $CTDI_{100}$ in the center location from the values shown in the $CTDI_{100}$ Head section, depends on the mAs for the minimum, maximum, and mid-range values as shown by the ratios below. The maximum deviations of these ratios are also provided.

	$CTDI_{100}$	
Minimum $CTDI_{100}$ (at 15 mAs)	$0.06 \pm 20\%$	...times the $CTDI_{100}$ at 250 mAs
Mid-range $CTDI_{100}$ (at 500 mAs)	$2.00 \pm 20\%$	...times the $CTDI_{100}$ at 250 mAs
Maximum $CTDI_{100}$ (at 1000 mAs)	$4.00 \pm 20\%$	...times the $CTDI_{100}$ at 260 mAs

Collimation Setting Dependence - Head Scan

Several collimation settings are available. The $CTDI_{100}$ in the center of the head phantom depends on the collimation setting, as shown below.

	$CTDI_{100}$	
The $CTDI_{100}$ of the 8 x 3 collimation is ...	$1.00 \pm 15\%$	... times the $CTDI_{100}$ of the 16 x 1.5 collimation
The $CTDI_{100}$ of the 16 x 1.5 collimation is ...	$1.00 \pm 15\%$	... times the $CTDI_{100}$ of the 16 x 1.5 collimation
The $CTDI_{100}$ of the 12 x 1.5 collimation is ...	$1.04 \pm 15\%$	... times the $CTDI_{100}$ of the 16 x 1.5 collimation
The $CTDI_{100}$ of the 16 x 0.75 collimation is ...	$1.12 \pm 15\%$	... times the $CTDI_{100}$ of the 16 x 1.5 collimation
The $CTDI_{100}$ of the 4 x 1.5 collimation is ...	$1.34 \pm 15\%$	... times the $CTDI_{100}$ of the 16 x 1.5 collimation
The $CTDI_{100}$ of the 4 x 0.75 collimation is ...	$1.82 \pm 20\%$	... times the $CTDI_{100}$ of the 16 x 1.5 collimation
The $CTDI_{100}$ of the 2 x 0.6 collimation is ...	$1.71 \pm 20\%$	... times the $CTDI_{100}$ of the 16 x 1.5 collimation

Voltage Dependence - Center

The X-ray voltage can be varied between 80 and 140 kV. The $CTDI_{100}$ in the center of the head phantom depends on the X-ray voltage as shown below.

CTDI ₁₀₀		
With 80 kV, the CTDI ₁₀₀ is...	0.28 ± 15%	... times the CTDI ₁₀₀ at 120 kV
With 100 kV, the CTDI ₁₀₀ is ...	0.59 ± 15%	... times the CTDI ₁₀₀ at 120 kV
With 140 kV, the CTDI ₁₀₀ is...	1.46 ± 15%	... times the CTDI ₁₀₀ at 120 kV

Voltage Dependence - Edge

The CTDI₁₀₀ at the peripheral or edge dose position in the head phantom depends on the X-ray voltage as shown below:

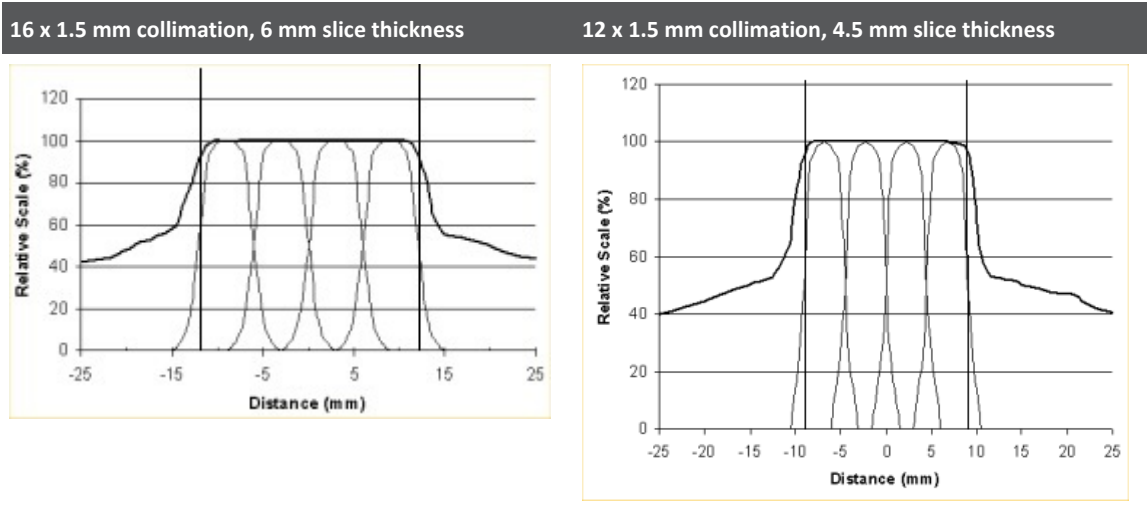
CTDI ₁₀₀		
With 80 kV, the CTDI ₁₀₀ is...	0.29 ± 15%	... times the CTDI ₁₀₀ at 120 kV
With 100 kV, the CTDI ₁₀₀ is ...	0.60 ± 15%	... times the CTDI ₁₀₀ at 120 kV
With 140 kV, the CTDI ₁₀₀ is...	1.44 ± 15%	... times the CTDI ₁₀₀ at 120 kV

Dose and Sensitivity Profiles - Head Scan

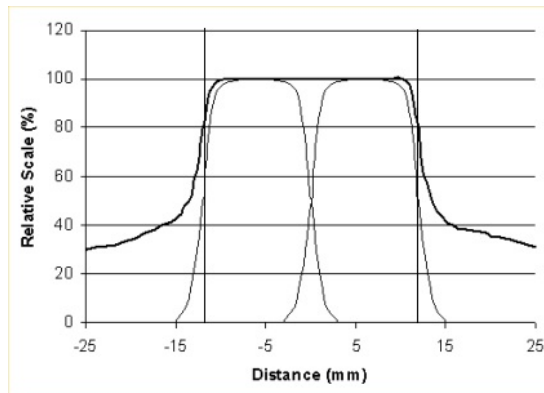
The dose profiles at the center of the phantom superimposed on the sensitivity profiles are presented in the following sections. The maximum deviations from the drawn curves are ±20%.

NOTICE

The sensitivity profiles were simulated under the typical conditions of operation presented in the Head Scan Information section, while changing the slice thickness only. The limited resolution of the image causes the thin slice thicknesses to appear thicker than they really are.



8 x 3 mm collimation, 12 mm slice thickness



2 x 0.6 mm collimation, 0.6 mm slice thickness

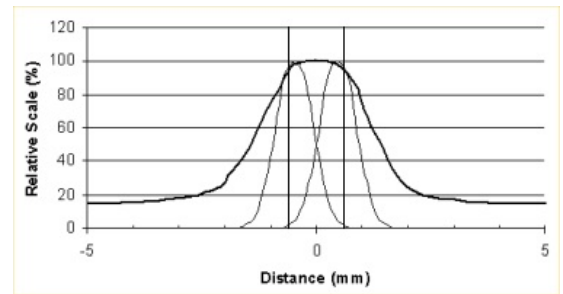
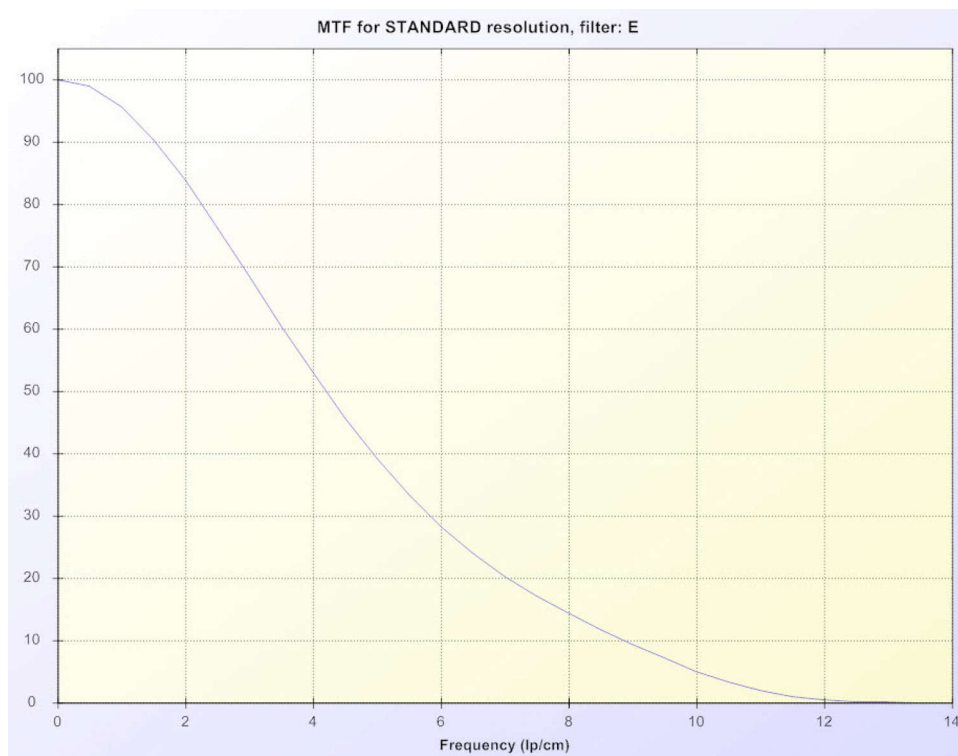


Image Quality - Head

The mean noise on a 200 mm diameter system phantom is $0.27\% \pm 0.04$ (16 x 1.5 mm scan mode, 120 kV, 250 mAs, 12 mm thickness, 0.75 sec rotation time, STD resolution, filter = UA).

The phantom and measurement method are described earlier.



The maximum deviation of the MTF curve is $\pm 15\%$.

These are the nominal tomographic section thicknesses:

- $0.6 \text{ mm} \pm 0.5 \text{ mm}$
- $0.75 \text{ mm} \pm 0.5 \text{ mm}$

- 1.5 mm ± 0.75 mm
- 3.0 mm ± 1.0 mm
- 4.5 mm ± 1.0 mm
- 6 mm ± 1.0 mm
- 12 mm ± 1.0 mm

Selected sensitivity profiles are drawn in Dose and Sensitivity Profiles - Head Scan.

CT Number Uniformity

The system CT Number uniformity is 0 ± 8 Hounsfield Units for Head scans on a system phantom.

CT Number Accuracy

The CT number accuracy for water as measured on the system phantom is typically in the following range:

Phantom Type	Phantom Area	Hounsfield Units Accuracy			
		80kVp	100kVp	120kVp	140kVp
Adult	Head	±6 HU	±4 HU	±4 HU	±6 HU
Infant	Head	±4 HU	±4 HU	±4 HU	--

Body-scan Information

Body dose - typical body CT scan conditions of operation:

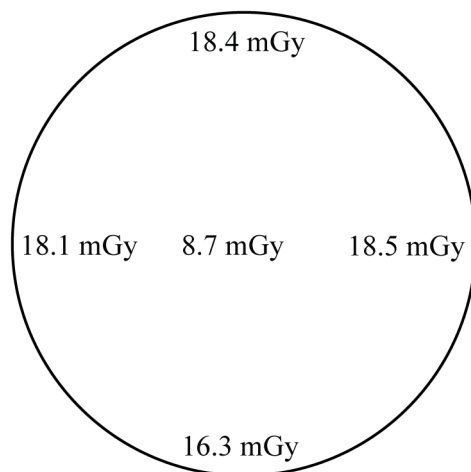
Main	
FOV	350
Storage	Local
Scan	
Collimation	16 x 1.5
Thickness	6
Increment	0
Rot Time	0.75
Voltage	120
mAs	250

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Recon	
Filter	B
SP Filter	No
Adaptive	No

CTDI100 Body



The maximum dose is delivered at the 12 o'clock position.

The 32 cm diameter phantom is placed in the center of the Gantry opening on the table with one of the dosimeters at the maximum dose position.

Maximum deviation from the values shown below is $\pm 30\%$. $CTDI_{VOL} = 14.8$ mGy.

The $CTDI_{100}$ is not dependent on the scan diameter.

The maximum deviation from the reported $CTDI_{VOL}$ for 2×0.6 is $\pm 35\%$.

Tube Current - Exposure Time Product (mAs) Dependence

The dose increases linearly with the tube current - exposure time product. The $CTDI_{100}$ in the center of the body phantom depends on the mAs, as shown below.

	$CTDI_{100}$	
Minimum $CTDI_{100}$ (at 15 mAs)	$0.06 \pm 20\%$	...times the $CTDI_{100}$ at 250 mAs
Mid-range $CTDI_{100}$ (at 500 mAs)	$2.0 \pm 20\%$	...times the $CTDI_{100}$ at 250 mAs
Maximum $CTDI_{100}$ (at 1000 mAs)	$4.0 \pm 20\%$	...times the $CTDI_{100}$ at 250 mAs

Collimation Setting Dependence - Body Scan

Several collimation settings are available. The $CTDI_{100}$ in the center of the body phantom depends on the collimation setting, as shown below.

	$CTDI_{100}$	
The $CTDI_{100}$ of the 8×3 collimation is ...	$1.00 \pm 15\%$	... times the $CTDI_{100}$ of the 16×1.5 collimation
The $CTDI_{100}$ of the 16×1.5 collimation is ...	$1.00 \pm 15\%$	... times the $CTDI_{100}$ of the 16×1.5 collimation

CTDI ₁₀₀		
The CTDI ₁₀₀ of the 12 x 1.5 collimation is ...	1.04 ± 15%	... times the CTDI ₁₀₀ of the 16 x 1.5 collimation
The CTDI ₁₀₀ of the 16 x 0.75 collimation is ...	1.12 ± 15%	... times the CTDI ₁₀₀ of the 16 x 1.5 collimation
The CTDI ₁₀₀ of the 12 x 0.75 collimation is ...	1.19 ± 15%	... times the CTDI ₁₀₀ of the 16 x 1.5 collimation
The CTDI ₁₀₀ of the 4 x 1.5 collimation is ...	1.34 ± 15%	... times the CTDI ₁₀₀ of the 16 x 1.5 collimation
The CTDI ₁₀₀ of the 4 x 0.75 collimation is ...	1.82 ± 20%	... times the CTDI ₁₀₀ of the 16 x 1.5 collimation
The CTDI ₁₀₀ of the 2 x 0.6 collimation is ...	1.71 ± 20%	... times the CTDI ₁₀₀ of the 16 x 1.5 collimation

Voltage Dependence - Center

The X-ray voltage may be varied between 80 and 140 kV. The CTDI₁₀₀ in the center of the body phantom depends on the X-ray voltage as shown below.

CTDI ₁₀₀		
With 140 kV, the CTDI ₁₀₀ is...	1.53 ± 15%	... times the CTDI ₁₀₀ at 120 kV
With 100 kV, the CTDI ₁₀₀ is...	0.56 ± 15%	... times the CTDI ₁₀₀ at 120 kV
With 80 kV, the CTDI ₁₀₀ is...	0.24 ± 15%	... times the CTDI ₁₀₀ at 120 kV

Voltage Dependence - Edge

The CTDI₁₀₀ at the peripheral or edge dose position in the body phantom depends on the X-ray voltage as shown below.

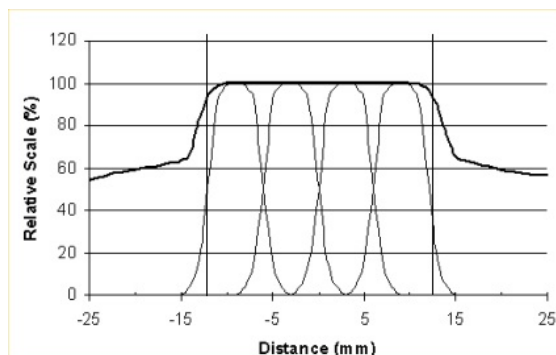
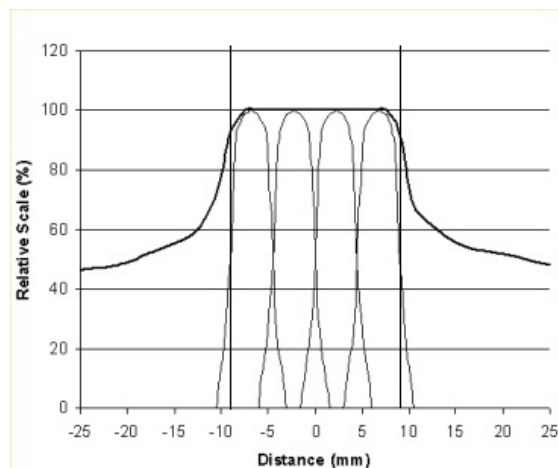
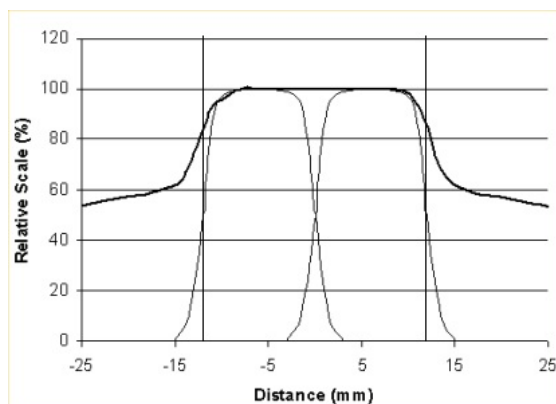
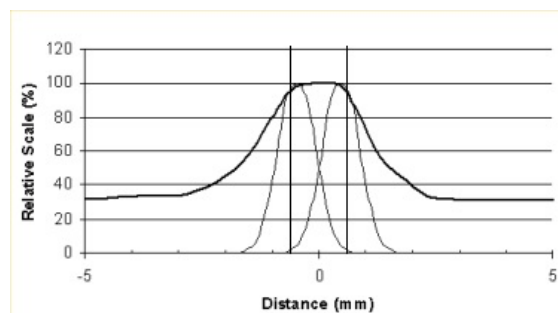
CTDI ₁₀₀		
With 140 kV, the CTDI ₁₀₀ is...	1.46 ± 15%	... of the CTDI ₁₀₀ at 120 kV
With 100 kV, the CTDI ₁₀₀ is...	0.59 ± 15%	... of the CTDI ₁₀₀ at 120 kV
With 80 kV, the CTDI ₁₀₀ is...	0.28 ± 15%	... of the CTDI ₁₀₀ at 120 kV

Dose and Sensitivity Profiles - Body Scan

The dose profiles at the center of the phantom superimposed on the sensitivity profiles are presented below. The maximum deviations from the drawn curves are ±20%.

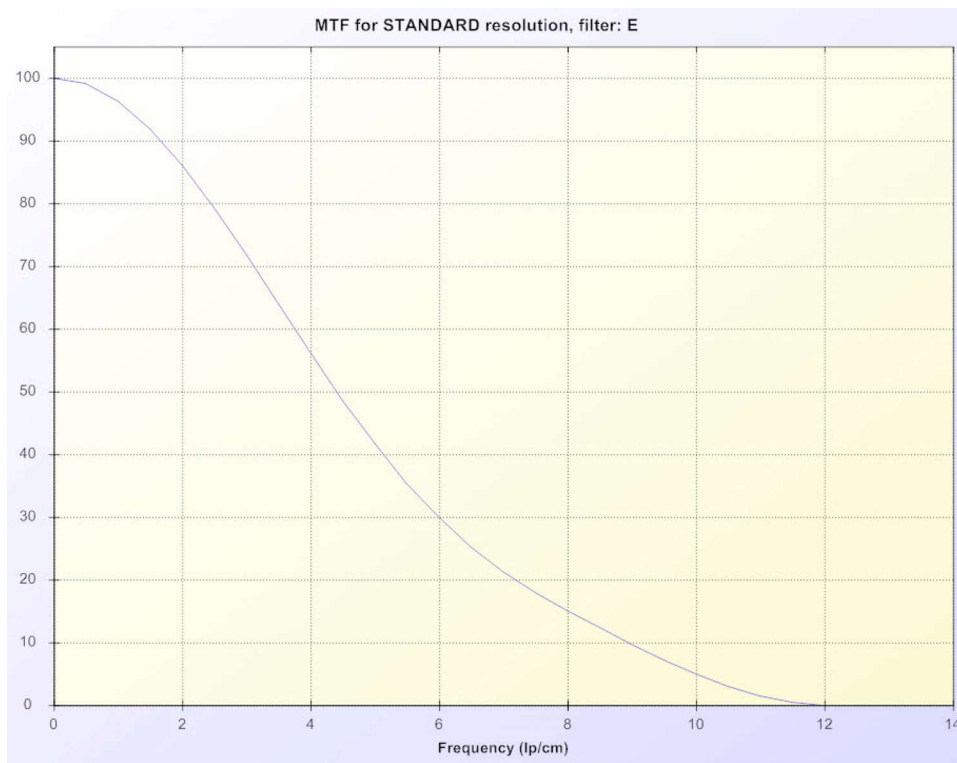
NOTICE

The sensitivity profiles were simulated under the typical conditions of operation presented in the Body Scan Information section, while changing the slice thickness only. The limited resolution of the image causes the thin slice thicknesses to appear thicker than they really are.

16 x 1.5 mm collimation, 6 mm slice thickness**12 x 1.5 mm collimation, 4.5 mm slice thickness****8 x 3 mm collimation, 12 mm slice thickness****2 x 0.6 mm collimation, 0.6 mm slice thickness****Image Quality - Body**

The mean noise on a 300 mm diameter system phantom is $1.00 \pm 0.14\%$ (16 x 0.75 mm collimation, 12 mm slice thickness, 120 kV, 315 mAs, filter = B).

The phantom and measurement method are described earlier.



The maximum deviation of the MTF curve is $\pm 15\%$.

These are the nominal tomographic section thicknesses:

- 0.6 mm $\pm$ 0.5 mm
- 0.75 mm $\pm$ 0.5 mm
- 1.5 mm $\pm$ 0.75 mm
- 3.0 mm $\pm$ 1.0 mm
- 4.5 mm $\pm$ 1.0 mm
- 6 mm $\pm$ 1.0 mm
- 12 mm $\pm$ 1.0 mm

Selected sensitivity profiles are drawn in Dose and Sensitivity Profiles - Body scan.

CT Number Uniformity

The system CT Number uniformity is 0 ± 8 Hounsfield Units for Body scans on a 30-cm water phantom.

CT Number Accuracy

The CT number accuracy for water as measured on the system phantom is typically in the following range:

Phantom Type	Phantom Area	Hounsfield Units Accuracy			
		80kVp	100kVp	120kVp	140kVp
Adult	Body	-4 to 10	-1.5 to 6.5	-0.5 to 7.5	0 to 8
Infant	Body	±4 HU	±4 HU	±4 HU	--

CTDI Free Air

The CTDI Free Air can be found in the following tables. The conditions of operation for Body scans, unless specified in the table, are QA Axial Body 2D, Standard Resolution, 16 x 1.5, 0.75 seconds, 120 kVp, 250 mAs.

CTDI Free Air for Body Conditions of Operation

kVp\NxT	4 x 0.75	4 x 1.5	12 x 0.75	16 x 0.75	12 x 1.5	16 x 1.5	8 x 3
80	--	--	--	--	--	10.5 mGy	--
100	--	--	--	--	--	21.4 mGy	--
120	64.5 mGy	47.5 mGy	42.2 mGy	39.8 mGy	36.9 mGy	35.5 mGy	35.5 mGy
140	--	--	--	--	--	51.7 mGy	--

CTDI Free Air for Other Conditions of Operation

Scan mode	CTDI free air
Adult Head, 16 x 1.5, 120 kVp, 450 mAs	62.2 mGy

The maximum deviation from the values shown is $\pm 25\%$.

If either the Head or Body scans at 16 x 1.5 collimation and 120 kVp is measured repeatedly, each value should be within $\pm 10\%$ of the mean of a set of 10 measurements.

Conditions to Achieve 1000 mGy CTDIvol (Peripheral)

It is impossible to achieve 1000 mGy in a single axial scan on the Big Bore. Helical scans cannot achieve 1000 mGy to the same position on the patient, as the table is continuously moving during the scan. It is possible to see 1000 mGy for repeated axial scans in the same location, such as interventional scans or perfusion scans, so those types of scans will be the focus here. Typically, these types of scans are done far below the maximum mAs settings at each kVp.

Adult Head

The maximum peripheral CTDI is seen at the 12:00 o'clock peripheral position. Using the maximum limits on tube mAs for each type of axial scan, the maximum peripheral CTDI which can be achieved for Head Mode can be summarized as follows. If the mAs is lower than the maximum listed here, the number of rotations to exceed 1000 mGy will be increased proportionally.

Head/Brain Perfusion Non-Jog

kVp	Collimation	Rotation Time	mAs	CTDI (12:00 o'clock position)	Number of Axial scans to exceed 1000 mGy
120	16 x 1.5	0.75 sec	375	43.1 mGy	24
100	16 x 1.5	0.75 sec	375	25.8 mGy	39
80	16 x 1.5	0.75 sec	375	12.5 mGy	81

Brain Axial 2D

This scan is typically done with 12 mm scan increment, but can be manually set to 0 mm scan increment, which will scan the same location repeatedly. This scan can be performed at 420 degree scan, which can increase the mAs even further. If so, then the following maximum 12:00-o'clock-position CTDI values can be achieved.

kVp	Collimation	Rotation Time	mAs	CTDI (12:00 o'clock position)	Number of Axial scans to exceed 1000 mGy
120	16 x 0.75	1.5 sec	875	112.5 mGy	9
140	16 x 0.75	1.5 sec	700	129.6 mGy	8
100	16 x 0.75	1.5 sec	875	67.5 mGy	15
80	16 x 0.75	1.5 sec	875	32.6 mGy	31

Head Axial

Similar to the Brain Axial, this scan is typically done with 12 mm scan increment, but can be manually set to 0 mm scan increment, which will scan the same location repeatedly. This scan can be performed at 420 degree scan, which can increase the mAs even further. If so, then the following maximum 12:00-o'clock-position CTDI values can be achieved. This scan can be done with 1.5 second rotation time, which increases the CTDI per scan.

kVp	Collimation	Rotation Time	mAs	CTDI (12:00 o'clock position)	Number of Axial scans to exceed 1000 mGy
120	16 x 0.75	1.5 sec	875	112.5 mGy	9
140	16 x 0.75	1.5 sec	700	129.6 mGy	8
100	16 x 0.75	1.5 sec	875	67.5 mGy	15
80	16 x 0.75	1.5 sec	875	32.6 mGy	31

Head/Sinus Limited HR

This scan is typically done with 10-15 mm scan increment, but can be manually set to 0 mm scan increment, which will scan the same location repeatedly. This scan can also be performed at 420 degree scan, which can increase the mAs even further. If so, then the following maximum 12:00-o'clock-position CTDI values can be achieved.

kVp	Collimation	Rotation Time	mAs	CTDI (12:00 o'clock position)	Number of Axial scans to exceed 1000 mGy
120	2 x 0.6	0.75 sec	435	85.4 mGy	12
140	2 x 0.6	0.75 sec	350	98.9 mGy	11
100	2 x 0.6	0.75 sec	435	51.2 mGy	20
80	2 x 0.6	0.75 sec	435	24.8 mGy	41

Adult Body Mode

The maximum peripheral CTDI is seen at the 12:00 o'clock peripheral position. Using the maximum limits on tube mAs for each type of axial scan, the maximum peripheral CTDI which can be achieved for Adult Body mode can be summarized as follows. If the mAs is lower than the maximum listed here, the number of rotations to exceed 1000 mGy will be increased proportionally.

Abdomen/Body Perfusion Axial

kVp	Collimation	Rotation Time	mAs	CTDI (12:00 o'clock position)	Number of Axial scans to exceed 1000 mGy
120	16 x 1.5	0.75 sec	375	27.6 mGy	37
100	16 x 1.5	0.75 sec	375	16.3 mGy	62
80	16 x 1.5	0.75 sec	375	7.7 mGy	130

Thorax/High Resolution Chest Axial

This scan is typically done with 10-15 mm scan increment, but can be manually set to 0 mm scan increment, which will scan the same location repeatedly. If so, then the following maximum 12:00-o'clock-position CTDI values can be achieved.

kVp	Collimation	Rotation Time	mAs	CTDI (12:00 o'clock position)	Number of Axial scans to exceed 1000 mGy
120	2 x 0.6	0.75 sec	375	47.2 mGy	22
140	2 x 0.6	0.75 sec	300	55.1 mGy	19
100	2 x 0.6	0.75 sec	375	27.8 mGy	36
80	2 x 0.6	0.75 sec	375	13.2 mGy	76

CCT Mode - Interventional/CCT Single or CCT Continuous

It is possible to achieve high CTDI with CCT modes. Utilizing similar limits as above, and choosing the collimation which gives this highest peripheral CTDI, results in the following table:

kVp	Collimation	Rotation Time	mAs	CTDI (12:00 o'clock position)	Number of Axial scans to exceed 1000 mGy
120	16 x 0.75	0.75 sec	270	22.3 mGy	45
140	16 x 0.75	0.75 sec	100	12.1 mGy	83
100	16 x 0.75	0.75 sec	375	18.2 mGy	55
80	16 x 0.75	0.75 sec	375	8.7 mGy	116

Interventional/CCT Fluoro

CCT Fluoro is a continuous scan mode, and displays CTDI rate in mGy per second (mGy/s). However, faster rotation times at the same mAs can yield higher CTDI rates. Utilizing similar limits as above, and choosing the collimation which gives this highest peripheral CTDI, results in the table below. Note that the default CCT Fluoro scan is limited to 30 seconds, but it is possible to extend or repeat the scan.

kVp	Collimation	Rotation Time	mAs	CTDI rate (12:00 o'clock position)	Number of scan seconds to exceed 1000 mGy
120	4 x 1.5	0.75 sec	185	48.7 mGy	21
140	4 x 1.5	0.75 sec	100	38.4 mGy	27
100	4 x 1.5	0.75 sec	185	28.7 mGy	35
80	4 x 1.5	0.75 sec	185	13.6 mGy	74

Size Specific Dose Estimate (SSDE)

The $CTDI_{vol}$ provided by the scanner is a measure of the absorbed dose, expressed in units of mGy, to either a 32 cm or 16 cm diameter acrylic phantom over the volume scanned with a specific Exam Card. The $CTDI_{vol}$ for a selected Exam Card, therefore, does not represent the absorbed dose to a patient. For infants, the $CTDI_{vol}$ underestimates the absorbed dose to the scan volume by up to a factor of 3. Conversely, the $CTDI_{vol}$ for large patients overestimates absorbed dose to the scan volume; for very large patients $CTDI_{vol}$ can overestimate absorbed dose by as much as 40%.

Through a series of experiments and models, the American Association of Physicists in Medicine devised conversion factors from $CTDI_{vol}$ to a new dose metric, Size Specific Dose Estimate (SSDE) also expressed in units of mGy. The appropriate SSDE conversion factor for a given patient depends on the attenuation of the patient and the top of the couch in the scanned area. The SSDE is the product of this patient attenuation-specific conversion factor and the $CTDI_{vol}$ for the selected Exam Card. For infant head and body scans, the conversion factors are typically larger than 1 because infant heads and bodies are smaller than the 16 cm and 32 cm phantom, respectively, used to calculate $CTDI_{vol}$ for the Exam Card; for these patients, SSDE values are higher than $CTDI_{vol}$ values. For adult body scans the conversion factor is typically smaller than 1, since patients are usually larger than the 32 cm phantom used to calculate $CTDI_{vol}$; for these patients, SSDE values are lower than $CTDI_{vol}$ values. For adult head scans, the conversion factors are usually closer to 1 such that SSDE and $CTDI_{vol}$ values are similar.

SSDE provides a better estimate of the average absorbed dose to the patient by taking into account both the radiation output of the CT scanner and the patient's size. Although SSDE is intended to describe dose for patients of all sizes, better estimates of dose are especially important for small pediatric patients since the actual absorbed dose to the patient is higher than indicated by the $CTDI_{vol}$ for a given Exam Card and because radiation exposure is of greatest concern in infants and children.

The IEC is formalizing the calculation of SSDE into a new standard so that all vendors can define and provide this new metric in the same way. When a Surview is performed and DoseRight is enabled, the scanner calculates a Water Equivalent Diameter (WED) for the patient, that is, the diameter of a water equivalent cylinder with the same X-ray attenuation as the patient and the couch top, over the entire Surview. This value is displayed as Patient Size with units of cm on the Surview image. When a shorter scan range is selected for the clinical scan, the Patient Size is still displayed but the scanner also calculates an Average Scan Size in cm, representing the average WED within the scan region only. Average Scan Size is the patient metric used to determine the appropriate conversion factor for SSDE calculation. If DoseRight is not enabled, only Average Scan Size is displayed.

Before each clinical scan, the scanner displays an estimated Average Scan Size and SSDE based on the planned scan region and the planned x-ray output. After a clinical scan, the scanner recalculates Average Scan Size and SSDE based on the actual scan region and the actual x-ray

output (estimated and actual values are usually the same). Updated values for Average Scan Size and SSDE are included in the preview display. Final values for all scans are also tabulated in the dose report, compiled at the completion of the exam.

General limitations of the Size Specific Dose Estimate (SSDE) methodology

It is important to recognize that SSDE is still an estimate of the absorbed dose to the scan volume even though it takes into account patient attenuation in the scanned region. The accuracy of this estimate, compared to the actual absorbed dose to the scan volume, is approximately $\pm 20\%$.

Limitations of SSDE in special clinical scenarios

Neck included in scanned anatomy

Additional uncertainty in SSDE of approximately 10% is expected for scans of the head and neck when the scan length in the head and neck region are approximately equal. This is because conversion factors for the head are applied to the entire scan length even though they are not as appropriate for the neck.

Single or bilateral extremities are scanned

In the case of bilateral lower extremity scans or bilateral upper extremity scans where the arms are above the head, patient size estimates from the surview image can be less accurate. This can have a minor impact on the SSDE but any additional uncertainty in the estimate is not expected to exceed 5%.

Patient is not positioned at the center of rotation along the source/detector direction

When patients are not properly centered, patient size estimates from the surview image can be less accurate. Any additional uncertainty in the estimate of SSDE is not expected to exceed 5%.

Patient anatomy outside the scan field of view

Patient anatomy outside the scan FOV will result in an underestimation of patient size from the surview and an overestimation of SSDE. However, at large patient sizes, the conversion factors vary slowly with changes in patient size. Except for morbidly obese patients, any additional uncertainty in the estimate of SSDE is not expected to exceed 5%.

Foreign Objects in the Scan Field

When foreign objects (e.g., metal implants, radiation therapy planning hardware, life support devices, bismuth shields) are in the scan FOV, patient size estimates from the surview image can be less accurate. The magnitude of uncertainty in estimation of patient size will depend on the physical size of the foreign object and the attenuation of that material relative to bone and soft tissue. This may result in overestimation of patient size and an underestimation of SSDE causing additional uncertainty in the SSDE that may exceed 5%.

Essential Performance for Interventional Imaging

Image Spatial Accuracy

Spatial Accuracy – XY	+/- 1 mm over a distance of 50 mm in plane
Spatial Accuracy - Z	+/- 1 mm over 300 mm of bed travel

Slice Thickness

The system achieves the following slice thicknesses while scanning with Head Scan Type using 120 kVp, at least 200 mAs, field of view of 250 mm, and 768 x 768 matrix size:

- For a requested slice thickness of 0.75 mm the measured slice thickness is between 0.5 mm and 1.5 mm using High resolution.
- For a requested slice thickness of 1.5 mm the measured slice thickness is between 1.0 mm and 2.0 mm using Standard resolution.

Low Contrast

The system provides a scanning mode that achieves Low Contrast Resolution so that a 4-mm diameter pin, that is 3-HU different than its background, is distinguishable on the image.

Noise

The system achieves a percent noise of no more than 0.45%, when irradiating with no more than 50 mGy CTDI_{vol}.

High Contrast Spatial Resolution

When scanning in Standard Resolution, the system achieves at least 10 lp/cm (measured at 0% MTF) in plane resolution (x-y) within a radius of 100 mm from the iso-center and in the central two slices.

Dose Management

The system computes the accumulated CTDI_{vol} and DLP values for all planned Acquisitions at each anatomic position throughout the exam. If the cumulative CTDI_{vol} or DLP at any anatomic position is expected to exceed the alert value when the next scan is performed, the Dose Alert pop-up message will be displayed. Dose Alerts are associated with complete studies, not individual Acquisitions, and are enabled in Preferences >Dose Management. Use the factory default values or enter values instituted for your site.

Default CTDI_{vol} values have been set at 1000mGy, which is consistent with values suggested by government regulators such as the FDA.

No DLP values are included in the factory settings.

An alert warrants more stringent review before proceeding than a notification and requires a higher level of action by the user. The FDA has suggested an alert value for CTDI_{vol} of 1000 mGy, which would deliver approximately half the dose associated with the onset of skin injury.

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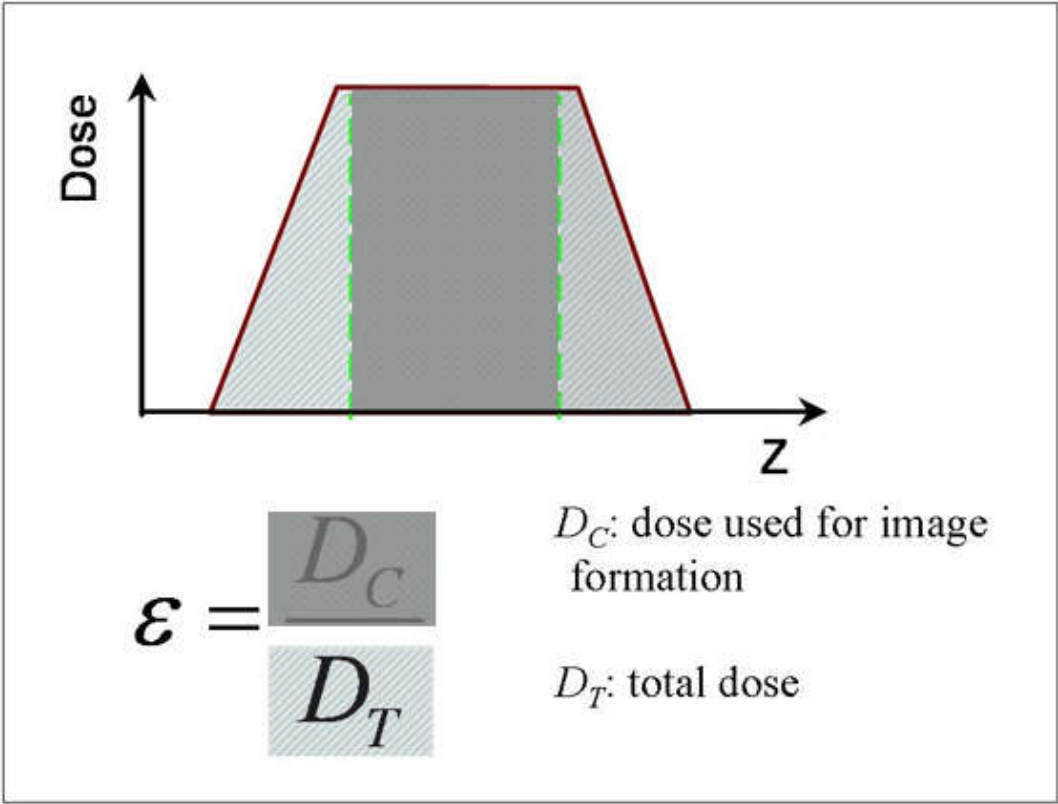
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NOTICE

For IEC60601-2-44:2016 Compliance, the DOSE ALERT Value shall not be set greater than 2000 mGy. CTDI_{vol} values at or above this limit can cause injury to the patient.

Geometric Efficiency Measurements

The Geometric Efficiency in the Z-Direction is one measure of how efficiently the radiation exposed to the patient is utilized by a scanner. It characterizes the ratio of the radiation that is used to generate images to the total radiation to which the patient is exposed. The Geometric Efficiency in the Z-Direction is expressed as the area of the dose profile in the Z direction subtended by the detector elements used for image generation to the area of the entire dose profile produced at that collimation. This is shown below in an illustrative diagram. The green line denotes the width of the radiation that is used for image generation.



The Geometric efficiencies listed here are established according to the techniques listed in the IEC 60601-2-44:2012 standard, Edition 3.1, Section 203.113. The dose profiles used to establish these efficiencies were measured using GafChromic XRQA2 radiochromic film. The films were digitized using a color flatbed scanner and the profile is extracted from the scan.

The factors that influence Geometric Efficiency in Z-Direction are Collimation Setting, and focal spot size and position. The table below lists the system parameters that influence these factors and the corresponding Geometric Efficiency in Z-Direction for those parameters.

NOTICE

The software displays a notification when the efficiency is lower than 70%.

The scan protocol is as follows: QA Axial Body 2D, variable collimation, 0.75 seconds rot speed, 120kV, 300mAs, variable resolution.

Resolution	Collimation	Efficiency in Z-Direction
Standard	16 x 1.5 mm	0.88
Standard	8 x 3 mm	0.88
Standard	12 x 1.5 mm	0.86
Standard	16 x 0.75 mm	0.79
Standard	12 x 0.75 mm	0.74
Standard	4 x 1.5 mm	0.66
Standard	4 x 0.75 mm	0.49
High	16 x 1.5 mm	0.88
High	12 x 1.5 mm	0.86
High	16 x 0.75 mm	0.79
High	4 x 0.75 mm	0.49
High	2 x 0.6 mm	0.52

The maximum deviation is $\pm 20\%$.

HU-Value Conversion

Gammex Tissue Characterization Phantom Model 467 from Gammex, Inc. was used to measure the conversion of measured HU-values to electron and electron density values relative to water. The CT number was measured for air, water, two different soft-tissue equivalent materials and two different bone-equivalent materials. For the Gammex Tissue Characterization Phantom Model 467, the air hole and five tissue mimicking material rods were identified for measurement. The CT number conversion factors of five materials for the head scan mode were measured: Air, Water, BRN-SR2 Brain (soft-tissue-equivalent material), CB2-30% CaCO₃ (bone-equivalent material) and CB2-50% CaCO₃ (bone-equivalent material). For body scan mode, LV1 Liver (soft-tissue-equivalent material) was measured instead of BRN-SR2 Brain (soft-tissue-equivalent material), and the other 4 materials were measured.

The Head measurements were performed with the phantom positioned at isocenter and using Onco Brain Axial exam card with the following scan parameters to represent the typical oncology head mode scan:

Collimation 16 x 0.75= 12mm, 1.0s rotation time, 3.0 mm slice width, UB filter, FOV 600mm, 120 kVp, 450 mAs, CTDI 53.5mGy.

The Body measurements were performed with the phantom positioned at isocenter and using Onco Abdomen exam card with the following scan parameters to represent the typical oncology body mode scan:

Collimation 16 x 0.75 = 12mm, 0.75s rotation time, 3.0 mm slice width, pitch 0.313, B filter, FOV 600mm, 120 kVp, 300 mAs, CTDI 19.9mGy.

The tables below represent the relationship between the relative electron density and CT number (in HU) of the five materials for head and body scan mode measured at 120kVp for the Big Bore scanner. The numbers presented below are representative numbers only based on a limited sample and are not intended for use as is, but rather as a reference.

The following table lists the typical electron density relative to water of materials and the corresponding CT number (in HU) as measured on the Big Bore scanner in oncology head scan mode.

Materials	Electron Density Relative to Water	HU-Value (HU)
Air	0.00	-989.3
Water	1.00	-1.2
BRN-SR2 Brain	1.04	26.1
CB2-30% CaCO ₃	1.28	410.4
CB2-50% CaCO ₃	1.47	774.8

The following table lists the typical electron density relative to water of materials and the corresponding CT number (in HU) as measured on the Big Bore scanner in oncology body scan mode.

Materials	Electron Density Relative to Water	HU-Value (HU)
Air	0.00	-977.3
Water	1.00	-4.5
LV1 Liver	1.06	69.3
CB2-30% CaCO ₃	1.28	445.9
CB2-50% CaCO ₃	1.47	809.3

The largest available reconstructed field-of-view of Big Bore is 700 mm, but the true field-of-view is 600 mm. The Big Bore does not have any inherent accuracy limitations within the 600 mm true field-of-view. CT values within the 600 mm field-of-view are generally accepted to be accurate within scanner specifications, however they may vary due to the specific circumstances such as beam hardening artifacts, etc. CT numbers between the edge of 600 mm true field-of-view and the edge of the 700 mm extended field-of-view are expected to exhibit large deviations, and are not intended to be used for treatment planning purposes.

It is important to understand that the conversion factors can be different from scanner to scanner, i.e. the CT number variation may be observed from scanner to scanner. It is important to have scanner-specific HU-value conversion calibrations of each CT-based treatment planning computers. The conversion factors are also subject to change with specific scan parameters that can affect measured HU values as well, for example kVp changes or changes to certain

reconstruction filters. Philips strongly recommends confirmation of all conversion factors before use. Factors to consider when evaluating conversion factors include but are not limited to: 1) Treatment Planning software manufacturer recommendations, 2) Industry guidance for treatment planning software and CT simulator commissioning, and 3) Any other unique considerations for each user's intended use.

The values in the tables here are only a representative of the conversion of measured HU-values to electron and electron density values relative to those of water. These CT numbers represent the calibration of a typical scanner and are not intended to be used as calibration data of any other CT scanner.

Half Value Layer (HVL)

For all HVL measurements, the CT system operates in a stationary X-ray tube position in the standard resolution mode, at 80, 100, 120 and 140 kVp, 200 mAs, 16 x 0.75 collimation, 12 mm slice width (one slice). Type 1100 aluminum filters of various thicknesses are used.

Aluminum Quality Equivalent filtration for different scan modes and kVps

Scan Mode	HVL
80 kVp	6.4 ± 0.5mm Al
100 kVp	7.8 ± 0.5 mm Al
120 kVp	8.9 ± 0.5 mm Al
140 kVp	9.9 ± 0.5 mm Al

Stray Radiation Dose Maps

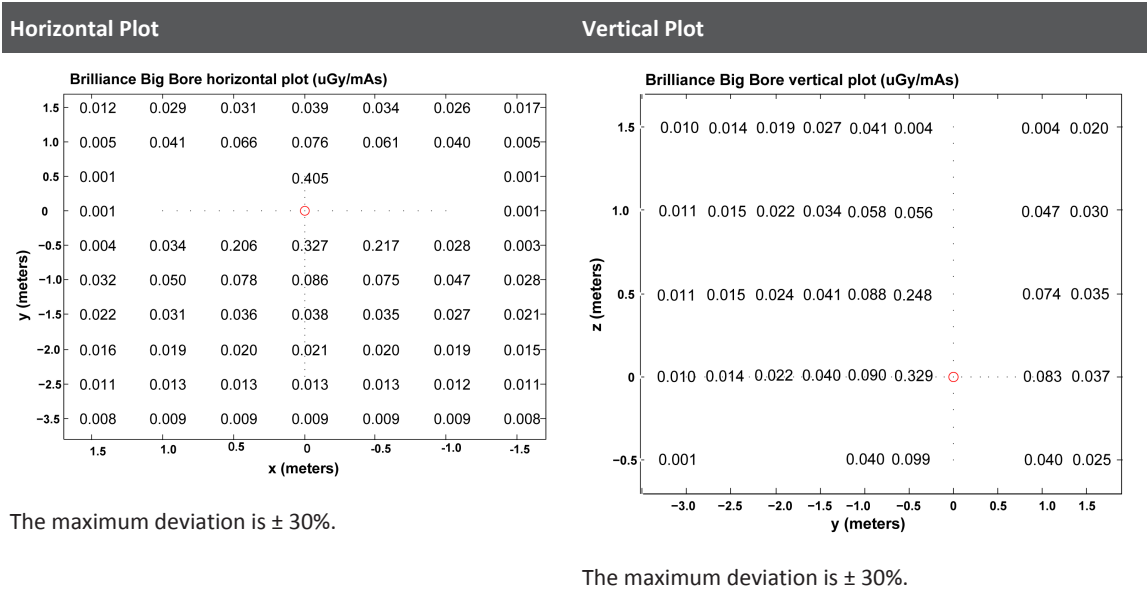
The map dose values units are $\mu\text{Gy} / \text{mAs}$, calculated from direct measurements of $\text{mGy} / 2500 \text{ mAs}$.

Measurements are made with the QA Axial Body 2D exam card at the maximum collimation of $16 \times 1.5 = 24 \text{ mm}$ and at 140 kVp, 100mAs, 2.0s Rotation Time, 25 cycles in the horizontal plane through the system axis and in vertical plane along the system axis. The room dimensions are 5.0 m wide, 6.0 m long and 3.3 m high.

The body CTDI phantom was centrally positioned in the tomographic scan plane and scanned as indicated to produce the near worst case scatter map values listed. This PMMA material phantom has a cylindrical shape with a diameter of 32 cm and a length of 15 cm. Any missing values on the charts were not measured because the location was not accessible with the measurement probe.

NOTICE

In X-ray radiation environments like CT, Sievert and Gray values are interchangeable; one μSv equals one μGy . By Philips convention, map values are given in $\mu\text{Gy}/\text{mAs}$ for ease of calculation.

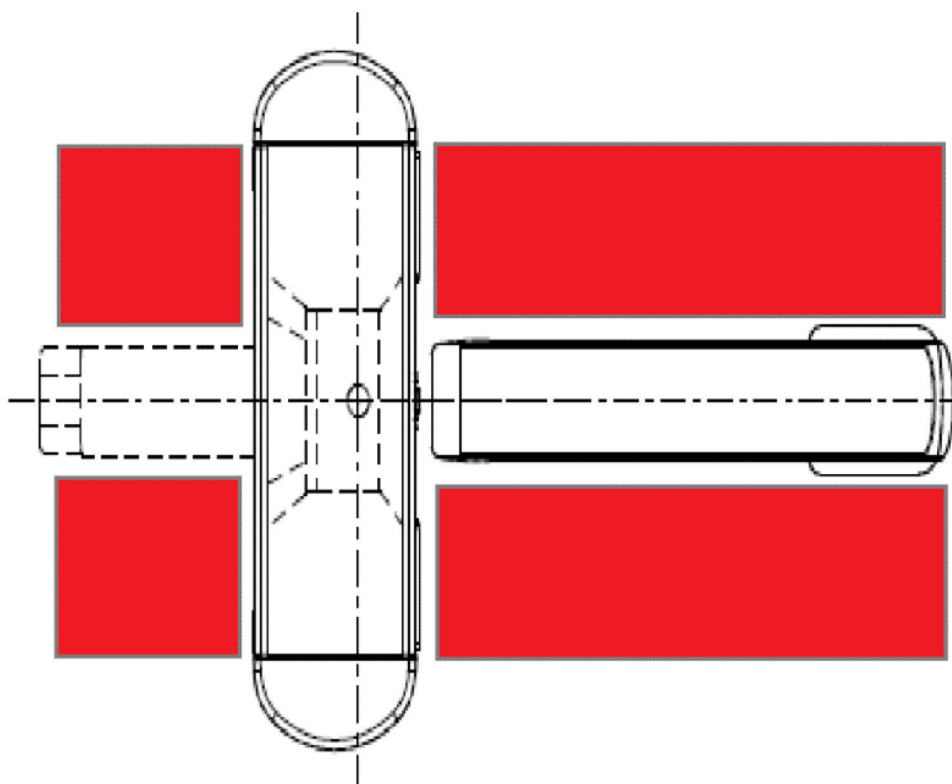


Zones of Occupancy

Permitted zones of occupancy are designated in the shaded areas.

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The zones of occupancy designated above are to be used for any CT examination where occupancy of the scanner room by medical staff is unavoidable.

The zones of occupancy are shown in the shaded outline on the diagram above. The two zones in the rear of the gantry are 120 cm x 120 cm and the zones alongside the table in front of the gantry are 120 cm wide and 280 cm long.

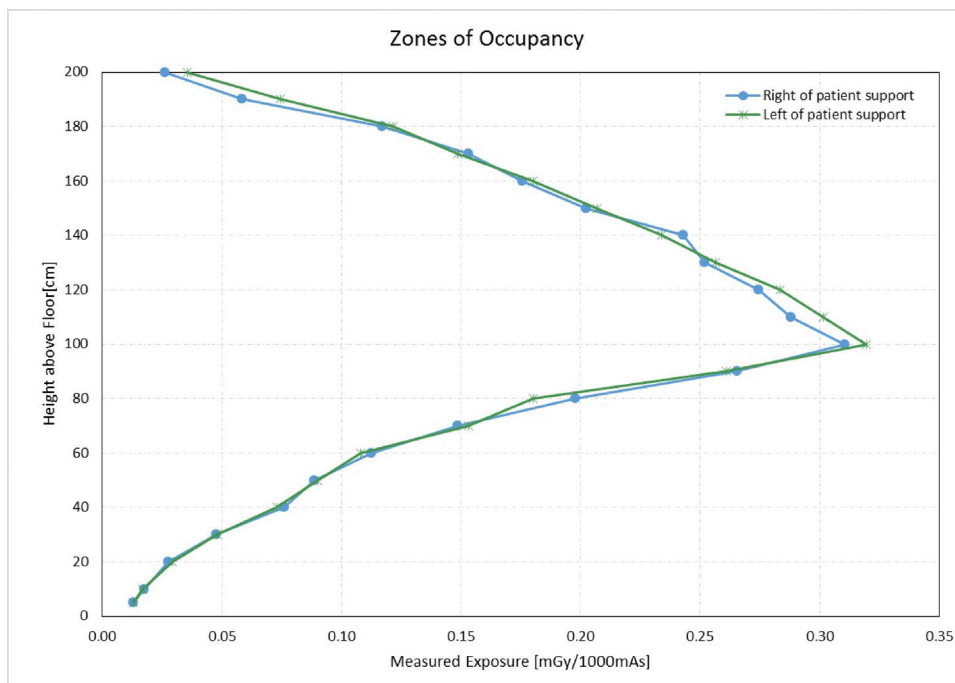
The radiation profiles (shown below) represent the exposure measured in the worst-case scenario for system use. Measurements were taken using a 32 cm diameter, 15 cm long PMMA material body CTDI phantom to simulate a large patient centered at isocenter. The phantom was scanned at 140 kVp to create the worst-case scatter possible.

For the profiles, measurements were made with a 230 cc ion chamber with a 100 cm² effective cross section. The measurements were made with the QA Axial Body 2D exam card at the maximum collimation of 16 x 1.5 = 24 mm and at 140 kVp, 100mAs, 2.0s Rotation Time, 25 cycles. The profiles were generated with measurements taken at 10 cm intervals from floor level to 200 cm above the floor. The profiles dose values units are mGy/1000mAs, calculated from direct measurement of uGy/2500mAs. The profiles shown do not represent the use of any protective devices.

CT is not designed to operate in a continuous mode, so for ease of use the measurements provided represent a single 1000 mAs shot. In order to scale these air kerma values to represent one hour at the conditions of loading that achieve the maximum X-ray tube continuous average power of 5500 W, the values must be multiplied by 141.5. The maximum continuous tube power at 140 kV is 5500 W which would be a tube current of 39.3 mA for one hour, or 141500 total mAs, which is 141.5 times the 1000 mAs used for the measured values.

Exposure from Scattered Radiation in the Front of the Gantry

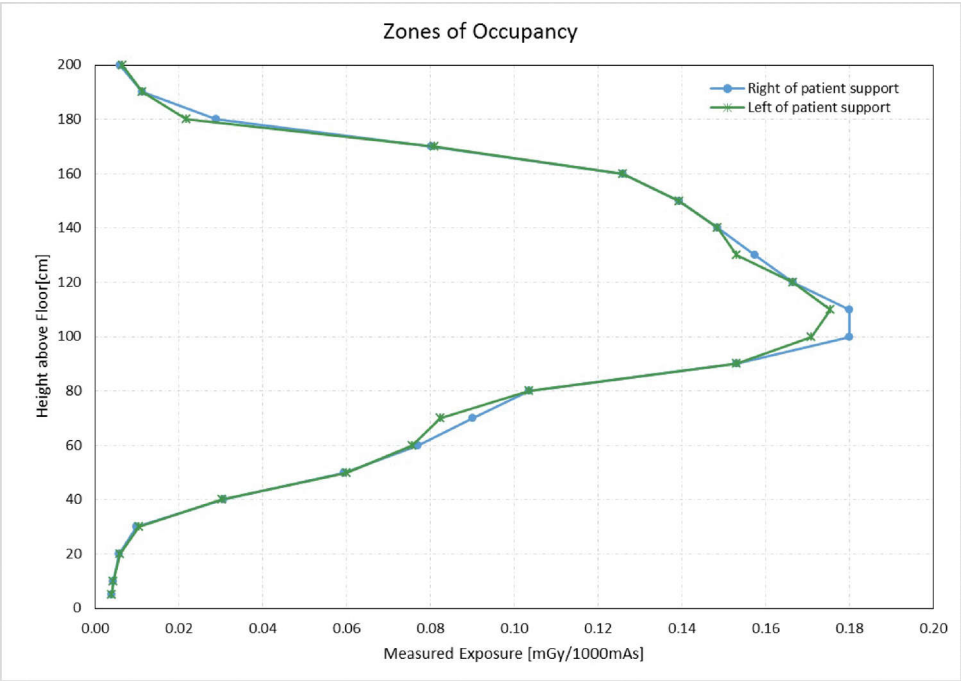
The blue line represents measurements taken on the right side of the patient support. The green line represents measurements taken on the left side of the patient support.



The maximum deviation is $\pm 30\%$.

Exposure from Scattered Radiation in the Rear of the Gantry

The blue line represents measurements taken on the right side of the patient support. The green line represents measurements taken on the left side of the patient support.



The maximum deviation is $\pm 30\%$.

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8 IEC Acceptance Testing for Big Bore

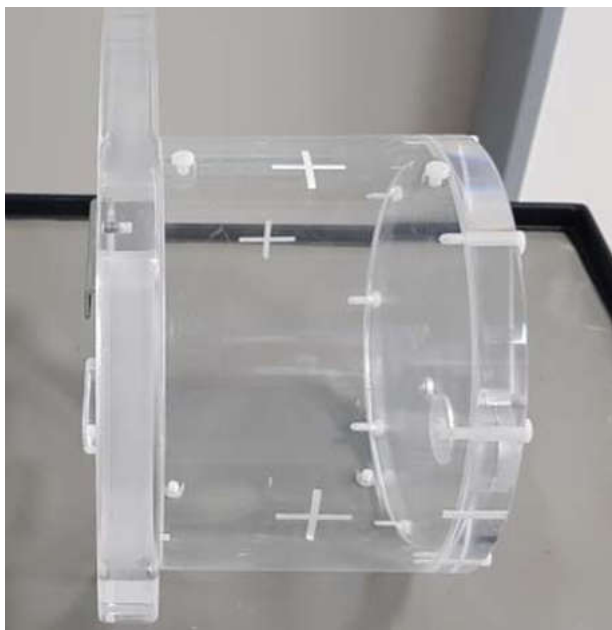
The IEC 61223-3-5 Standard requires sites to perform Acceptance and Constancy testing at regular intervals. This CT scanner has been tested at the factory for the various tests listed in the standard, and again upon installation. The Service Engineer can provide you with the baseline values set at installation at the site. However, if the site prefers to perform the testing interpedently, and to maintain their own baseline repository, the following test procedures describe the tests and tolerances for this scanner.

Couch Accuracy

The IEC 61223-3-5 standard describes the test for Couch movement accuracy. Please refer to that document, as there are no changes for Philips CT Scanners.

Laser Alignment Accuracy

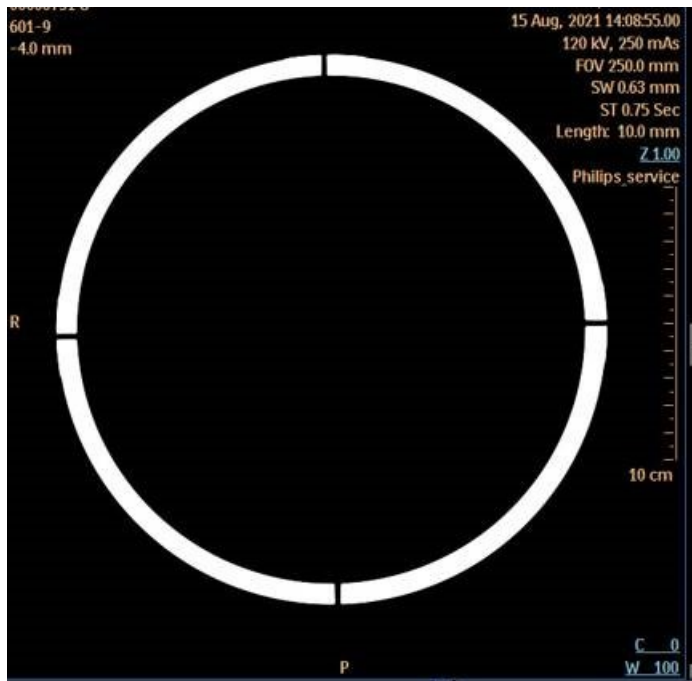
The Physics section of the system phantom has white crosshairs on the sides and the top. Beneath these crosshairs are small (1 mm diameter) holes drilled into the shell, to test laser alignment.



Make sure that the Physics section is level, and then align the lasers on these crosshairs in the center of the bore.

Perform a scan with the parameters: Standard QA Axial Body 2D, Default parameters except 16 x 0.625, Standard Resolution, 0.75 seconds, 120 kVp, 100 mAs, slice thickness 0.75 mm, B Filter.

In one of the center four slices (slice 7 through 10), an image of the shell with holes in it should look like the following:



If all four position air holes are not visible in the same slice, adjust the phantom z-position or/and tilt or/and swivel to make them all four visible. The repeat the test with the slice lasers in the middle of the crosshairs.

Note: The phantom swivel should be correct, if the couch is properly adjusted relative to the gantry rotation plane but to make small correction for phantom swivel you can do this using small free play of phantom holder which should be enough.

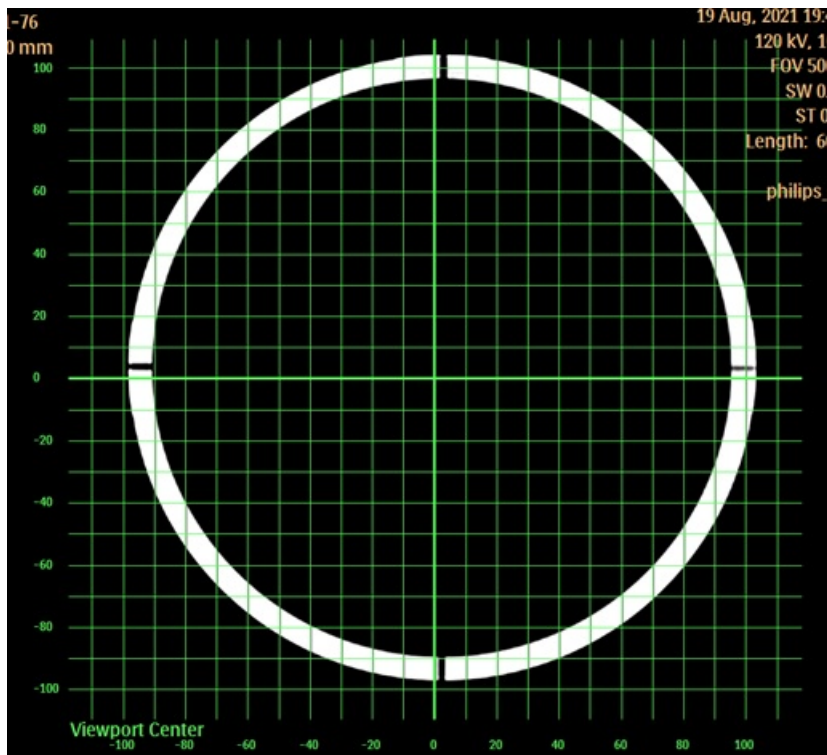
To test the external lasers, align the lasers on the crosshairs of the system phantom away from the bore, as shown below:



When aligned this way, move the phantom into the bore, so that the central laser is aligned on the top crosshair.

Perform a scan with the parameters: Standard QA Axial Body 2D, Default parameters except 16 x 0.625, Standard Resolution, 0.75 seconds, 120 kVp, 100 mAs, slice thickness 0.625 mm, B Filter.

In the same central image as before, the holes at 3:00, 12:00 and 9:00 should be aligned with the grid on the image, to within ± 2 mm.

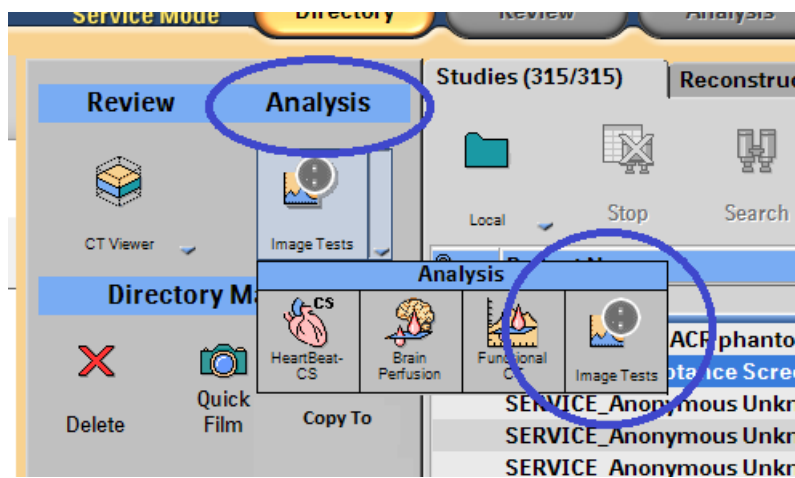


Reconstructed Section Thickness

This test uses the Slice Thickness wires of the Physics Section of the chapter “Harmonized System Phantom” on page 93, and the Slice Thickness tool described below.

To Measure Reconstructed Section Thickness:

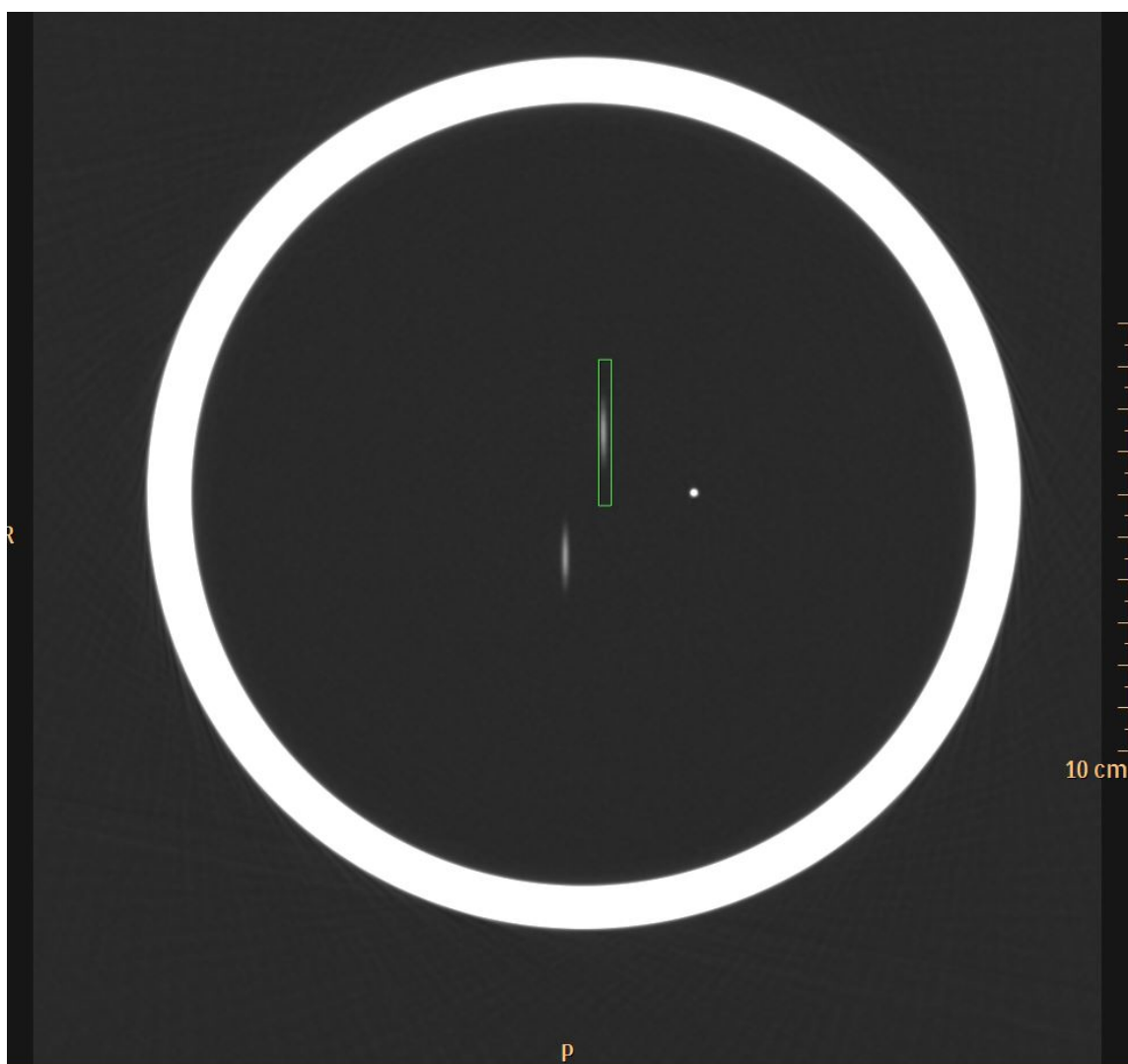
1. Adjust the table so that the phantom physics layer centerline aligns with the laser lights.
2. Set up a standard patient scan (refer the following table for details).
Note: The gantry tilt should be set at zero degrees (perpendicular).
3. Perform the first axial scan using the parameters for first slice width scan (e.g. 16 x 0.75, Axial) from the following table.
4. From the **Directory**, select the scan series.
5. Select **Image Tests** from the **Analysis** options.



6. Select the middle image of the first scan.
7. In the Slice thickness row, select the **Slice Thickness Position** Rectangle to place a rectangle on the image.



8. Move and resize the **Position Rectangle ROI** to symmetrically position it to surround one image of the wire.



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9. To view the sensitivity profile and calculate slice thickness, click  **Calculate Slice Thickness**. The appropriate measured value is the “Slice Width at 50% FWHM (mm)” value.

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Slice Width	
Protocol 2D, 120kV, 12x1.5(SW 4.5), STANDARD, 0.75s, 200mAs, B, FOV 250	
Image Number 1	
Physical Slice Number 12	
Nominal Slice Width 4.500	
--- Slice Thickness ---	
Slice Width at 50% FWHM (mm)	4.341
Slice Width at 10% FWTM (mm)	5.276
--- Details ---	
Max CT (HU)	95.394
Background CT (HU)	22.062
Center of slice (mm)	2.390

10. Repeat steps 7-9 to measure both vertical wires, and average the two results.
11. Repeat steps 6-10 using the first slice.
12. Verify the results are within the allowed tolerance.
13. Exit the **Image Tests** program.

Scan Parameters and Results for Slice Thickness Measurements:

Measurement Element	Scan parameters	Slices to measure	Low Limit (mm)	Upper limit (mm)	Comments
Slice Thickness 1	Axial 2D, Body, 120kV, 16 x 0.75 (SW 0.75 mm), High, 0.75s, 200mAs, E, FOV 250, 1024 x 1024	1-st slice and one of middle slices	0.25	1.25	0.75 ± 0.50
Slice Thickness 2	Axial 2D, Body, 120kV, 16 x 1.5 (SW 1.5mm), STD, 0.75s, 200mAs, YA, FOV 250, 512 x 512	1-st slice and one of middle slices	0.75	2.25	1.50 ± 0.75
Slice Thickness 3	Axial 2D, Body, 120kV, 12 x 1.5 (SW 4.5mm), STD, 0.75s, 200mAs, B, FOV 250, 512 x 512	1-st slice and one of middle slices	3.50	5.50	4.50 ± 1.00

Dose

Dose can be measured as either $CTDI_{vol}$ with an ion chamber inserted into large phantoms made with PMMA, or by measuring $CTDI_{free\ air}$ with an ion chamber suspended in air at the isocenter.

To measure $CTDI_{vol}$, it is important that the CTDI phantoms be centered and leveled to within ± 3 mm in all three axes. Also, with Philips scanners, the system is designed to start generating X-rays within a short time from pressing the **Scan** button, but then the tube can be anywhere in the 360° as it spins. To account for the effect that this has on the peripheral CTDI measurements, we do repeated scans with a fixed cycle time. The intention is to reliably sample the motion around the gantry in a technique sufficient to get good repeatability. For this scanner, we need to measure 8 scans separated by a cycle time of 4.5 seconds. If the Ion chamber is set to accumulate total dose, then divide the measured dose by 8, otherwise write down 8 measurements and average them. $CTDI_{vol}$ is a weighted average of five measurements in the five holes of the CTDI phantoms.

The typical Head and typical Body scan parameters and $CTDI_{100}$ specifications are as listed in chapter “Head-scan Information” on page 121 and chapter “Body-scan Information” on page 126.

To measure $CTDI_{free\ air}$, it is important that the ion chamber be centered in air at the isocenter (within ± 5 mm is usually sufficient), but it must also be level.

After the probe is positioned, the $CTDI_{free\ air}$ is calculated from the measured dose as:

$$CTDI_{free\ air} = \frac{(Measured\ Dose) \times 100}{N \times T}, \text{ Where } N \times T \text{ is Collimation (e.g. } 16 \times 0.75 \text{ is } 12 \text{ mm)}$$

$CTDI_{free\ air}$ Test and Tolerances:

Stage #	Scan Parameters Scan type, kV, Collimation, Resolution	Expected Value (mGy)	Maximum deviation
1	Axial 2D Body, 120kV, 16 x 1.5, STD, SW 3, RT 0.75s, 250 mAs, YB	35.48	25%
2	Axial 2D Body, 120kV, 8 x 3, STD, SW 3, RT 0.75s, 250 mAs, YB	35.48	25%
3	Axial 2D Body, 120kV, 12 x 1.5, STD, SW 4.5, RT 0.75s, 250 mAs, YB	36.91	25%
4	Axial 2D Body, 120kV, 16 x 0.75, STD, SW 3, RT 0.75s, 250 mAs, YB	39.79	25%
5	Axial 2D Body, 120kV, 12 x 0.75, STD, SW 3, RT 0.75s, 250 mAs, YB	42.19	25%
6	Axial 2D Body, 120kV, 4 x 1.5, STD, SW 3, RT 0.75s, 250 mAs, YB	47.46	25%

Stage #	Scan Parameters Scan type, kV, Collimation, Resolution	Expected Value (mGy)	Maximum deviation
7	Axial 2D Body, 120kV, 4 × 0.75, STD, SW 3, RT 0.75s, 250 mAs, YB	64.48	30%
8	Axial 2D Body, 120kV, 2 × 0.6, High, SW 1.2, RT 0.75s, 250 mAs, YD	60.64	35%
9	Axial 2D Body, 80kV, 16 x 1.5, STD, SW 3, RT 0.75s, 250 mAs, YB	10.48	25%
10	Axial 2D Body, 100kV, 16 x 1.5, STD, SW 3, RT 0.75s, 250 mAs, YB	21.4	25%
11	Axial 2D Body, 140kV, 16 x 1.5, STD, SW 3, RT 0.75s, 250 mAs, YB	51.67	25%
12	Axial 2D Adult Head, 120kV, 16 x 1.5, STD, SW 6, RT 0.75s, 250 mAs, YB	50.04	25%
13	Axial 2D Pediatric Head, 120kV, 16 x 1.5, STD, SW 6, RT 0.75s, 200 mAs, YB	28.38	25%
14	Axial 2D Pediatric Body, 120kV, 16 x 1.5, STD, SW 6, RT 0.5s, 200 mAs, YB	28.38	25%

CT number, Uniformity and Noise

The Mean CT number, CT number Uniformity and Image Noise are measured on the Head or Body sections of the Harmonized System Phantom. Acceptance measures these on the Head Section of the phantom for all modes except Adult Body, 120 kVp, which is also measured on the large Body section. This allows Constancy to run only with the Head Section, for convenience.

Head phantom:

To measure CT number, use an ROI of area $300 \pm 50 \text{ mm}^2$ (effective diameter 20 mm), centered in the phantom. To measure CT number uniformity, use ROI's of the same size, located at approximately 12:00, 3:00, 6:00 and 9:00 in the phantom, 1 cm inside the plastic shell. Image noise uses an ROI of $5000 \pm 50 \text{ mm}^2$ (effective diameter 80 mm), also centered in the phantom.

Body phantom:

To measure CT number, use an ROI of area $700 \pm 75 \text{ mm}^2$ (effective diameter 30 mm), centered in the phantom. To measure CT number uniformity, use ROI's of the same size, located at approximately 12:00, 3:00, 6:00 and 9:00 in the phantom, 1 cm inside the plastic shell. Image noise uses an ROI of $11,300 \pm 100 \text{ mm}^2$ (effective diameter 120 mm), also centered in the phantom.

Scan Parameters and Tolerances for CT number, Uniformity and Noise Measurements:

Scan n	Protocol Element	Scan Parameters	Mean CT limits (HU)	Uniformity limits (HU)	Noise Low Limit (HU)	Noise Up Limit (HU)
1	Adult Head	Axial 2D Head, 120kV, 16 × 1.5 (SW 6mm), STD, 0.75s, 200mAs, UB, FOV 250, 512 x 512	[-4.0, +4.0]	[-4.0, +4.0]	3.5	4.7
2	Adult Body Large phantom	Axial 2D Body, 120kV, 16 × 1.5 (SW 6mm), STD, 0.75s, 315mAs, B, FOV 350, 512 x 512	[-6.0, +6.0]	[-8.0, +8.0]	9.0	12.2
3	Adult Body on Head	Axial 2D Body, 120kV, 16 × 1.5 (SW 6mm), STD, 0.75s, 300mAs, B, FOV 250, 512 x 512	[-6.0, +6.0]	[-8.0, +8.0]	3.3	4.5
4	Adult Body with varied tube voltage	Axial 2D Body, 80kV, 16 × 0.75 (SW 6mm), STD, 0.75s, 315mAs, B, FOV 250, 512 x 512	[-6.0, +6.0]	[-8.0, +8.0]	6.9	9.3
5		Axial 2D Body, 100kV, 16 × 1.5 (SW 6mm), STD, 0.75s, 300mAs, B, FOV 250, 512 x 512	[-6.0, +6.0]	[-8.0, +8.0]	4.3	5.9
6		Axial 2D Body, 140kV, 16 × 1.5 (SW 6mm), STD, 0.75s, 300mAs, B, FOV 250, 512 x 512	[-6.0, +6.0]	[-8.0, 8.0]	2.8	3.8
7	Pediatric Head	Axial 2D Head, 120kV, 16 × 1.5 (SW 1.5mm), STD, 0.75s, 200mAs, UB, FOV 250, 512 x 512	[-4.0, +4.0]	[-4.0, +4.0]	6.6	9.0
8	Pediatric Body	Axial 2D, Body, 120kV, 16 × 1.5 (SW 3mm), STD, 0.5s, 200mAs, B, FOV 250, 512 x 512	[-4.0, +4.0]	[-8.0, +8.0]	5.7	7.8
9	Pediatric Body with varied tube voltage	Axial 2D, Body, 80kV, 16 × 0.75 (SW 3mm), STD, 0.5s, 200mAs, B, FOV 250, 512 x 512	[-6.0, +6.0]	[-8.0, +8.0]	12.1	16.4

Scan	Protocol Element	Scan Parameters	Mean CT limits (HU)	Uniformity limits (HU)	Noise Low Limit (HU)	Noise Up Limit (HU)
10		Axial 2D, Body, 100kV, 16 ×1.5 (SW 3mm), STD, 0.5s, 200mAs, B, FOV 250, 512 x 512	[-6.0, +6.0]	[-8.0, +8.0]	8.6	10.3
11		Axial 2D, Body, 140kV, 16 × 1.5 (SW 3mm), STD, 0.5s, 200mAs, B, FOV 250, 512 x 512	[-6.0, +6.0]	[-8.0, +8.0]	4.8	6.5

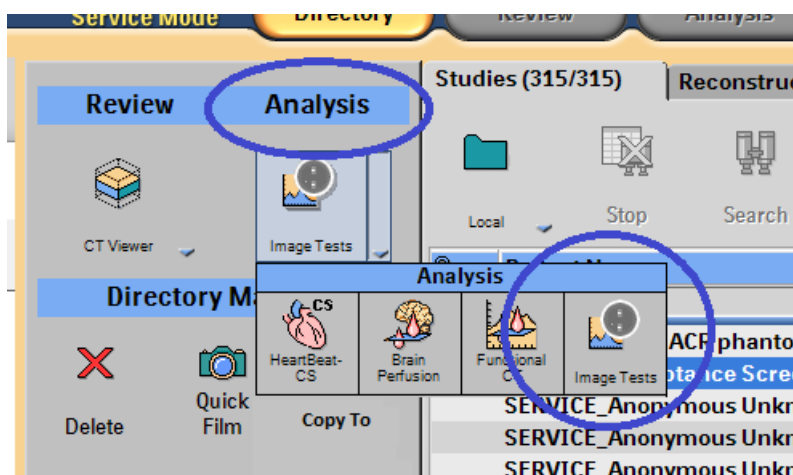
All slices should meet the CT number, Uniformity and Noise requirements. All scans use the Head Section of the system phantom, except Scan 2. Scan 2 uses the large Body Section of the system phantom.

Spatial Resolution

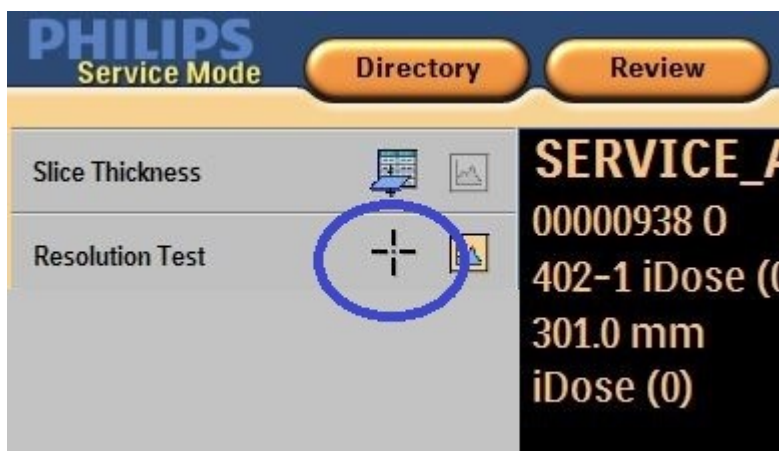
Spatial Resolution is measured with the Modulation Transfer function. This is accessed in the Image Tools menu of the Analysis tab in the Patient Directory view. This test uses the horizontal wire in the Physics Section of the system phantom, which is located approximate 25 mm from the center of the phantom. The Physics Section of the system phantom is described in chapter “Harmonized System Phantom” on page 93.

To Measure MTF:

- Scan the physics layer of the system phantom using the protocol and settings specified in the appropriate table.
 - For Head scans, use **Head > Axial > Impulse Response Head Protocol**.
 - For Body scans, use **Abdomen > Axial > Impulse Response Body Protocol**.
- From the **Directory**, select the scan series.
- Select **Image Tests** from the **Analysis** options.



4. Select the middle image of the scan.
5. In the Slice thickness row, select the Position on the pin to place a cursor on the image near the pin.



6. To view the spatial resolution result, click **Calculate Resolution** . The appropriate results are in the lines MTF at 10% (lp/cm) and MTF at 50% (lp/cm).

Impulse Response Manual Mode	
Protocol 2D, 120kV, 16x1.5(SW 3), STANDARD, 0.75s, 200mAs, E, FOV 50	
--- MTF ---	
MTF at 10 % (lp/cm)	9.449
MTF at 50 % (lp/cm)	4.727
--- Impulse Response ---	
Width at 10 % (mm)	1.560
Width at 50 % (mm)	0.864
--- Details ---	
Max CT at center of pin	299.000
Background CT	-998.542
X(c.g) (mm)	{X=261,Y=255}
Wire Cor (mm)	-0.012

- Repeat steps 5-6 on the other central image.
- Compare each recorded value to the specification provided within the limits listed in the table below.

Scan Parameters and Tolerances for Spatial Resolution Measurements:

Resolution measurement			10% MTF (lp/cm)		50% MTF (lp/cm)	
Measurement Element	Scan parameters	Slices to measure	Low Limit (mm)	Upper limit (mm)	Low Limit (mm)	Upper limit (mm)
1. Resolution, Adult Head	Axial Head 2D, 120kV, 16 × 1.5 (SW 3.0mm), STD, 0.75s, 200mAs, E, FOV 50, 512 x 512 Image center at x=25, y=0	Two middle slices	7.9	9.9	3.6	5.0

Resolution measurement			10% MTF (lp/cm)		50% MTF (lp/cm)	
2. Resolution, Pediatric Head	Axial Head 2D, 120kV, 16 x 0.75 (SW 1.5mm), High, 1.5s, 200mAs, E, FOV 50, 1024 x 1024 Image center at x=25, y=0	Two middle slices	10.3	14.3	4.7	7.7
3. Resolution, Adult Body	Axial Body 2D, 120kV, 16 x 1.5 (SW 3.0mm), STD, 0.75s, 220mAs, E, FOV 50, 512 x 512 Image center at x=25, y=0	Two middle slices	8.0	10.0	3.8	5.2
4. Resolution, Pediatric Body	Axial Body 2D, 120kV, 16 x 1.5 (SW 3.0mm), STD, 0.75s, 200mAs, E, FOV 50, 512 x 512 Image center at x=25, y=0	Two middle slices	8.0	10.0	3.8	5.2

9 Third Party Devices Compatibility Matrix

Philips Medical Systems Nederland B.V. has performed compatibility tests utilizing a limited number of samples of Product. The scope of this compatibility statement is strictly limited to 3rd party devices, upgrades and or accessories as included in CT/AMI Accessories list as presented in appendix 1.

CT/AMI Accessories list as presented in appendix 1, include 3rd party devices, upgrades and or accessories along with the following information:

- Name of the accessory with reference to 6NC/12 NC
- Classification of the accessories
- Legal Manufacturer name

Through the basic compatibility testing with the Product, Philips found the CT/AMI products performed as intended and specified, with no detrimental degradation of CT/AMI products efficacy or safety when used per the conditions as stated in their Instructions for Use. Basic compatibility testing indicates compatibility of the CT/AMI products along with the 3rd party devices, upgrades and or accessories.

Safety and efficacy of the Product is the sole responsibility of Philips Medical Systems Nederland B.V.

This compatibility statement does not guarantee assurance that compatibility will be maintained with future changes to Product, including incorporated software releases, modifications and upgrades.

Accessories as Medical Devices

6NC/ 12NC	Name	Applicable Product	Risk Class	UDI (yes/no)	Legal Manufacturer	Category
45501800105x	ECG MONITOR ENGLISH	Brilliance CT Big Bore, Philips CT Big Bore, Ingenuity CT	IIb	Yes	PMS Boeblingen	Accessory
45356748489x	ECG Monitor English	Vereos	IIb	Yes	PMS Boeblingen	Accessory
45980158427x	DORADOnova 3 w/ CARINANav, Red Wall	Brilliance CT Big Bore, Philips CT Big Bore	I	Yes	LAP GmbH Laser Applikationen	Accessory
45980158428x	DORADOnova 3 w/ CARINANav, Red Floor	Brilliance CT Big Bore, Philips CT Big Bore	I	Yes	LAP GmbH Laser Applikationen	Accessory

45980158431x	DORADOnova3 w/CARINANav, Red Bridge	Brilliance CT Big Bore, Philips CT Big Bore	I	Yes	LAP GmbH Laser Applikationen	Accessory
45980158429x	DORADOnova3 w/ CARINANav, Green Wall	Brilliance CT Big Bore, Philips CT Big Bore	I	Yes	LAP GmbH Laser Applikationen	Accessory
45980158430x	DORADOnova3 w/CARINANav, Green Floor	Brilliance CT Big Bore, Philips CT Big Bore	I	Yes	LAP GmbH Laser Applikationen	Accessory
45980158432x	DORADOnova3 w/CARINANav, Green Bridge	Brilliance CT Big Bore, Philips CT Big Bore	I	Yes	LAP GmbH Laser Applikationen	Accessory
45980109682x	Certegra SyncRight upgrade kit	Ingenuity CT, Brilliance CT Big Bore, Philips CT Big Bore, Vereos, Brilliance iCT Spectral CT	IIb	Yes	Bayer healthcare LLC	Accessory
98960520105x	SyncRight/ Medrad P3T Cardiac	Ingenuity CT, Brilliance CT Big Bore, Philips CT Big Bore, Vereos, Brilliance iCT IQon Spectral CT	IIb	Yes	Bayer healthcare LLC	Accessory
98960520106x	SyncRight/ Medrad P3T Abdomen	Ingenuity CT, Brilliance CT Big Bore, Philips CT Big Bore, Vereos, Brilliance iCT IQon Spectral CT	IIb	Yes	Bayer healthcare LLC	Accessory

98960520126x	SyncRight Certegra Injector - Pedestal	Ingenuity CT, Brilliance CT Big Bore, Philips CT Big Bore, Vereos, Brilliance iCT IQon Spectral CT	IIb	Yes	Bayer healthcare LLC	Accessory
98960520125x	SyncRight Certegra Injector – OCS Long Arm	Ingenuity CT, Brilliance CT Big Bore, Philips CT Big Bore, Vereos, Brilliance iCT IQon Spectral CT	IIb	Yes	Bayer healthcare LLC	Accessory
98960520124x	SyncRight Certegra Injector – OCS Long Arm	Ingenuity CT, Brilliance CT Big Bore, Philips CT Big Bore, Vereos, Brilliance iCT IQon Spectral CT	IIb	Yes	Bayer healthcare LLC	Accessory
98960520123x	SyncRight Certegra Injector – OCS Short Arm	Ingenuity CT, Brilliance CT Big Bore, Philips CT Big Bore, Vereos, Brilliance iCT IQon Spectral CT	IIb	Yes	Bayer healthcare LLC	Accessory
45980109684x	Certegra SyncRight Kit iCT – P3T PA	Ingenuity CT, Brilliance CT Big Bore, Philips CT Big Bore, Vereos, Brilliance iCT IQon	IIb	Yes	Bayer healthcare LLC	Accessory
45980105060x	Pinnacle3 Tumor LOC	Brilliance CT Big Bore, Philips CT Big Bore	IIb	Yes	PROS	Accessory

Third Party Devices Compatibility Matrix

45980053073x	Breath Hold ES, Patient Display	Vereos	Class I	Yes	Medspira	Accessory
45980053072x	Breath Hold ES, Base Unit	Vereos	Class I	Yes	Medspira	Accessory
45980066324x	ASSY, PULMO CHEST BELLOWS, TUBING ES	Vereos	Class I	Yes	Medspira	Accessory

10 EURATOM Compliance Statement

For basic safety standards for protection against exposure to ionising radiation. (Directive 2013/59/EURATOM)

Product Name: Refer to the System Label.

Device class: II b

Company Name and Address: Philips Medical Systems Nederland B.V.

Veenpluis 6

5684 PC Best

The Netherlands.

Contact Information: Refer to System labels and IFU.

Target Users and Training: Refer to the IFU.

Information on the residual risks, any adverse event and precaution for use: The Philips Risk Management Process comply with ISO 14971 and is applicable to all stages of the life cycle of the device. IFU contains the warnings and precautions for use, applicable to safety of the product. The Risk Management Process recognized hazards associated with the device to estimate and evaluate all the associated risks. Identified risks associated with the use of the device are mitigated and deemed acceptable when weighed against the benefits to the patient. Information pertains to residual risks, adverse events and precautions identified through the instruction for use.

Device description: Refer to IFU

The summary of the clinical evaluation results as mentioned in article R.5211-36-1: With respect to safety of the device, Clinical evaluations conducted thru clinical evaluation planning, identify equivalent devices, a comprehensive analysis of available pre- and postmarket clinical data to ensure the safety and performance of the device intended use. Furthermore, reviewed clinical data did not identify any risks specific to the device, not assessed in the risk analysis. Therefore, this clinical evaluation concludes that the device will not compromise the clinical condition or the safety of patients, or the safety of the users.

Conclusion:

1. The clinical safety and performance of the device was demonstrated with the clinical evaluation;
2. Conformity with the relevant essential requirements is demonstrated through technical documentation.

With respect to post market clinical follow up, no specific device features or other aspects were identified that require special attention during the post market phase. Post market surveillance monitoring activities (i.e., conducting a search in the literature and clinical experience databases) related to the use of the device in the market are planned to obey our internal processes.

For the list of the applicable Harmonized Standards, you can refer to the Declaration of Conformity.

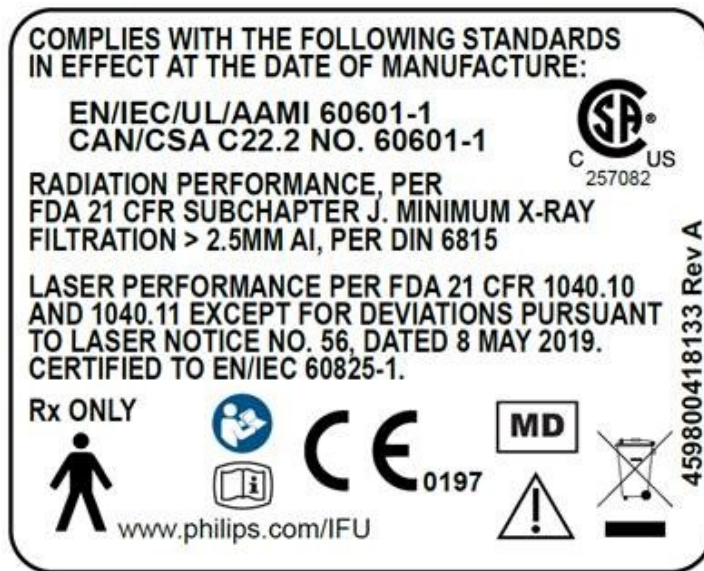
11 CE Mark Information Sheet

See the meaning of the CE mark that is applicable to your system.

Circumstance	Applicable Mark
Where these Instructions for Use (IFUs) have been delivered in association with a device that bears a CE Mark on the system labelling for a device placed on the market on 26 May 2020 or later, or as part of an upgrade where the system being upgraded was placed on the market on 26 May 2020 or later, then the CE Mark indicates that this device complies with the provisions of the Medical Device Directive (93/42/EEC, as amended), the ROHS II Directive (2011/65/EU) and Medical Device Regulations (MDR 2017/745). The Notified Body number adjacent to the CE Mark claims compliance to 93/42/EEC, Annex II, and EU MDR 2017/745, Annex IX Conformity assessment based on a Quality Management System and on assessment of Technical Documentation.	
Where these Instructions for Use (IFUs) have been delivered in association with a device that bears a CE Mark on the system serial tag or regulatory tag (adjacent to the serial tag) and these IFUs are for a device placed on the market on 22 July 2014 or later, or as part of an upgrade where the system being upgraded was placed on the market on 22 July 2014 or later, then the CE Mark indicates that this device complies with the provisions of the Medical Device Directive (93/42/EEC, as amended) and the ROHS II Directive (2011/65/EU). The Notified Body number adjacent to the CE Mark is exclusive to 93/42/EEC, Annex II, and indicates the Notified Body for the system when it was originally placed on the market.	
Where these Instructions for Use have been delivered as part of an upgrade to a system placed on the market prior to 22 July 2014, then the CE Mark shown on the system serial tag or regulatory tag (adjacent to the serial tag) indicates that this device complies with the provisions of the Medical Device Directive (93/42/EEC, as amended). The Notified Body number adjacent to the CE Mark indicates the Notified Body for the system when it was originally placed on the market. The upgraded device is exempt from the RoHS II Directive 2011/65/EU, per Article 4, 4.(b).	
Where these Instructions for Use are associated with a device that does not bear a CE Mark on the system serial tag or regulatory tag (adjacent to the serial tag), then no claim of compliance to either the Medical Device Directive (93/42/EEC, as amended) or the RoHS II Directive (2011/65/EU) is made for the particular device.	No mark

NOTICE

In the event of a serious incident involving a patient and/or operator during use of the system, immediately report the incident to the manufacturer and the competent authority of the Member State in which the operator and/or patient is established.



The system product label is typically on or near the gantry.

12 Reference Training Checklist

Table and Gantry Controls

- ☐ I can locate and recognize when and how to use the following gantry and table controls:
- ☐ Control panels
 - ☐ Gantry covers
 - ☐ Head holder and accessories – positioning and intended use
 - ☐ Foot board (not to be used for any body part other than feet)
 - ☐ Couch pad alignment
 - ☐ Emergency Stops
 - ☐ Table Motion controls and settings
 - ☐ Laser Lights- Internal/External
 - ☐ Load/Unload Foot Pedal
 - ☐ Maximum Patient Load
 - ☐ Scan-able Range
 - ☐ Patient positioning aids
- ☐ I can locate the following safety devices and recognize when to use them:
- ☐ Gantry Emergency Stops
 - ☐ Emergency patient releases
 - ☐ Radiation warning lamps on the scanner
- ☐ I can reset from an Emergency Stop.
- ☐ I can comply with the following patient safety standards:
- ☐ Using the foot board for feet only (no other body parts)
 - ☐ Keep the patient under constant observation during all movements of the gantry and patient table
 - ☐ Avoid collision and ensuring patient safety by checking to make sure there is nothing under the table when lowering the patient couch
 - ☐ Instructing patients NOT to stare into the laser beam

- ☐ I am familiar with the following safety requirements regarding the equipment:
- ☐ Do not stare into the laser beam.
 - ☐ Do not remove covers or cables from the equipment.
 - ☐ Do not use explosive disinfecting sprays while cleaning the CT scanner.

Power-up and Shutdown

- ☐ I can perform System Power-up and Shutdown, including the following:
- ☐ Daily Logout/Login
 - ☐ Weekly Complete System Shutdown and Power-up
 - ☐ Console and System UPS shutdown if applicable
- ☐ I will review, understand and follow the system maintenance quick reference, including the following:
- ☐ System Power-up and Shutdown procedures
 - ☐ Scanner's Do's and Do not's
 - ☐ Appropriate Scanning and Control room conditions

Quality Assurance – Daily/Monthly

- ☐ I can perform the following quality assurance functions:
- ☐ Short Tube Conditioning
 - ☐ Air Calibrations
 - ☐ Phantom Placement
 - ☐ Quick IQ Check
 - ☐ Constancy
- ☐ I am familiar with the following safety advisories:
- ☐ The X-ray unit may be dangerous to patient and operator unless safe exposure factors, operating instructions and maintenance schedules are observed.
 - ☐ Do not use the CT scanner for any application until you are sure that the image Performance QA is complete and that the Preventive Maintenance program is up to date.

User Interface/GUI Overview

- ☐ **I can use the user interface.**
- ☐ *Access the right mouse functions*
 - ☐ *Select folders and devices (verify connection- Free space)*
 - ☐ *Use the Queue Manager*
 - ☐ *Bug Report Submission*
 - ☐ *Monitor Calibration*
 - ☐ *Error Message and Recovery*

Patient Data Entry and Exam Card Selection

- ☐ **I can enter patient data and select exam cards, including:**
- ☐ *Entering or selecting new/current/anonymous/worklist*
 - ☐ *Completing Mandatory Fields*
 - ☐ *Selecting Voice Language*
 - ☐ *Selecting Patient Position*
 - ☐ *Selecting Exam Card (EC)*
 - User EC's
 - Oncology Reference EC's
 - Diagnostic Reference EC's
 - Suggested EC's
- ☐ **I am familiar with the following safety advisories:**
- ☐ *Make sure to use radiation protection devices correctly, e.g., adding bismuth shielding after the Surview acquisition and dedicated pediatric protocols to reduce radiation dose.*
 - ☐ *Make sure to enter the patients' age correctly, when using pediatric protocols.*

Exam Card Manager

☐ I can build Exam Cards, and can determine when and how to use the following acquisition and result parameters.

☐ *Survview Including:*

- Length
- View Angle
- Direction
- Frontal
- Lateral
- Dual

☐ *Collimation*

☐ *Sampled Collimation*

☐ *DoseRight Index (DRI)*

☐ *3D Modulation*

☐ *Z Modulation*

☐ *Automatic Scan Time*

☐ *Resolution*

☐ *kVp*

☐ *Dose Notification*

☐ *Slice Thickness*

☐ *Increment*

☐ *Filters*

☐ *Edit before final Recon*

☐ *Injection Settings*

☐ *O-MAR*

General Safety Notifications

- ☐ I am familiar with the following safety notifications:
- ☐ Portable radio transmitting devices such as mobile phones should not be used around the CT scanner as they can interfere with proper functioning of the system.
 - ☐ Do not use the CT scanner for any application until you have received adequate and proper training in its safe and effective use.
 - ☐ Never attempt to remove, modify, over-ride or forcibly move any safety device on the equipment.
 - ☐ Do not use the CT scanner for any purpose other than those for which it is intended.
 - ☐ Exposing an implanted medical device such as a pacemaker or neuro-stimulator to medical radiation has the potential to cause the device to malfunction.
 - ☐ Are you familiar with the regulations for radiation safety for your state, province or country?
 - ☐ Philips recommends that when HU measurements are needed, reconstruct the raw data using filters which do not affect the HU numbers, such as the A, B or C filters. Absolute Hounsfield Units should never be used as the sole basis for any diagnosis.
 - ☐ Do NOT use O-MAR in the following instances:
 - External metals
 - Bismuth shields
 - Metal in or near intra-body air spaces
 - Small surgical implanted devices, i.e. screws, pins, clips, etc.

Patient Scanning with Dose Management

- ☐ I can scan patients using the Exam Cards including the following features:
- ☐ Scan Ruler
 - ☐ Result Direction/Acquisition Direction
 - ☐ Auto Enable
- ☐ I can perform the following functions:
- ☐ End exam
 - ☐ End exam and continue working
 - ☐ Work with Recon Queue Manager

- ☐ I understand how to produce acceptable quality images at an optimized radiation dose.
- ☐ I understand how to use the Dose Management tools for scanning adult and pediatric patients.
- ☐ I can scan patients using the Exam Cards and understand how the Acquisition and Results parameters assist in producing optimal quality images and affect patient dose.
- ☐ I can scan patients using exam cards that include the following dose management tools:
 - ☐ *DoseRight*
 - ☐ *DoseRight Index (DRI)*
 - ☐ *3D Modulation*
 - ☐ *Z- Modulation*
 - ☐ *Absolute Min and Max mAs*
 - ☐ *Reference Size*
 - ☐ *Patient Size*
 - ☐ *Liver Area DoseRight Index*
 - ☐ *Brain Area DoseRight index*
 - ☐ *Dose Notification – CTDI & DLP*

Scan Control Box

- ☐ I can explain how and when to use the following Scan Control Box options:
 - ☐ *Manual-Auto-Pause-Enable*
 - ☐ *Gantry Controls- Up/Down-In/Out*
 - ☐ *Gantry Key On-Off*
 - ☐ *Volume Controls- Console and Gantry Speakers*
 - ☐ *Patient Intercom*

Patient Directory, Archive Manager, and System Settings

- ☐ I can explain how and when to use the following Patient Directory tasks:
 - ☐ *Original vs. Derived Images*

- ☐ *Sub-selection*
- ☐ *Sort*
- ☐ *Recon*
- ☐ *Queue Manager*
- ☐ *Multimedia Viewer*
- ☐ *Quick Review*

☐**I can perform the following Directory Manager tasks**

- ☐ *Sort*
- ☐ *Copy to*
- ☐ *Recon*
- ☐ *Delete Studies*
- ☐ *Archive notification*
- ☐ *Hard Drive Maintenance*
- ☐ *Burn a CD (CDR)*
- ☐ *View a CD*

☐**I can identify when and how to select Scanning Option Preferences including the following:**

- ☐ *Display overlap warning*
- ☐ *Automatically move to "View" when the last scan is complete*
- ☐ *Enable SyncRight*
- ☐ *Switch to overview layout for editing before final recon*
- ☐ *Automatic Dual Surview*
- ☐ *Play sound along with blue arrow*
- ☐ *Automatically display the mAs profile for each scan*
- ☐ *Bolus Tracking help images*
- ☐ *Restore Default Helper Images*
- ☐ *Enable auto alert before X-Ray*

☐**I can identify when and how to select Reconstruction Preferences, including the following:**

- ☐ *Axial view convention*
- ☐ *Decubitus image view convention*

- ☐ *Automatic matrix*
- ☐ *Remove blank images*
- ☐ *Enable metal reduction*
- ☐ *Oncology features including:*
 - Enable Extended FOV
 - Enable 'CT Simulation Exam Card' and 'Crosscheck Slice locations' checkboxes in Exam Card Manager
- ☐ *CT Simulation Exam Card Presets including:*
 - Force Image Center X,Y to 0,0
 - Force FOV of all results in an acquisition to be equal
 - Disable Gantry Tilt
 - Disable Result Rotation
- ☐ **I can identify when and how to select Exam Summary Preferences:**
 - ☐ *Patient Information*
 - ☐ *Exam Information*
 - ☐ *Results*
 - ☐ *Executed Surview*
 - ☐ *Reference Surview*
 - ☐ *Injection Summary*
- ☐ **I can identify when and how to select Cardiac and Pulmonary Preferences**
 - ☐ *Heart-rate dependent phases*
 - ☐ *Always Detect Wave*
 - ☐ *Automatic Retrospective arrhythmia detection*
 - ☐ *Pulmonary Sensor Device*
 - Bellows
 - Varian
- ☐ **I can identify when and how to select Dose Management preferences:**
 - ☐ *Dose Alert DLP & CTDI*
 - ☐ *Display Dose Efficiency warning*
 - ☐ *Enable Reference Noise*
 - ☐ *Display mA*

- ☐ *Enable DoseRight*
- ☐ *Dose Modulation: enable special Head area DoseRight Index*
- ☐ *Dose Modulation: enable special Liver area DoseRight Index*
- ☐ *Enable DoseRight in head for Infant*
- ☐ *Enable DoseRight in head for Child*
- ☐ *Enable DoseRight in head for Adult*
- ☐ *Create Dose Structured Report*
 - Select Devices

☐ **I am familiar with the following additional preference options:**

- ☐ *Patient Data*
- ☐ *Connectivity*
- ☐ *Patient Directory*
- ☐ *Windowing Presets*
- ☐ *Image Titles*
- ☐ *Measurements*
- ☐ *Save Images*
- ☐ *Segmentation Preset*
- ☐ *Reporting*
- ☐ *Film Header/Footer*
- ☐ *Viewing Applications*
- ☐ *Institute Information*
- ☐ *Licensing*
- ☐ *Regional Settings*

Bolus Tracking and SyncRight

☐ **I am comfortable performing Bolus Tracking and SyncRight (if applicable) with patients, including:**

- ☐ *Building the Injection in the EC*

- ☐ *Modifying the injection from the Scan Ruler*
- ☐ *Using P3T (if applicable)*

Brain Perfusion

- ☐ **I can perform perfusion studies. (if applicable)**
 - ☐ *I understand that dose values for brain perfusion should be set at lower values than routine brain imaging.*

Reconstruction Modes, Physics Overview and Workflow Options

- ☐ **I can explain how to use the available Recon Modes (iDose4, IMR)**
- ☐ **I can explain the IMR image definitions for brain including:**
 - ☐ *Brain Routine*
 - ☐ *Sharp*
 - ☐ *SharpPlus*
- ☐ **I can explain the IMR image definitions for body including:**
 - ☐ *Soft Tissue*
 - ☐ *Routine*
 - ☐ *SharpPlus*
- ☐ **I can explain the IMR Image definition for cardiac including:**
 - ☐ *Cardiac Routine*
 - ☐ *Cardiac Sharp*
 - ☐ *Body Routine*
 - ☐ *Body Soft Tissue*
 - ☐ *Body SharpPlus*
- ☐ **I can recognize the clinical indications for using Cardiac IMR**
- ☐ **I can select appropriate parameters to customize Exam Cards to include iDose4, and IMR**

- ☐ I can determine which parameters to adjust to improve low contrast and spatial resolution
- ☐ I can use the IMR worksheet to improve dose optimization
- ☐ I can add Auto MPR's, MinIP, MIP and Volume to an Exam Card. how to select Dose Settings

Administrative Functions

- ☐ I can recover from errors.
- ☐ I can create a bug report.

IT/CT Administrator Role

- ☐ The Philips representative has assisted me in doing the following:
 - ☐ Establish security requirements
 - ☐ Create accounts and passwords for CT users
- ☐ I can identify my role as administrator and perform the following user management functions:
 - ☐ Lock and unlock accounts
 - ☐ Access the Audit Trail
 - ☐ Access the ECM
 - ☐ Create users
 - ☐ Reset passwords
 - ☐ Remove users
- ☐ I have been informed of the password policy
 - ☐ Login attempts settings
 - ☐ Password expiry settings
 - ☐ Password complexity requirements
- ☐ I can enable, disable, and configure Screen Blanking
- ☐ I can back up and restore user accounts
- ☐ I can use the audit trail viewer

Overview of Remaining CT Viewers and Applications

- ☐ I can select and use the following viewers:
 - ☐ *CT Viewer*
 - ☐ *2D*
 - ☐ *Slab*
 - ☐ *Volume*
 - ☐ *Endoscopy*
- ☐ I can select and use the following Analysis applications (as available):
 - ☐ *Coronary Calcium*
 - ☐ *Dental Planning*
 - ☐ *Brain Perfusion*
 - ☐ *Functional CT*
 - ☐ *Bone Mineral Density*

Lecture: Biopsy/CCT and Perfusion (if applicable)

- ☐ I can perform interventional/CCT and perfusion
- ☐ I can make the following system adjustments:
 - ☐ *1 or 3 image display*

CCT Tools (if applicable)

- ☐ *Biopsy mode*
- ☐ *CCT Single mode*
- ☐ *CCT Continuous mode*
- ☐ *CCT Fluoro mode*
- ☐ *Changing parameters during procedures*

- ☐ *Reference series*
- ☐ *Volume mode*
- ☐ *Interventional Console Controls*
 - ☐ *Saving Work Position*
 - ☐ *Save Needle Position*
 - ☐ *Move to Selected Position*
 - ☐ *Set incremental slice/table movement*
- ☐ *Interventional Hardware Controls*

Cardiac Overview

- ☐ **I can explain the anatomy and functionality of the heart including:**
 - ☐ *Chambers*
 - ☐ *Coronary vessels*
 - ☐ *Heart Rate effect on the Image Quality of Cardiac Scans*

Patient Prep, Skin Prep, and Proper ECG Lead Placement

- ☐ **I can prepare the patient for the exam, including:**
 - ☐ *Explain the cardiac exam*
 - ☐ *Providing breathing instructions*
 - ☐ *Performing skin prep*
 - ☐ *Properly positioning the patient for the for cardiac exam*
 - ☐ *Placing electrodes properly*
 - ☐ *Evaluate the patients HR during Breath hold*

ECG Viewer

- ☐ I can explain and demonstrate how to use the ECG Viewer tools during the scan:
 - ☐ Display ECG wave or HR graph
- ☐ I can explain and demonstrate how to use the ECG Viewer tools after acquisition:
 - ☐ ECG Editing tools
 - ☐ Additional Right Click Menus Options
- ☐ I can explain how to respond to the following warning messages:
 - ☐ ECG Disconnect Warning Message
 - ☐ ECG HR below or above ECG range Warning Message

Calcium Scoring

- ☐ I can perform a Calcium Score
 - ☐ Plan on Calcium Score/Plan on Previous Scan
 - ☐ Set Start/End for the next Cardiac Scan from the Preview images
- ☐ I can load to Coronary Calcium Scoring Analysis.
- ☐ I can perform a calcium score analysis and save the results.

Retrospective Cardiac

- ☐ I can perform a cardiac helical exam, including:
 - ☐ Expected heart rate (HR)
 - ☐ Auto Pitch and Rotation Time based on HR
 - ☐ Identify Coronary vs. Functional Phases
 - ☐ Use DoseRight and Cardiac DoseRight
 - ☐ Use Phase Tolerance with Cardiac DoseRight (iCT and IQon only)
 - ☐ Perform Single Cycle Reconstruction
 - ☐ Handle Irregularities Online

ECG Editing

☐ I can use arrhythmia handling tools, including:

☐ *Move an ECG tag*

☐ *Delete an ECG tag*

Cardiac Viewer

☐ I can use the following viewers and tools to analyze cardiac images:

☐ *2D/Slab*

☐ *Compare*

☐ *Link*

☐ *Paddle Wheel*

☐ *MIP*

☐ *Cardiac axes*

☐ *ECG evaluation*

☐ *Reset crosshairs*

☐ *Planar*

☐ *Area Length Ejection Fraction*

☐ *Creating Short Axis Series*

☐ *Cardiac cine*

☐ *Echo views*

Oncology - Pulmonary Toolkit and 4DCT

- ☐ **Sensor Device Preference**
 - ☐ *Bellows Device*
 - ☐ *3rd Party (i.e. RPM™)*
- ☐ **Hardware interface**
 - ☐ *Philips Bellows*
 - ☐ *3rd Party (i.e. RPM™)*
- ☐ **Acquisition Types**
 - ☐ *Resp 4D CT*
 - ☐ *Resp Breath Hold*
 - ☐ *Resp Free Breath*
 - ☐ *Resp Single Phase*
- ☐ **Generation and use of Pulmonary/4DCT Exam Cards**
 - ☐ *Reference Exam Cards*
 - ☐ *User Exam Cards*
- ☐ **Pulmonary Acquisition/Results/Reconstructions**
 - ☐ *Helical Scan*
 - ☐ *Phased Data*
 - ☐ *Untagged Series*
 - ☐ *Incorporation of iDose with 4DCT*
 - ☐ *I4DCT off-Line Reconstructions*
 - ☐ *Clinical use of 4DCT*
- ☐ **Binning Type**
 - ☐ *Phase vs. Amplitude*
- ☐ **Respiratory Waveform**
 - ☐ *Waveform Review*
 - ☐ *Waveform Tools*
 - ☐ *Tags and Tag Editing*

- ☐ *Breathing Statistics*
- ☐ **Pulmonary Viewer**
 - ☐ *2D/Slab/Planar/Volume*
 - ☐ *Waveform / Image Correlation*
 - ☐ *Batch and Cine*
 - ☐ *Image Review*

Oncology – Tumor LOC

- ☐ **Hardware/Operating System**
 - ☐ *Tumor LOC Hardware/Server*
 - ☐ *Tumor LOC – Scanner Interface*
 - ☐ *System Maintenance*
- ☐ **Patient Directory**
 - ☐ *Import Images*
 - ☐ *Edit / Delete Patient*
 - ☐ *Transfer Patient (between institutions)*
 - ☐ *Available Images – Series Select*
 - ☐ *Image Set Details / Edit Image Set Details*
 - ☐ *Primary and Secondary Datasets*
 - ☐ *Add / Delete Plan*
 - ☐ *Launch Tumor LOC*
- ☐ **Launch Pad Overview**
 - ☐ *Institutions*
 - ☐ *Physics*
 - ☐ *Simulator Machine Commissioning*
 - ☐ *Configure*
 - ☐ *Backup*

- ☐ ☐ *Restore*
- ☐ **Tumor LOC - General**
- ☐ *Preferences*
- ☐ *2D / 3D Visualization Tools*
- ☐ *2D / 3D Images, Optimization and Layouts*
- ☐ *Scripting*
- ☐ *Maximize Viewing Window*
- ☐ *DRRs and DCRs*
- ☐ *Common Tools*
- ☐ *Hotkeys*
- ☐ *Right Mouse Tools*
- ☐ *Help*
- ☐ *Save*
- ☐ **Tumor LOC - Workflow**
- ☐ *Initialize*
- ☐ *Regions of Interest*
- ☐ *Points of Interest*
- ☐ *Beams*
- ☐ *Finalize*
- ☐ **Tumor LOC – Simulation Clinical Workflow**
- ☐ *Simulation procedures*
- Point and Click isocenter localization
 - Beam geometry isocenter localization
 - Contour based isocenter localization
 - User defined isocenter localization
- ☐ *Absolute vs. Relative Marking*
- ☐ *Laser Export*

- ☐ *Tumor LOC Report*
 - Generate Report
 - Report Comments
 - Print Report
 - Export Report
- ☐ *Life Size Filming (if applicable)*
- ☐ *Clinical use of Tumor LOC for CT simulation*

**Tumor LOC – 4DCT Tools**

- ☐ *Generation of Intensity Projections (MIP, MinIP, AVE)*
- ☐ *IP Begin / End / Type*
- ☐ *IP Add / Delete*
- ☐ *Waveform Display*
- ☐ *Waveform Tools*
- ☐ *Waveform / CT Image correlation and review*
- ☐ *Cine Phases of Respiration*
- ☐ *Breathing Statistics*
- ☐ *Image data labeling*
- ☐ *Clinical use of Tumor LOC 4DCT tools*

**DICOM Image / RT Save and Export**

- ☐ *CT Images*
- ☐ *RT Plan*
- ☐ *RT Structure Set*
- ☐ *RT Image*

Oncology – Image Quality / Iterative Reconstruction

**Oncology Image Quality**

- ☐ *iDose (patient exposure reduction)*
- ☐ *iDose (image quality improvements)*

- ☐ *IMR (patient exposure reduction)*
- ☐ *IMR (image quality improvements)*
- ☐ *OMAR (image quality improvements)*
- ☐ *Offline Reconstructions with iDose / OMAR/ Phase editing*

Sign-In Sheet

Site#:

Site Name:

Training Dates:

Addendum to system configuration:

Yes

No

☐
☐

Off-site training packet received and reviewed?

☐
☐

Attended off-site training?

☐
☐

CEU criteria met?

☐
☐

Continuing Education training materials supplied to the customer?

☐
☐

Clinical Education line discussed with customer?

☐
☐

Service call procedure discussed with customer?

☐
☐

Do you feel the training objectives were met?

"Philips Clinical Education complies with ASRT Accreditation guidelines, and must document that attendees receiving CEUs have received and have been present for at least 80% of this training event. Signatures below acknowledge 80% or greater attendance at this Philips Clinical Education training event."

Customer signature/title:

Date:

Customer comments:

CES signature:

CES comments:

[illegible]

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Philips

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